

EMERGENCY CRICOTHYROTOMY

Clinical situation

Typically, this procedure is performed in a hypoxic patient with an obstructed airway who cannot be ventilated with a mask or LMA and cannot be intubated.

Emergency “Stab” Cricothyrotomy

Equipment

- Scalpel, preferably with no. 11 blade
- ETT no. 6
- Syringe for inflating cuff
- Prep solution such as alcohol, chlorhexidine, or povidone-iodine

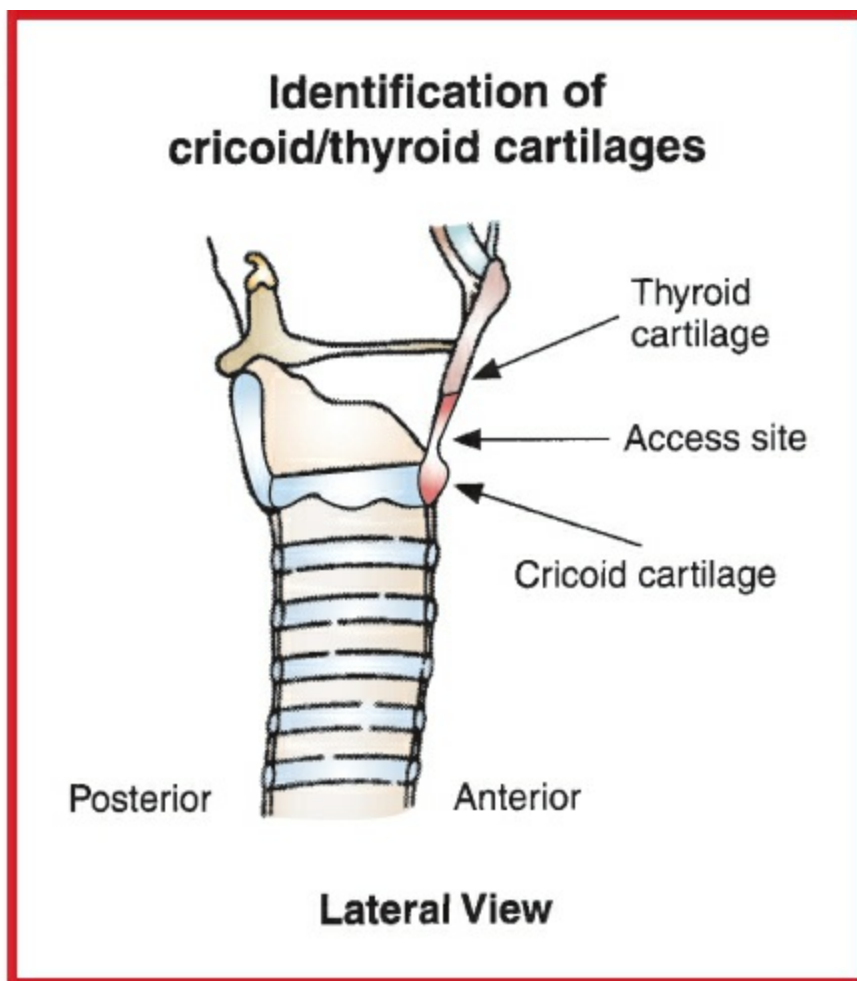


Figure 15.1 Identify cricoid membrane.



Incision for cricothyroidotomy

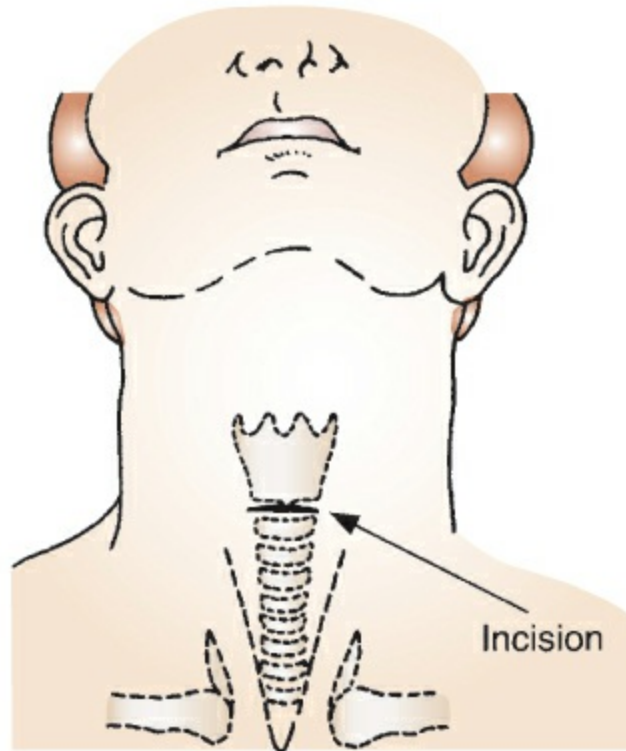


Figure 15.2 If possible, extend head and place bolster between scapula. Palpate thyroid and cricoid cartilages and then identify cricothyroid membrane (CTM). Skin incision may be vertical if unsure of CTM location, as a finger inserted through a vertical incision may have more ability to palpate the CTM by blunt pressure in a vertical or horizontal search.

Procedure

1. Prep skin if time permits.
2. Identify cricothyroid membrane between cricoid and thyroid cartilages ([Fig. 15.1](#)).
3. Make 2- to 3-cm transverse or vertical incision through skin.
4. Make 2.5-cm incision through cricothyroid membrane with single stab ([Fig. 15.2](#)).
5. Reverse scalpel, place handle into wound, and turn 90° to expand incision.
6. Pass tracheostomy tube (or standard ETT) into trachea.
7. Inflate cuff on tracheostomy/ETT.
8. Ventilate patient with capnography if possible.
9. Secure tube.

Emergency “Four-Step” Cricothyrotomy

Equipment	• Scalpel with no. 11 blade • Tracheal hook (may be improvised by bending a 16-ga needle into a hook with a hemostat) • ETT no. 6 • Syringe for inflating cuff • Prep solution such as alcohol, chlorhexidine, or povidone-iodine
Procedure	1. Right hand dominant operator stands on patient’s left side. Position with head and neck extended. 2. Prep skin if time permits. 3. Nondominant hand identifies cricothyroid membrane while stabilizing thyroid cartilage. Dominant hand makes 2.5-cm transverse incision through skin and cricothyroid membrane with single stab, keeping blade in place. 4. Nondominant hand places tracheal hook into incision, lifting cricoid cartilage ventrally, and then scalpel is removed. 5. Dominant hand inserts ETT and then removes tracheal hook. 6. Inflate cuff. 7. Ventilate patient with capnography if possible. 8. Secure tube.

Emergency “Bougie-Guided” Cricothyrotomy

Equipment	• Scalpel and hemostat (central line kit) • ETT no. 6 • Syringe for inflating cuff • Bougie (pediatric 10 Fr for ETT 4.0–5.5 or adult 15 Fr for ETT 6.0) • Prep solution such as alcohol, chlorhexidine, or povidone-iodine
Procedure	1. Operator may stand at head or either side of the patient. Position with head and neck extended. 2. Prep skin if time permits.

3. Nondominant hand identifies cricothyroid membrane while stabilizing thyroid cartilage.
 4. Dominant hand makes a 3-cm incision (vertical or transverse) through skin.
 5. Nondominant hand index finger placed into incision, palpates cricothyroid membrane with a side-to-side motion, and then finger is removed from incision.
 6. Dominant hand makes a 2.5-cm horizontal incision in the cricothyroid membrane.
 7. Dominant hand inserts coude tip end of bougie via incision seeking tracheal clicks and hang-up sign.
 8. Pass ETT over bougie into trachea. If difficult, use hemostat to dilate cricothyroid membrane with bougie in place and then pass ETT ([Fig. 15.3](#)).
 9. Remove bougie.
 10. Inflate cuff.
 11. Ventilate patient and monitor capnogram if possible.
 12. Secure tube ([Fig. 15.4](#)).
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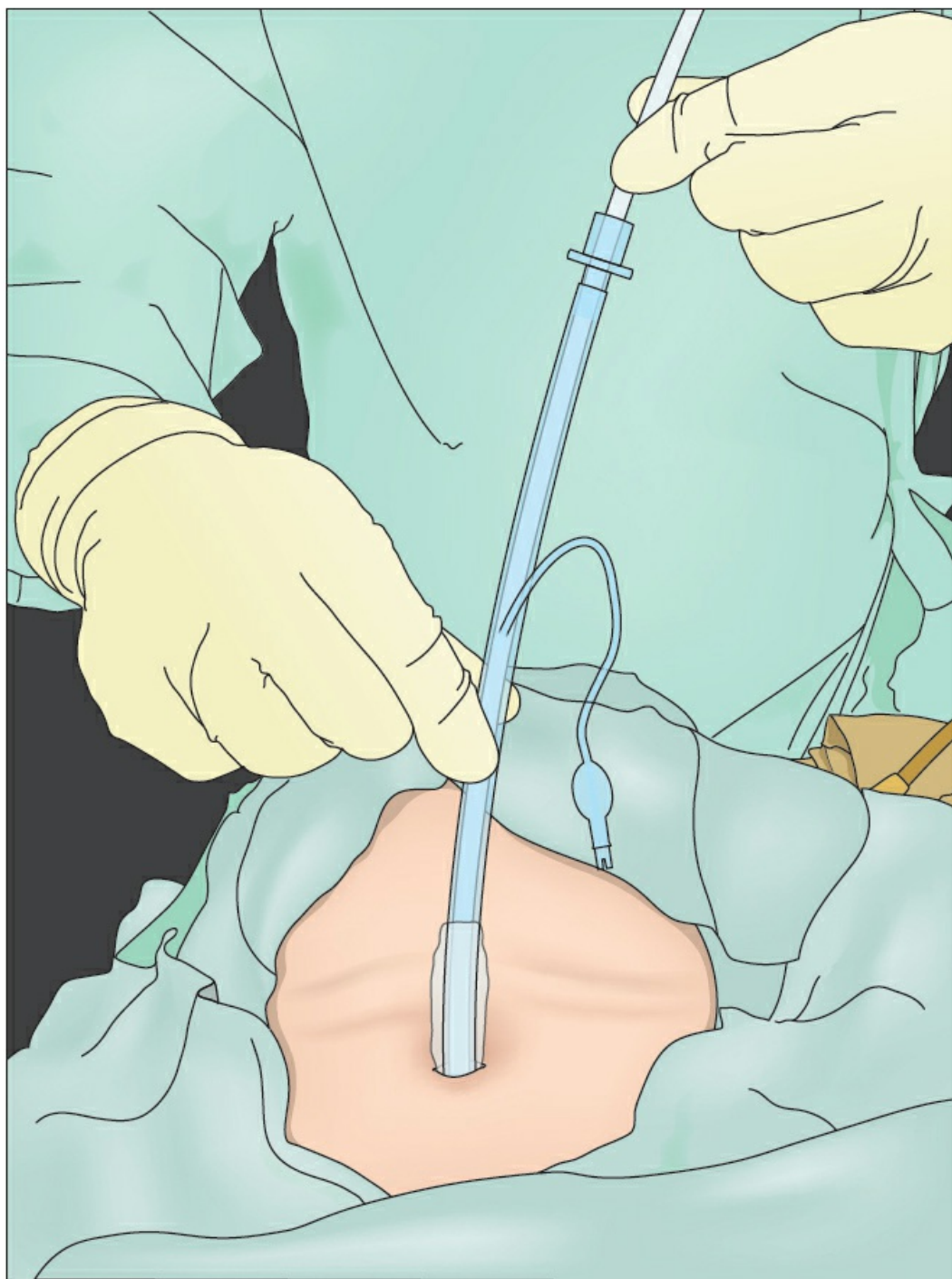


Figure 15.3 Adult bougie 15 Fr passed via incision with coude tip directed caudally attempting to feel for tracheal clicks and/or carinal hang-up sign. Average depth from skin to carina is 12–15 cm. A 6.0 or greater ETT will pass over an Adult 15 Fr bougie.

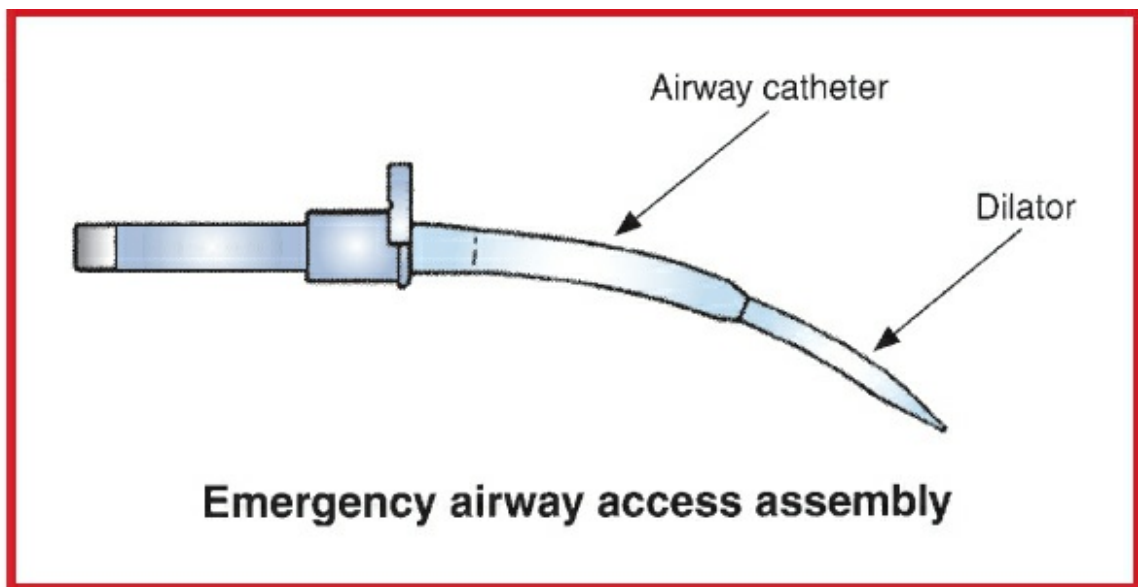


Figure 15.4 Melker cricothyroidotomy kit with dilator passed firmly into airway catheter prior to accessing the CTM.

EMERGENCY PERICARDIOCENTESIS

Clinical situation

The patient typically has severe ↓BP unexplained by any other causes (e.g., anesthetic drugs, autoPEEP, tension pneumothorax) and consistent with acute cardiac tamponade (dyspnea, tachypnea, ↓BP, ↑HR, electrical alternans, ↓voltage EKG, ↓pulse pressure or pulsus paradoxus, ↑CVP/JVD, distant heart sounds) ± equalization of pressures (RAP ~ RVEDP ~ PAD ~ PAOP) ± confirmation by TEE/TTE. Because of the high intrapericardial pressures, all “filling pressures” of both right and left heart appear high when preload is actually very low. In severe cases, patients will experience cardiac arrest with pulseless electrical activity (PEA).

Equipment	• 10-mL syringe • 18-ga spinal needle
Procedure	1. Identify xiphoid process and point 1 inch below and 1 inch left of midline (Fig. 15.5). 2. Prep skin below xiphoid. 3. An 18-ga spinal needle is directed toward the left shoulder and inserted at a $\leq 45^\circ$ angle to the skin. 4. Attach needle to syringe, and direct needle at $\leq 45^\circ$ to the skin under rib toward left shoulder (Fig. 15.5).

5. If pericardial fluid is withdrawn, BP will ↑ immediately.

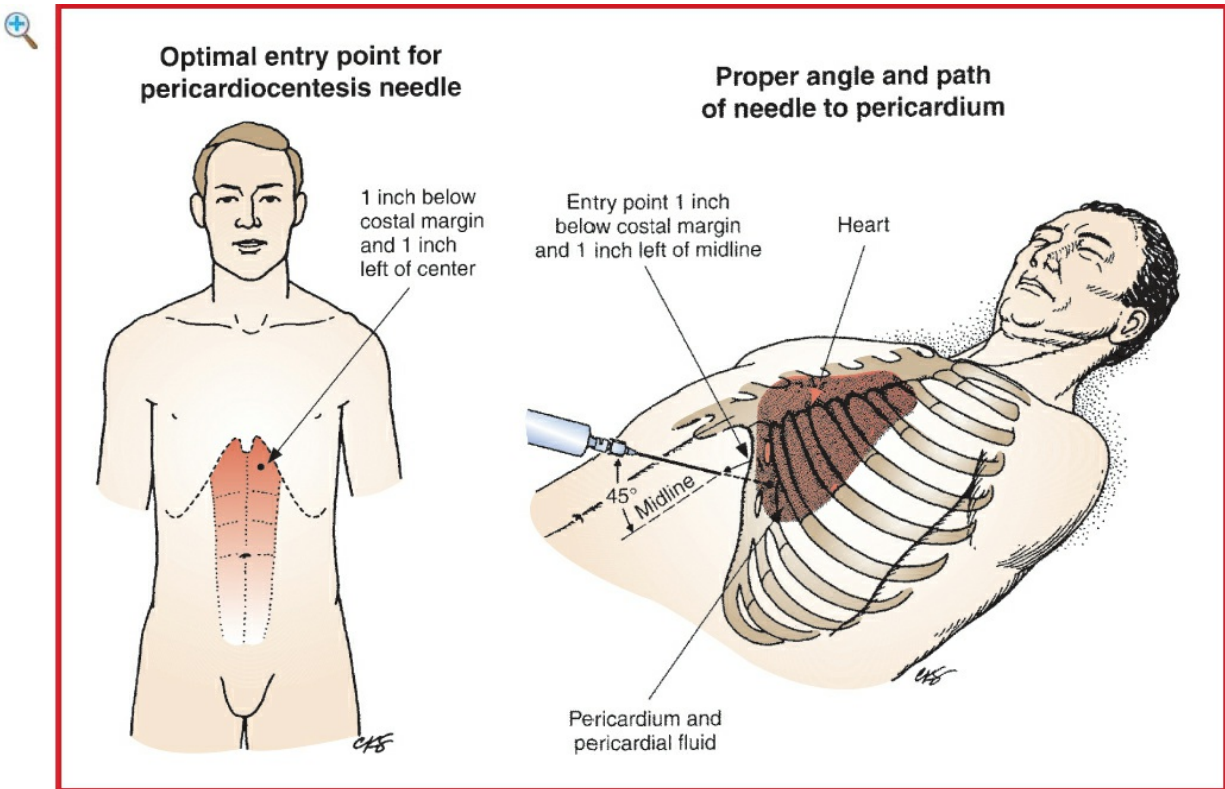


Figure 15.5 Emergency blind pericardiocentesis. Once diagnosis of cardiac tamponade is made, elevate head of bed to 30–45° to allow gravity to assist in fluid access. If hemodynamics do not allow this, place the patient into a supine position. The access site is 2.5 cm below the xiphoid process and 2.5 cm to the patient's left. An 18-ga spinal needle is directed toward the left shoulder and inserted at a $\leq 45^\circ$ angle to the skin. The stylet is removed, a syringe with stopcock attached, and the apparatus advanced with aspiration. If time allows, anesthetize the route with lidocaine 1–2% 5–10 mL.

EMERGENCY ULTRASOUND-GUIDED PERICARDIOCENTESIS^{1,5}

In the above-described setting of respiratory and hemodynamic stability, ultrasonography may be used to aid in diagnosis and treatment of cardiac tamponade. If you are unfamiliar with basic cardiac ultrasonography, a stat consult with a skilled ultrasonographer is necessary for performance of an ultrasound-guided pericardiocentesis. Precepted hands-on training must be sought prior to using ultrasound for diagnosis or treatment of cardiac tamponade.

Cardiac tamponade is a clinical diagnosis. Ultrasound findings include pericardial fluid with a “swinging heart” (ultrasound visualization of electrical alternans); increasing

pericardial pressure causes right atrial systolic collapse, right ventricle diastolic collapse, septal shift toward left ventricle in systole; respiratory variation across tricuspid and mitral valves; in addition, IVC plethora with no respiratory variation may be seen. Patients with normal, stable hemodynamics and pericardial effusion do not require emergent pericardiocentesis.

Equipment	Minimum • Skilled ultrasonographer • Ultrasound machine with phase array, curvilinear, or vascular access probe • Ultrasound gel or lubricant • Syringe • 18-ga 5-inch spinal needle or 18-ga Tuohy needle or longest 18-ga angiocatheter/needle available Preferred • Alcohol, chlorhexidine, or povidone-iodine skin prep • Sterile gloves • Skilled ultrasonographer • Ultrasound machine with phase array, curvilinear, or vascular access probe • Sterile ultrasound probe sheath with gel • 10-, 20-, or 60-mL syringe • 18-ga spinal needle • 3-way stopcock • Lidocaine 1–2% 5–10 mL with 5-mL syringe and 3-inch 25-ga needle
Subxiphoid Approach	1. Identify xiphoid process and point 1 inch below and 1 inch left of midline (Fig. 15.5). 2. Prep skin, then attach needle to 3-way stopcock on a syringe. 3. Use ultrasound probe with sterile sheath to find largest accessible collection of pericardial fluid in the subxiphoid space. A subxiphoid four-chamber view or short-axis midpapillary view should be attempted. Note that you may need to pass the needle through the liver to get to the pericardial space. If no chest compressions are being performed, consider positioning the patient in slight head-up or left lateral decubitus position to

optimize fluid accessibility. If the patient is conscious, consider local anesthetic injection along anticipated needle path under ultrasound visualization.

4. Place needle next to probe in same alignment to the skin and pericardial fluid (Fig. 15.6).
5. Advance needle tip under direct ultrasound visualization (Fig. 15.6).
6. When needle tip enters fluid collection, withdraw fluid. If flowing freely, continue to withdraw fluid until syringe is full, or you may turn the 3-way stopcock open to atmosphere to allow pericardial pressure to decompress. (If unsure of needle tip position, abort attempt or consider injection of 1- to 3-mL agitated saline via needle to identify location of tip relative to pericardial fluid.)
7. Remove needle once hemodynamics have improved with fluid removal.
8. Monitor patient for reaccumulation of pericardial fluid and consult for formal evaluation and possible pericardial drain placement.

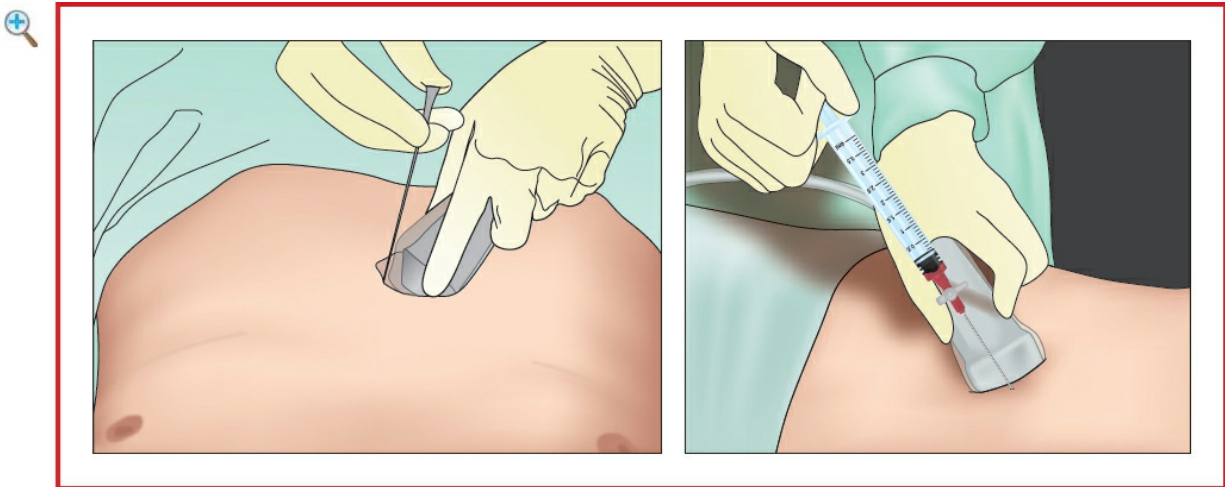


Figure 15.6 Emergency ultrasound-guided pericardiocentesis. Subxiphoid approach. Once ultrasonographic diagnosis of cardiac tamponade is made, elevate head of bed to 30–45° to allow gravity to assist in fluid access. If hemodynamics do not allow this, place the patient into a supine position. The subxiphoid access site is 2.5 cm below the xiphoid process and 2.5 cm to the patient's left. With the ultrasound probe in place, visualizing the effusion and the heart, an 18-ga spinal needle is directed toward the left shoulder and inserted at a $\leq 45^\circ$ angle to the skin. The stylet is removed, a syringe with stopcock attached, and the apparatus advanced with aspiration under direct visualization. If unsure of needle tip location, stop advancement and wait for expert help or consider injection of 1–3 mL of agitated saline to visualize tip location. If time allows, anesthetize the route with lidocaine 1–2% 5–10 mL. Both in-plane

and out-of-plane techniques are acceptable.

Parasternal Approach (Fig. 15.7)

Prep skin, then attach needle to 3-way stopcock onto syringe.

Use ultrasound probe with sterile sheath to find largest accessible collection of pericardial fluid in the parasternal area; 4th or 5th rib interspace along left sternal border. Note that you should not pass the needle through the pleural space/lung/vessels to enter the pericardial space. If patient is conscious, consider local anesthetic injection along anticipated needle path under ultrasound visualization.

Place needle next to probe in same alignment to the skin and pericardial fluid. You may use color flow Doppler to visualize vessels in intended needle path. Advance needle tip under direct ultrasound visualization.

When needle tip enters fluid collection, withdraw fluid. If flowing freely, continue to withdraw fluid until syringe is full, or you may turn the 3-way stopcock open to atmosphere to allow pericardial pressure/fluid to decompress. (If unsure of needle tip position, abort attempt or consider injection of 1- to 3-mL agitated saline via needle to identify location of tip relative to pericardial fluid.)

Remove needle once hemodynamics have improved with fluid removal.

Monitor patient for reaccumulation of pericardial fluid and consult for formal evaluation and possible pericardial drain placement.

Suggested Reading

1. Fitch MT, et al: Videos in clinical medicine. Emergency pericardiocentesis. N Engl J Med 2012; 366(12):e17.

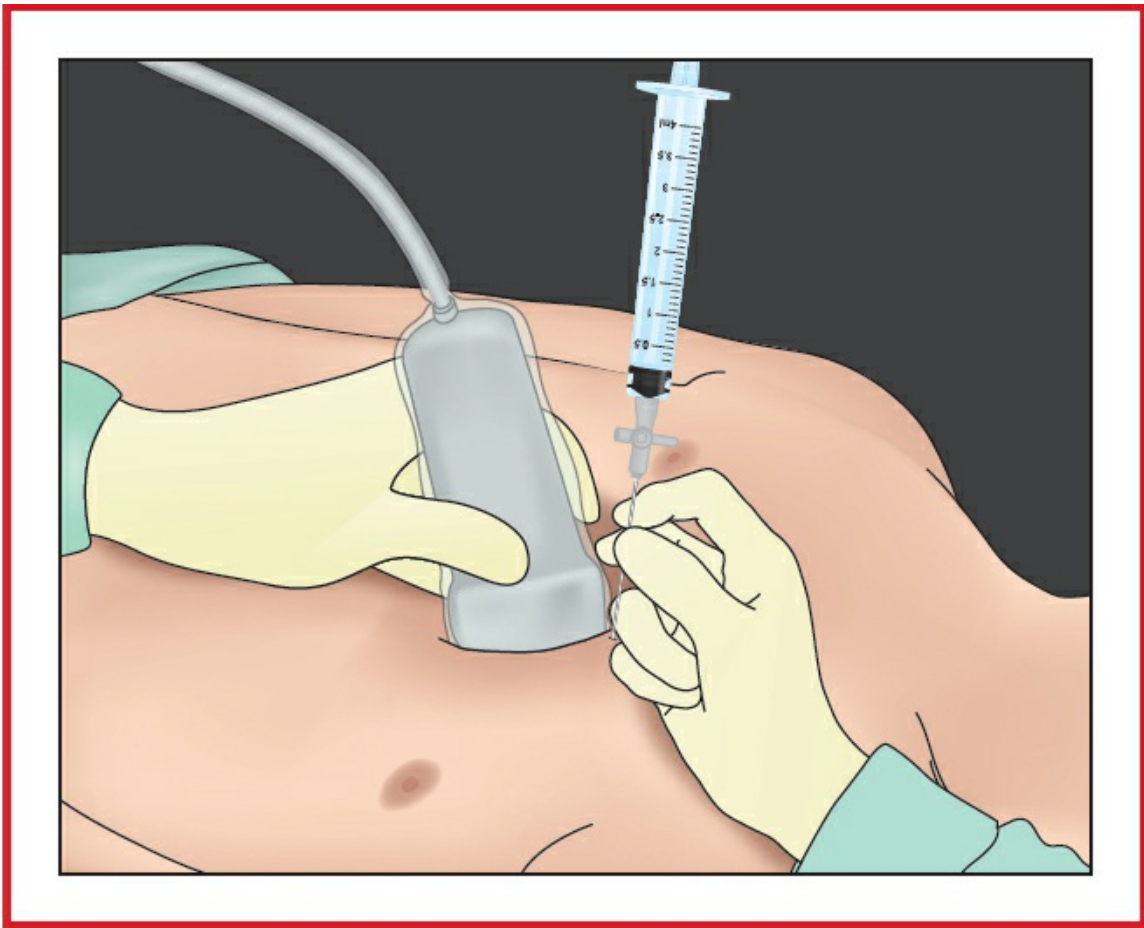


Figure 15.7 Emergency ultrasound-guided pericardiocentesis. Parasternal approach. Place the patient supine or in Trendelenburg to maximize fluid collection size in intended needle path. Identify the heart and fluid collection in the 4th or 5th intercostal space between the sternum and the midclavicular line. Be sure that the lung is not entering into ultrasonographic view. An 18-ga spinal needle is directed toward the patient's back and inserted at a 90° angle to the skin. The stylet is removed, a syringe with stopcock attached, and the apparatus advanced with aspiration under direct visualization. If unsure of needle tip location, stop advancement and wait for expert help or consider injection of 1–3 mL of agitated saline to visualize tip location. If time allows, anesthetize the route with lidocaine 1–2% 5–10 mL.

Suggested Viewing

Links are available online to the following videos:

Chest tube insertion: <https://www.youtube.com/watch?v=hQlt57AyQmg>

Emergency surgical airway: <https://www.youtube.com/watch?v=BwWlvJNecSg>

Cricothyrotomy: https://www.youtube.com/watch?v=_VFJGkJy1G8

Ultrasound-guided pericardiocentesis procedure—SonoSite ultrasound: <https://www.youtube.com/watch?v=T6Z9DvPPmXg>

Needle thoracostomy—www.proceduresconsult.com: <https://www.youtube.com/watch?v=4cuotNQPRNc>
(subscription required to view complete video)

Needle decompression (thoracostomy): <https://www.youtube.com/watch?v=TZpq8EPJ13Q&list=PLAB3A880AAEFB4992&index=2>

Needle decompression hazards: https://www.youtube.com/watch?v=fQqPktj_gFc

Venous cutdown.mp4: <https://www.youtube.com/watch?v=AHuJZHvTNGE>

EMERGENT NEEDLE/CATHETER THORACOSTOMY

Clinical situation

A patient is experiencing severe hypotension and hypoxemia unexplained by any other cause (e.g., autoPEEP, cardiac tamponade) and consistent with acute tension pneumothorax (dyspnea, tachypnea, tachycardia, syncope, $\uparrow P_{aw}$, $\downarrow$ movement of involved chest, $\downarrow$ breath sounds, $\uparrow$ resonance to percussion, and $\uparrow$ rib interspace distance compared to uninvolved side) and cardiovascular Sx related to $\uparrow$ thoracic pressures and $\downarrow$ preload ($\downarrow$ BP, $\uparrow$ HR, $\uparrow$ pulse pressure, $\uparrow$ CVP). The CVP is artificially elevated, reflecting the high intrathoracic pressures. Actual (transmural) CVP is very low. In severe cases, patients may experience cardiac arrest with pulseless electrical activity (PEA) because high intrathoracic pressures effectively stop venous return to the heart.

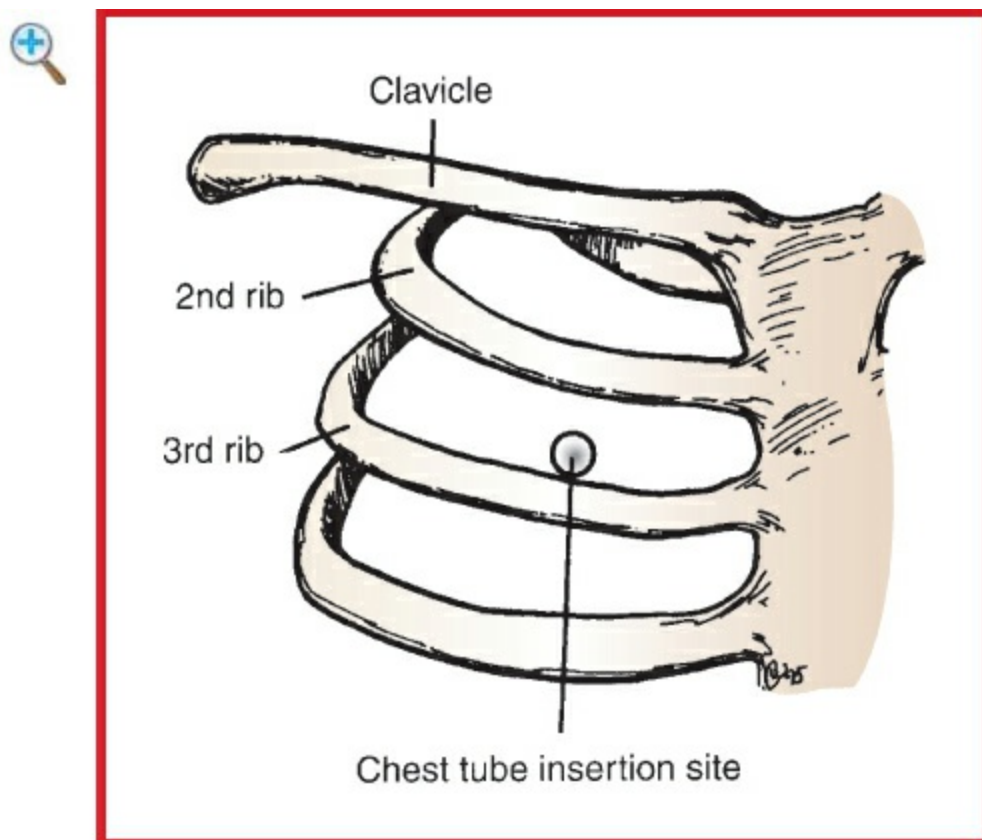


Figure 15.8 Midclavicular 2nd intercostal space. Lateral needle decompression in the 4th or 5th midaxillary line. To avoid injury to neurovascular bundle, insert needle along upper border of the inferior rib at the selected space.

Note: AutoPEEP MUST be ruled out first, particularly when the diagnosis is bilatera

tension PTX. AutoPEEP is effectively ruled out (except with a ball valve airway lesion) by disconnecting the ETT from ventilator/Ambu® bag and allowing patient to exhale. If the patient's condition immediately improves, you have made the diagnosis of autoPEEP.

Note: With a pneumothorax, release of the pressure that has built up in the chest is a lifesaving maneuver. A large-bore chest tube does not need to be placed emergently. The much simpler needle/catheter thoracostomy is effective and less demanding for the nonsurgeon.

Equipment	One of the following: • 16-ga needle or iv catheter/needle (preferred) with a length of 5 cm • 16-ga single-lumen CVP kit • Alcohol, chlorhexidine, or povidone-iodine skin prep
Procedure	1. Identify the space. Identify the 2nd intercostal space by palpating the angle of Louis (union of manubrium and sternum); then move laterally to find the insertion of the 2nd rib. The interspace directly below this rib is the 2nd intercostal space (Fig. 15.8). Access should occur in the midclavicular line in this space. Alternatively, the anterior or midaxillary line in the 4th–5th intercostal space may be used. Identify the anterior axillary line by drawing an imaginary line caudally from the anterior axillary fold to rib space at or above the nipple line. 2. Prep skin. 3. Enter the selected intercostal space anteriorly over the upper border of the inferior rib to avoid neurovascular bundle running along the underside of the superior rib. 4. With entry into the pleural space, there may be an audible rush of air and with immediate hemodynamic improvement. 5. Remove needle. 6. The angiocatheter may be held in place until a chest tube is inserted. 7. Note: A more permanent chest tube will still need to be placed (i.e., the patient no longer has a tension

pneumothorax, but still has a pneumothorax). This can be done in a more controlled situation by someone skilled in that procedure.

EMERGENT ULTRASOUND-GUIDED NEEDLE/CATHETER THORACOSTOMY

Tension pneumothorax diagnosis and management have been described earlier. If progressive pneumothorax with worsening cardiopulmonary function is suspected, a skilled ultrasonographer is needed to rule out pneumothorax prior to tension pneumothorax development. Lung ultrasound has >90% sensitivity and >90% specificity for pneumothorax.⁶ A vascular access, curvilinear, or phase array probe with a depth field of 6 cm using 2D imaging may be utilized to detect pneumothorax. 2D and M mode may be used to identify signs to rule in or out a pneumothorax. Precepted hands-on training must be sought prior to using ultrasound to diagnose or treat pneumothorax.

Equipment

- Skilled ultrasonographer
- Ultrasound machine with phase array, curvilinear, or vascular access probe
- Ultrasound gel or lubricant
- 16-ga needle or angiocatheter/needle

Procedure

1. Identify pneumothorax with ultrasound.⁷
2. Scan least dependent area of chest with ultrasound probe (in supine patients—3rd–5th rib interspace midclavicular line) at 6 cm in 2D with indicator toward the patient's head and held perpendicular to chest wall (Fig. 15.9).
3. Identify the pleural line and seek lung sliding. Lung sliding confirms that the lung is touching the chest wall. Presence of lung sliding rules out large tension pneumothorax.
4. Seek B lines in 2D. Presence of B line rules out large tension pneumothorax in the scanned space.
5. Check for seashore sign in M mode at site of lung sliding (Fig. 15.10). Presence of seashore sign rules out large tension pneumothorax in the scanned space.

6. Alternatively, absence of seashore sign and presence of bar code (Fig. 15.11) sign indicate pneumothorax. The lung point sign is a moving dark point seen at the interface of normal lung sliding to the air of the pneumothorax. If lung point sign is present, this indicates pneumothorax.
7. If all the above signs are absent and A lines are present, this does not rule out or in pneumothorax. Other modalities such as CXR or CT for diagnosis or continuing with diagnostic/therapeutic needle thoracostomy will be required.
8. Prep skin. Place angiocath/needle to side of probe.
9. Enter the selected intercostal space anteriorly over the upper border of the inferior rib to avoid neurovascular bundle; attempt to visualize needle tip.
10. With entry into the pleural space, there may be an audible rush of air, and there should be immediate hemodynamic improvement.
11. Remove needle once hemodynamics or ventilation improves to avoid lung laceration. The angiocatheter may be held in place until a chest tube is inserted.

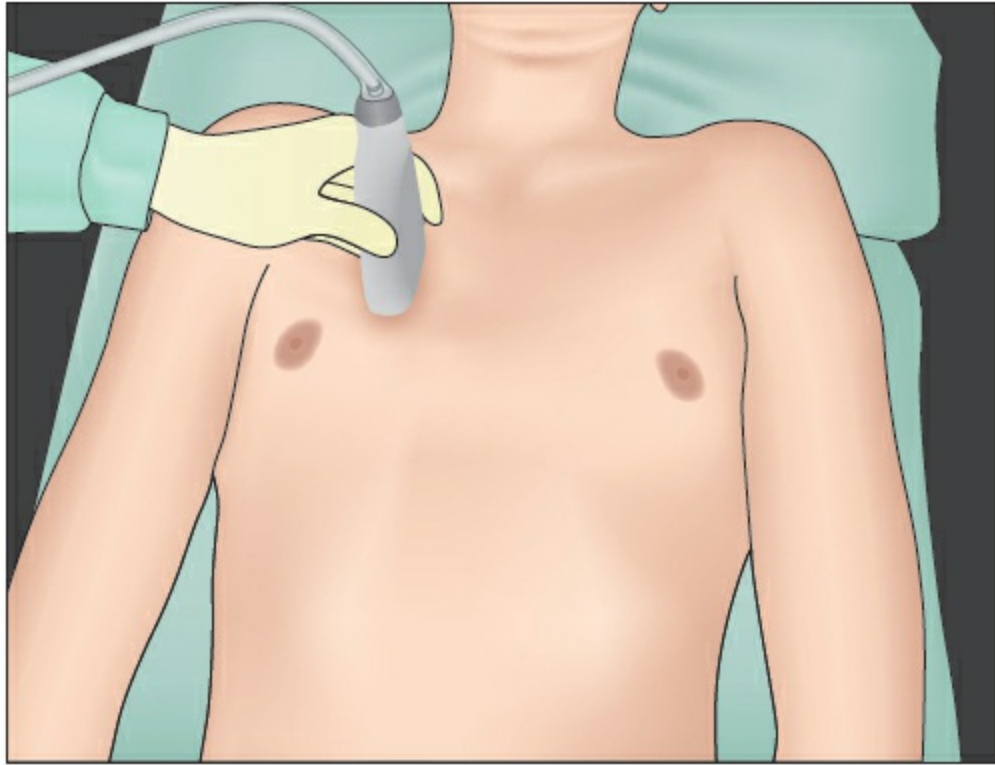


Figure 15.9 Place probe with indicator toward patient's head, perpendicular to chest wall, at the 3rd–5th intercostal space midclavicular line. In a supine patient, gas typically rises to this least dependent area of the chest cavity. You may move the probe caudally to scan multiple rib spaces in the midclavicular and axillary lines. (Ueda K, Ahmen W, Ross AF: Intraoperative pneumothorax identified with transthoracic ultrasound. *Anesthesiology* 2011; 115(3):653–5.⁸)

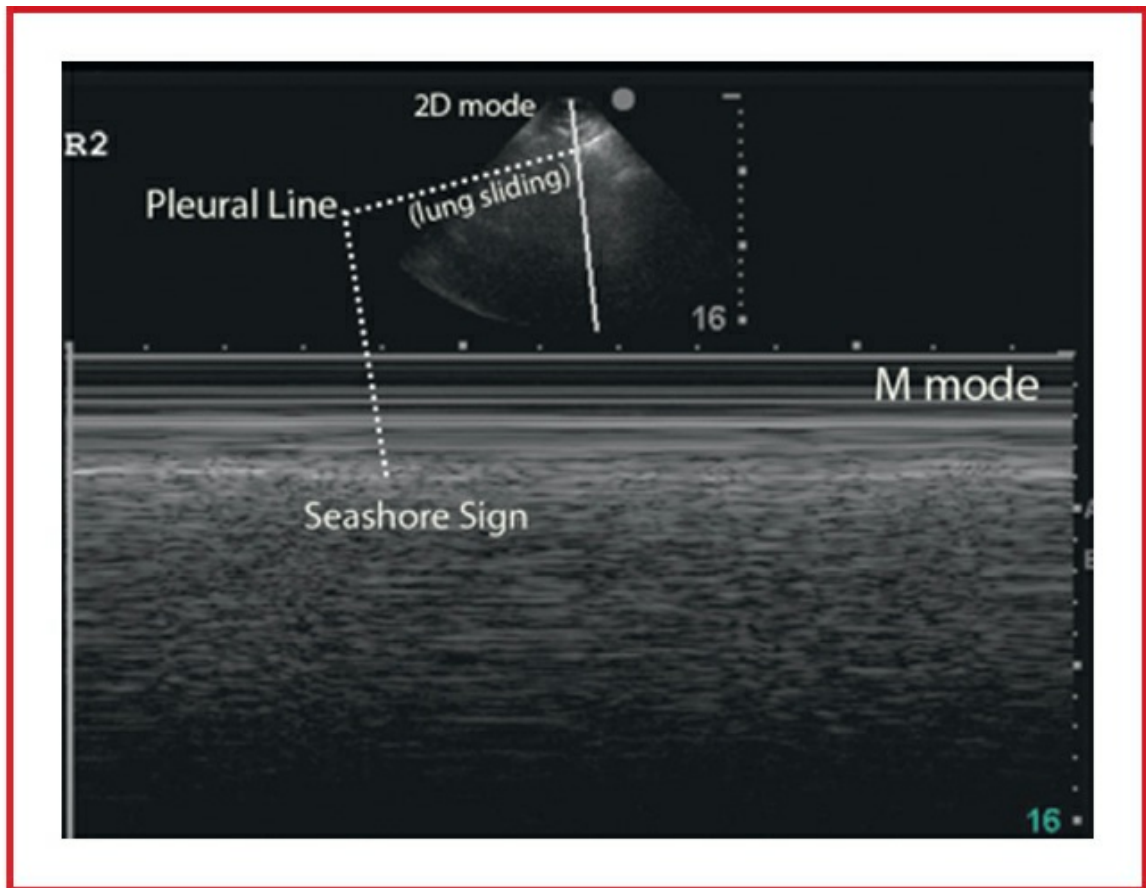


Figure 15.10 Seashore sign. M mode between the rib interspace with normal lung motion. The less mobile subcutaneous tissues and pleural tissues are seen as horizontal lines implying “waves” landing on a “sandy beach,” the moving lung tissue represented by granular pattern below the pleural line.

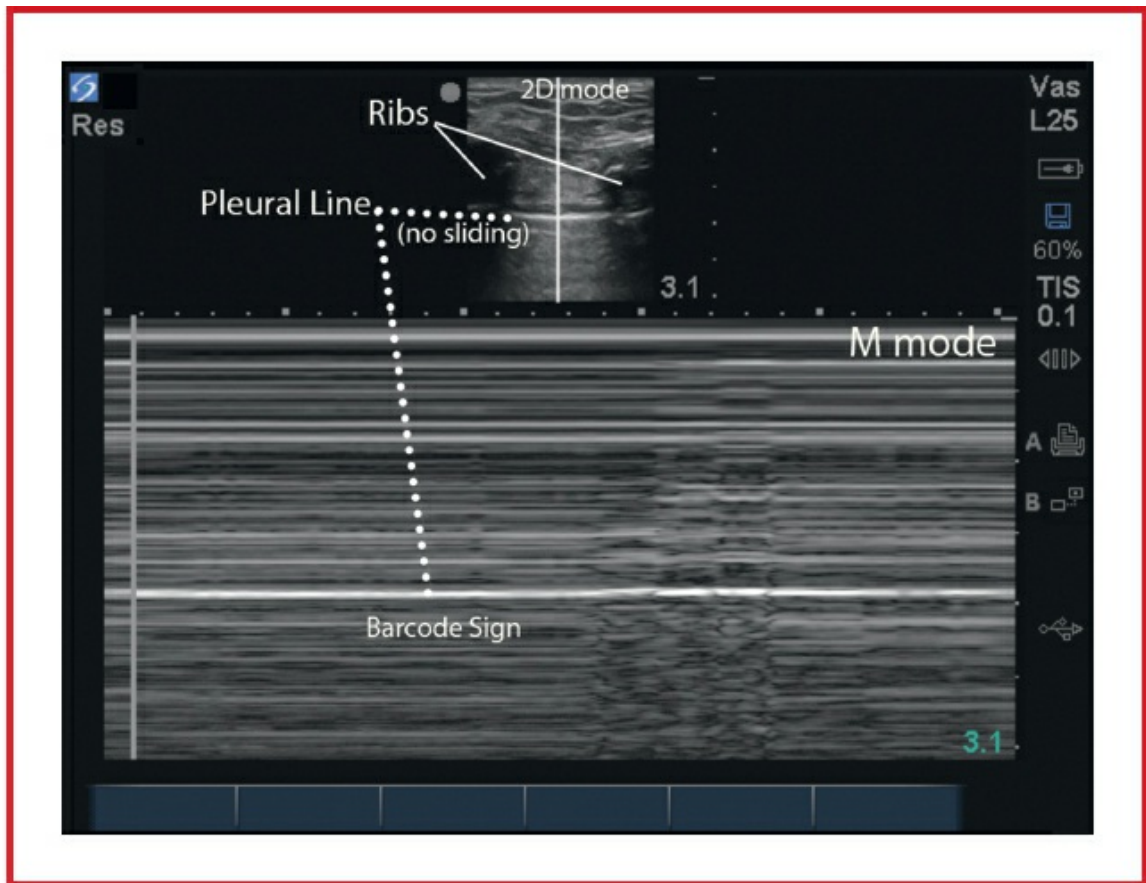


Figure 15.11 Bar code sign. During respiratory cycle, the “sand” (normal granular pattern) under the pleural line is replaced by “bar code” horizontal lines, indicating no lung sliding. (You-Ten KE, Siddiqui N, Teoh WH, et al: Point-of-care ultrasound (POCUS) of the upper airway. *Can J Anaesth* 2018; 65(4):473–84.⁹)

EMERGENCY INTRAOSSEOUS ACCESS

Emergency intraosseous (IO) access is indicated when immediate vascular access is needed, and peripheral/central access is not possible, and endotracheal medication administration will not suffice. We discuss the use of the EZ-IO system of IO access. Contraindications include burn, infection, or fracture of bone selected for access, joint replacement at selected site, or inability to identify landmarks.

Procedure

1. Prior to attaching the EZ-Connect®: ensure clamp is open. Prime set and purge air. Identify and clean insertion site (Fig. 15.12).
2. Snap EZ-IO needle set with plastic hub onto EZ-IO power driver (Fig. 15.13); (Red 15 ga, 15 mm for <40 kg. Blue 15 ga, 25 mm for >40 kg. Yellow hub 15 ga,

- 45 mm, >40 kg and excessive tissue or humerus site).
3. Remove needle safety cap.
 4. Gently press needle set through skin until tip touches bone.
 5. At least 5 mm of the catheter must be visible ([Fig. 15.14](#)).
 6. Squeeze driver trigger and apply gentle, steady pressure. Release trigger when sudden “give” or “pop” is felt, indicating entry into medullary space. If driver stalls or will not penetrate the bone, you may be applying too much downward force. Humerus site: to improve stability, needle should be advanced 2 cm after contact with bone. In adults, advance needle until catheter hub is against the skin. Change sites for access rather than have repeated attempts at the same site.
 7. Remove power driver and stylet; confirm catheter stability with at least three of the following four methods:
 - Stability of catheter
 - Ability to aspirate
 - Physiologic or pharmacologic changes with medication administration
 - Adequate flow rate
 9. Secure with EZ-stabilizer or have an assistant manually stabilize hub until secured, roll 2 sets of 4 × 4s into tight cylinders, apply cylinders in a parallel orientation around catheter, “sandwiching” the hub, then secure with tape or transparent adhesive film with hole cut out for hub access.
 9. Attach EZ-Connect® extension set. **DO NOT ATTACH A SYRINGE DIRECTLY TO THE EZ-IO CATHETER HUB EXCEPT WHEN DRAWING BLOOD FOR LABORATORY ANALYSIS (STABILIZE NEEDLE SET).**
 10. Flush the EZ-IO catheter with normal saline (10 mL for adults; use appropriate weight-based dose for smaller patients). For patients responsive to pain, consider lidocaine up to 0.2–1 mg/kg io slowly prior to saline

flush.

EZ-IO Removal

Attach Luer-lock syringe. Rotate syringe and catheter clockwise while using traction to withdraw catheter—DO NOT ROCK OR BEND THE CATHETER DURING REMOVAL ([Fig. 15.15](#)).

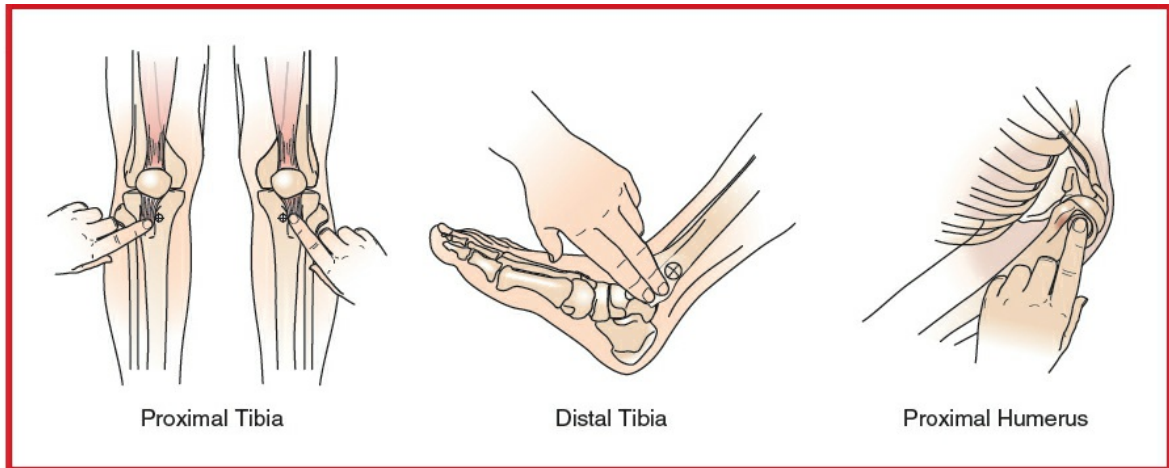


Figure 15.12 Identify and clean insertion site (proximal or distal tibia or humeral head).

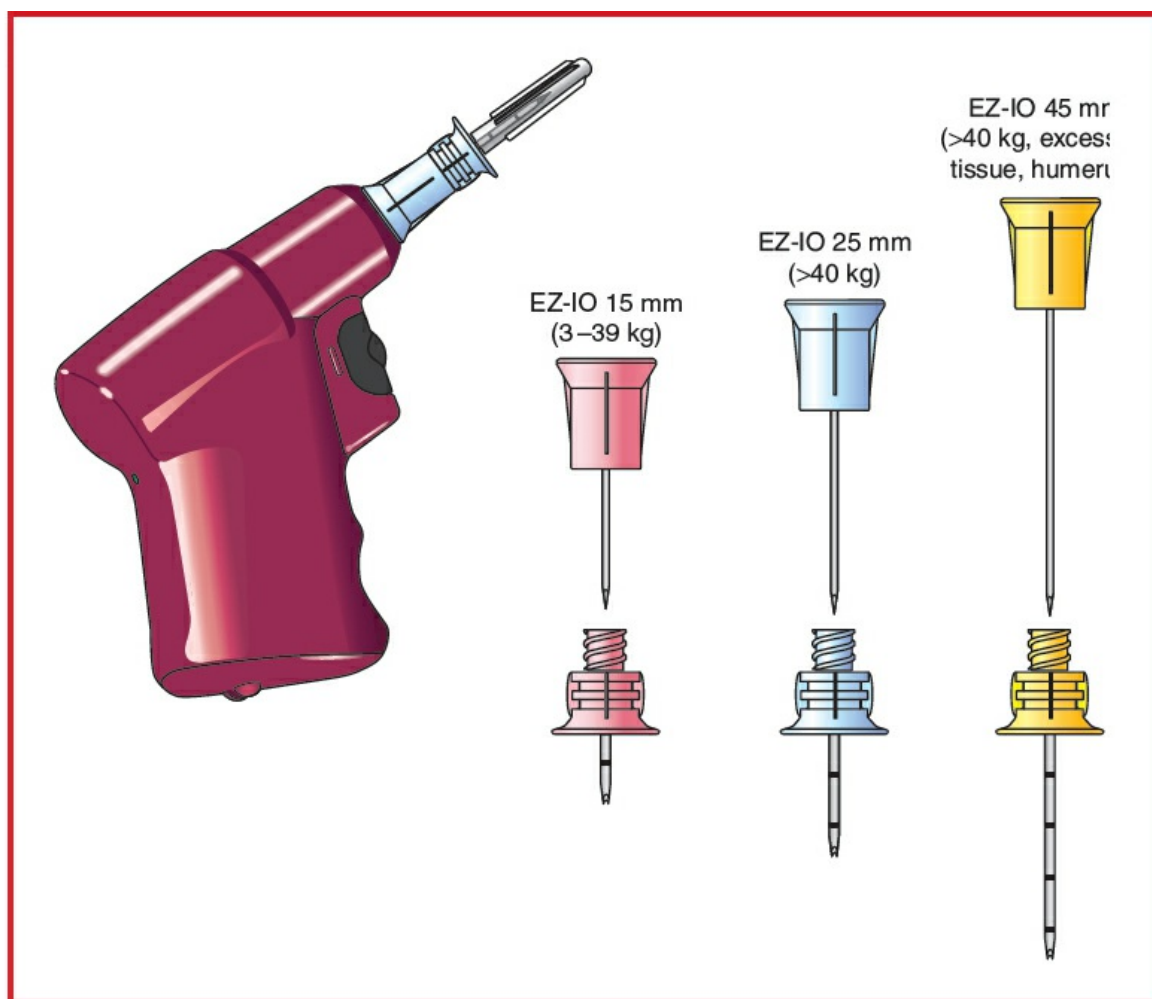


Figure 15.13 EZ-IO power driver and needle set sizes.

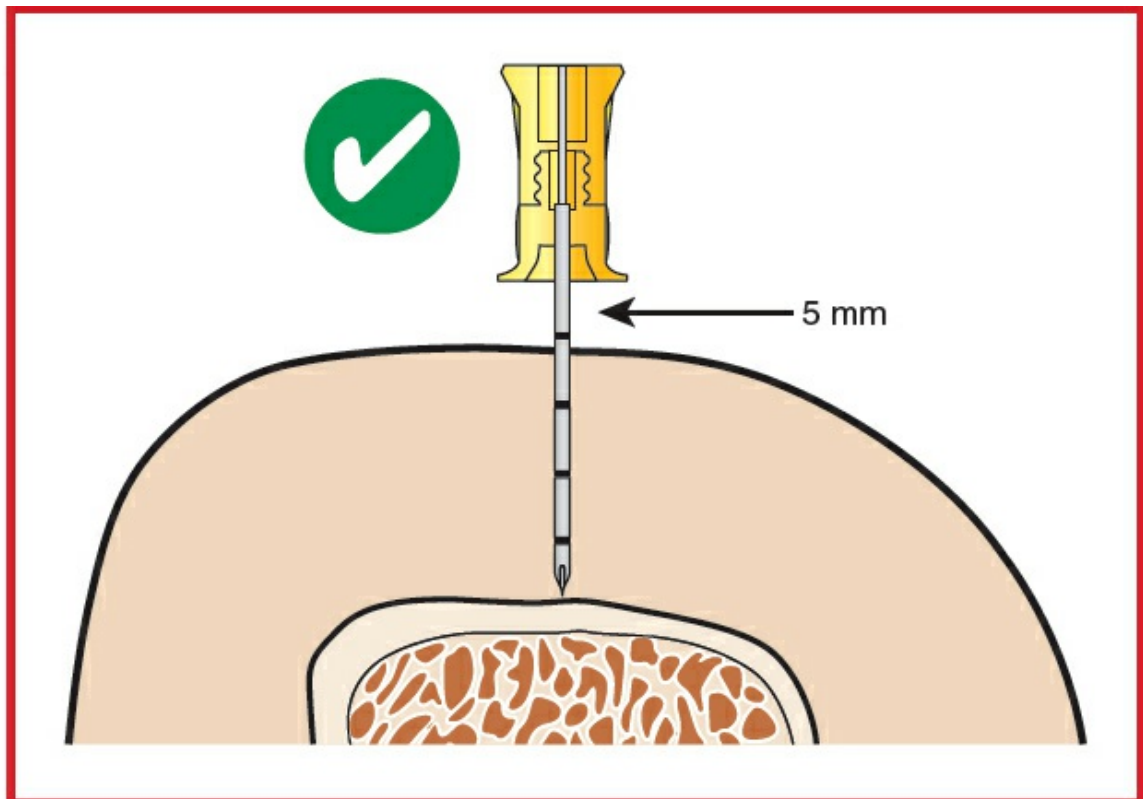


Figure 15.14 At least 5 mm of the catheter must be visible after pressing needle through skin to contact bone and prior to squeezing trigger.

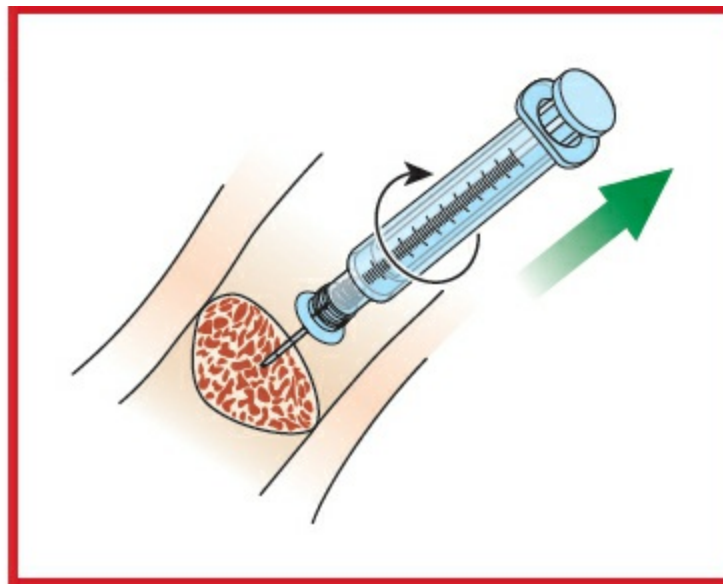


Figure 15.15 To remove EZ-IO catheter, attach Luer-lock syringe. Rotate syringe and catheter clockwise while using traction to withdraw catheter.

References

1. Lichtenstein D: Whole Body Ultrasonography in the Critically Ill Springer, New York: 2010.

2. Alrajhi K, Woo MY, Vaillancourt C: Test characteristics of ultrasonography for the detection of pneumothorax: a systematic review and meta-analysis. *Chest* 2012; 141(3):703–8.
3. Brofeldt BT, Panacek EA, Richards JR: An easy cricothyrotomy approach: the rapid four-step technique. *Acad Emerg Med* 1996; 3(11):1060–3.
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3. Ueda K, Ahmen W, Ross AF: Intraoperative pneumothorax identified with transthoracic ultrasound. *Anesthesiology* 2011; 115(3):653–5.
9. You-Ten KE, Siddiqui N, Teoh WH, et al: Point-of-care ultrasound (POCUS) of the upper airway. *Can J Anaesth* 2018; 65(4):473–84.

SECTION 16.0

REGIONAL BLOCKS—A BRIEF OVERVIEW

ANESTHESIOLOGISTS

Jean-Louis Horn, MD

Mastoor Nasiri, MD

UPPER EXTREMITY BLOCKS

INTERSCALENE

Indications

Shoulder surgery

Anatomy

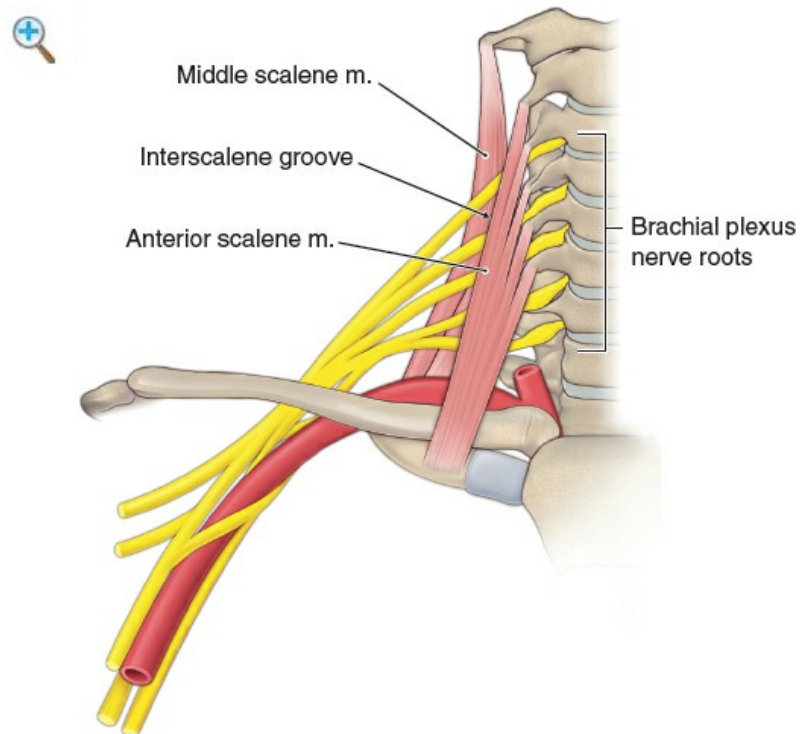


Figure 16.1 Interscalene anatomy. Superior trunk of the brachial plexus (C5-C6).

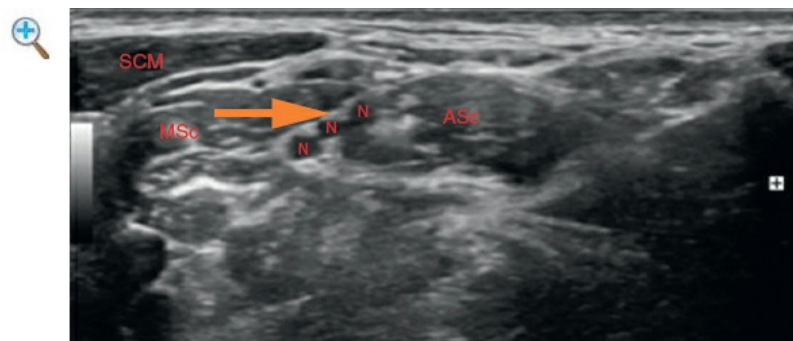


Figure 16.2 Interscalene ultrasound. SCM, sternocleidomastoid; MSc, middle scalene; ASc, anterior scalene; N, nerve.

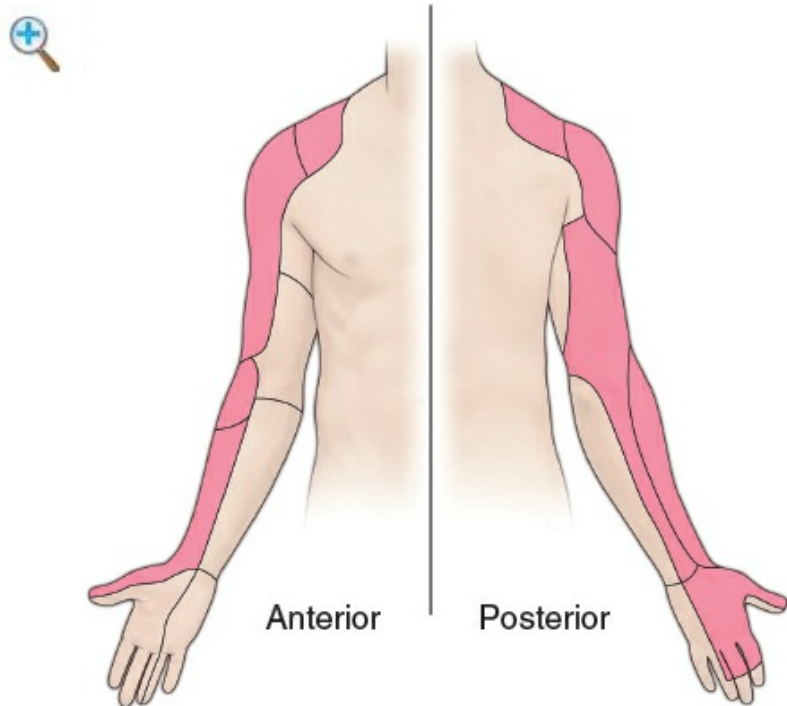


Figure 16.3 Interscalene anatomy.

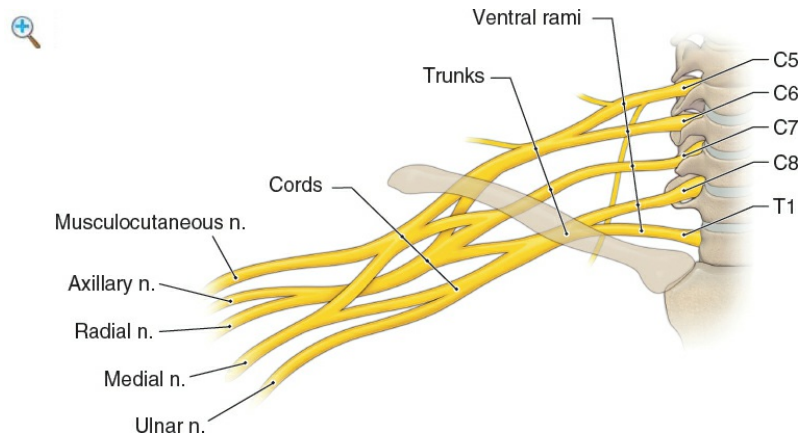


Figure 16.4 Interscalene anatomy.

Dosage	10–20 mL (0.25–0.5% ropivacaine/bupivacaine or 1–1.5% mepivacaine)
Nerve stimulation	Deltoid, distal extensor or lateral flexor
Side effects	Phrenic nerve block (nearly 100% with higher volume local anesthetic) Horner syndrome (rare, risk increases with high volume, fast injection) Recurrent laryngeal nerve block (rare, risk increases with high volume, fast injection) Superficial cervical plexus block extending to the ear

Complications	Permanent phrenic nerve palsy → hemidiaphragm paralysis: very rare Pneumothorax: low risk compared to more distal brachial plexus blocks Dorsal scapular nerve injury → winged scapula Long thoracic nerve injury → both DSN and LTN injury resulted in winged scapula can/or pain
Pearls	• Patient can be placed in the supine or lateral decubitus position • Ultrasound probe placed transversely with a 15–30° caudal angle, 3–4 cm above clavicle (level of cricoid cartilage) over external jugular vein with an in-plane or out-of-plane approach • Avoid injecting between C5 and C6 as this can increase risk of intraneural injection and side effects • Supplement with an intercostobrachial nerve block to cover the medial upper arm when needed (infrequent) • Special contraindications: contralateral phrenic nerve paresis; severe pulmonary disease
Alternative	Combined distal suprascapular and axillary nerve blocks Supraclavicular block Proximal suprascapular nerve block
Further reading	Mian A, et al: Brachial plexus anesthesia: a review of the relevant anatomy, complications, and anatomical variations. Clin Anat 2014; 27(2):210–21.

SUPRACLAVICULAR

Indications	Shoulder, arm, elbow, forearm, and hand surgery
Anatomy	

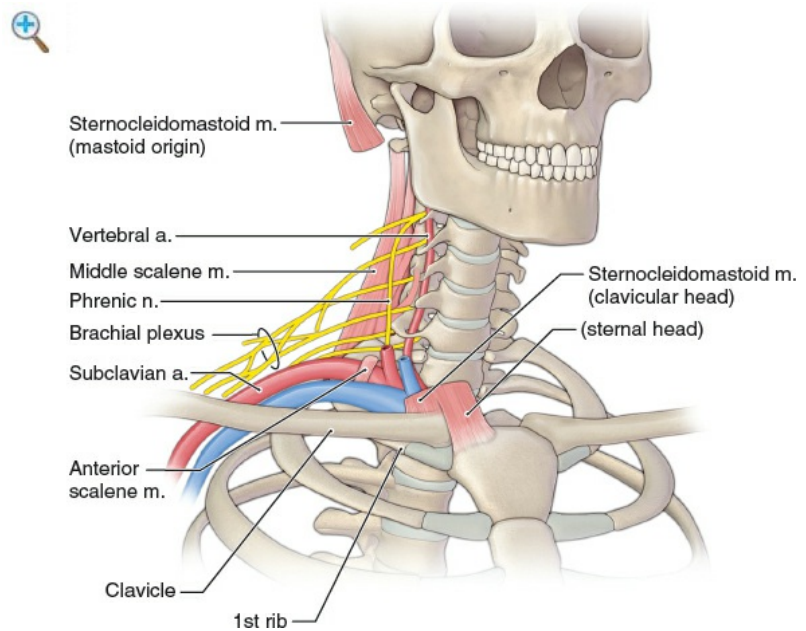


Figure 16.5 Supraclavicular anatomy. Often branches of the thyrocervical trunk cross the divisions creating a fascia preventing spread of local anesthetic. Divisions of the brachial plexus (anterior and posterior) lie mostly posterior to the artery.

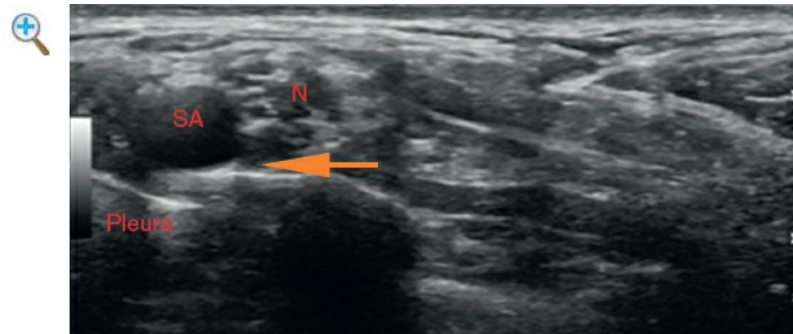


Figure 16.6 Supraclavicular ultrasound. SA, subclavian arter; N, nerve.

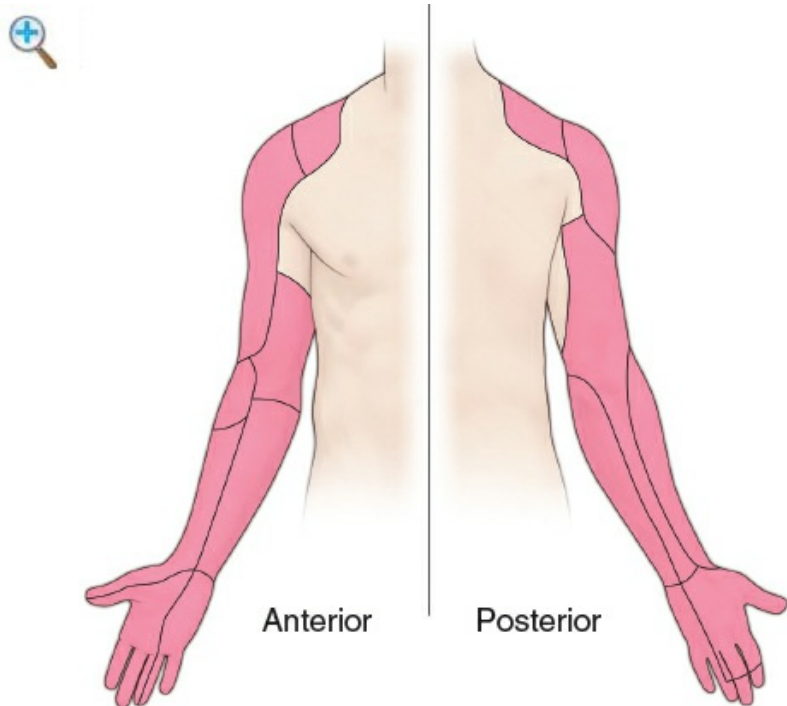


Figure 16.7 Supraclavicular anatomy.

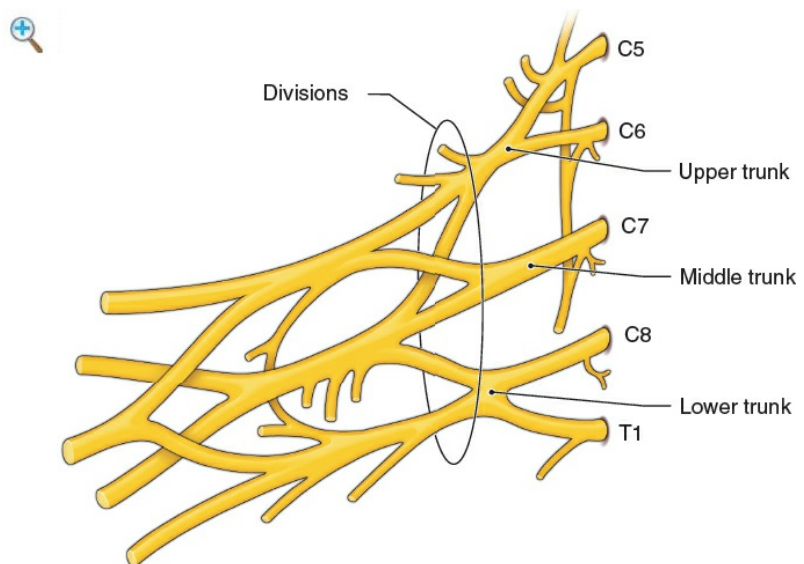


Figure 16.8 Supraclavicular anatomy.

Dosage	15–30 mL (0.5% ropivacaine/bupivacaine or mepivacaine 1–1.5%)
Nerve stimulation	Elbow, forearm, distal arm, hand
Side effects	Phrenic nerve block (40%) → hemidiaphragm paralysis Horner syndrome (very rare)

Complications	Pneumothorax: rare with ultrasound guidance Puncture of overlying vasculature: suprascapular, superficial cervical, transverse cervical, and dorsal scapular arteries (<1%) Suprascapular nerve injury → ↓ abduction and external rotation (supraspinatus + infraspinatus muscles)
Pearls	• Slightly prop the spine up to give space for injection. • Place ultrasound probe immediately parallel to the clavicle in the supraclavicular fossa and adjust the beam to place the target above the first rib to shield the lung. • Use a short needle to minimize chance of pneumothorax. • Success is associated with LA spread under the divisions next to the subclavian artery described as the corner pocket; however, fasciae often present barriers necessitating multiple site injections. • This is not the best site for catheter placement. • Supplement with an intercostobrachial nerve block to cover the medial upper arm when needed (infrequent). • Special contraindications: contralateral phrenic nerve paresis; severe pulmonary disease
Alternative	Axillary or infraclavicular block (especially with significant pulmonary disease to avoid phrenic nerve block)
Further reading	Vermeylen K, et al: Supraclavicular brachial plexus blocks: review and current practice. Acta Anaesthesiol Belg 2012; 63(1):15–21.

INFRACLAVICULAR

Indications	Arm, elbow, forearm, and hand surgery
Anatomy	

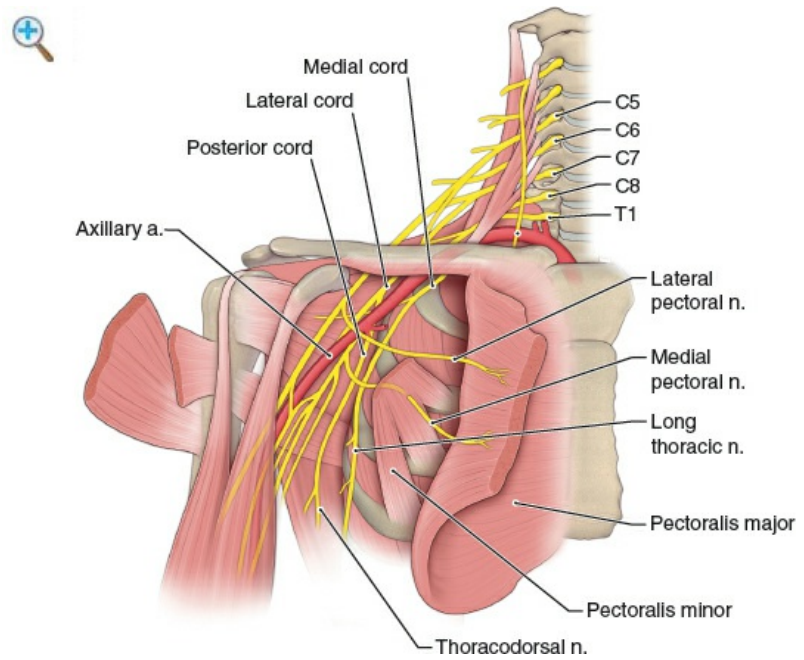


Figure 16.9 Infraclavicular anatomy.

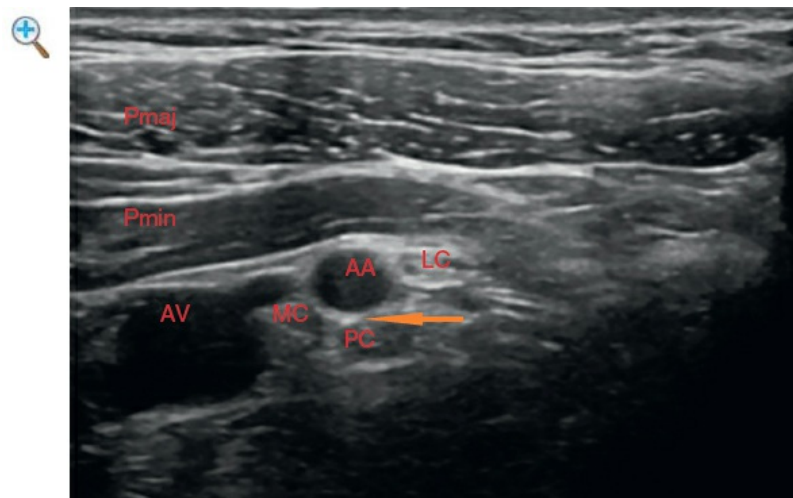


Figure 16.10 Infraclavicular ultrasound. Pmaj, pectoralis major; Pmin, pectoralis minor; AV, axillary vein; AA, axillary artery; LC, lateral cord; PC, posterior cord; MC, medial cord.



Anterior

Figure 16.11 Infraclavicular anatomy.

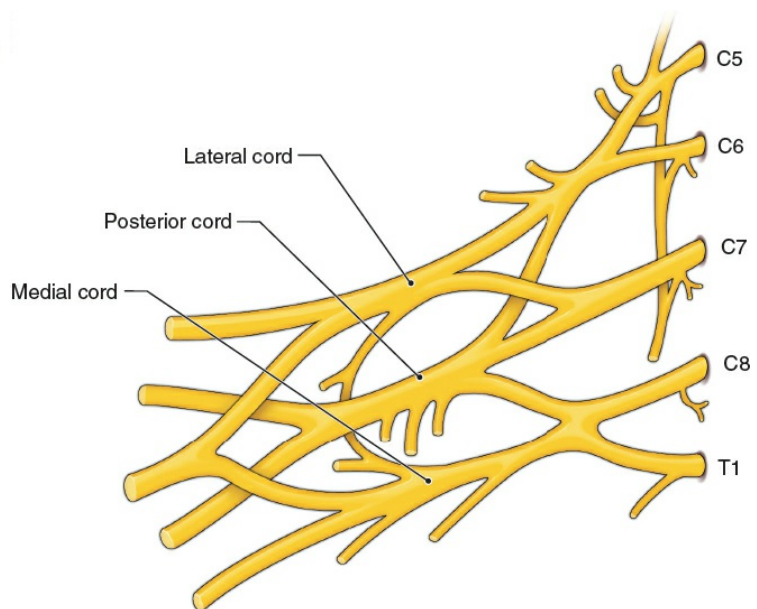


Figure 16.12 Infraclavicular anatomy.

Dosage	20–40 mL (0.5% ropivacaine/bupivacaine or mepivacaine 1–1.5%)
Nerve stimulation	Distal stimulation at the hand

Side effects	Minimal
Complications	Pneumothorax: rare with ultrasound guidance Vascular injury: <1% Lateral pectoralis nerve injury → ↓ pectoralis major muscle strength
Pearls	• Abduct the arm at the shoulder 90° to elevate the clavicle and open access to the cords. Place the probe in a sagittal plane. • For painful shoulder or arm, keep the arm adducted, rotate and translate the probe against the clavicle for a costoclavicular approach. Then tilt the probe slightly under the clavicle. • Success is associated with LA spread under the artery often described as the double bubble or eight ball for the sagittal approach. For the costoclavicular approach, the target is the corner pocket similar to the supraclavicular approach. • Supplement with an intercostobrachial nerve block to cover the medial upper arm when needed (infrequent).
Alternative	Supraclavicular block Axillary block
Further reading	Chin KJ, et al: Infraclavicular brachial plexus block for regional anaesthesia of the lower arm. Anesth Analg 2010; 111(4):1072.

AXILLARY BLOCK

Indications	Forearm and hand surgery
Anatomy	

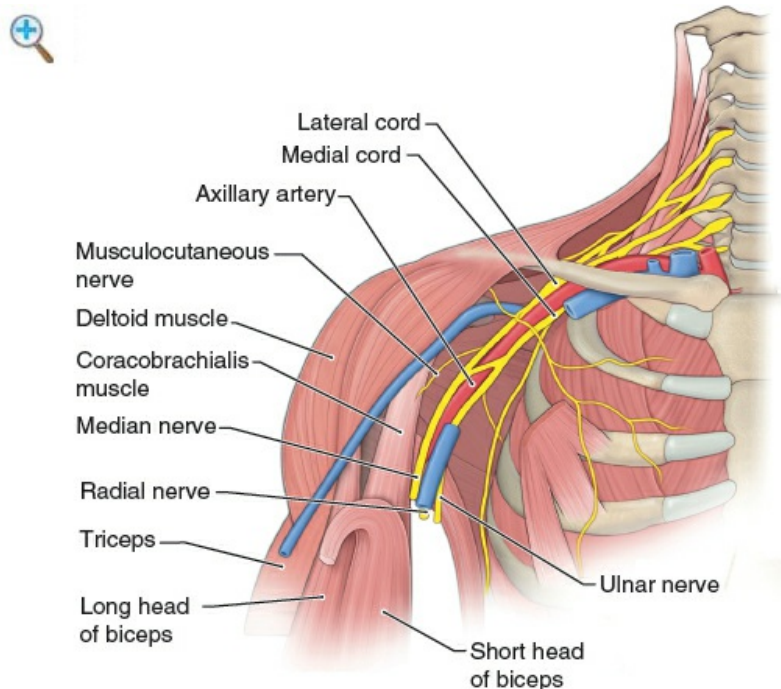


Figure 16.13 Axillary block anatomy. Terminal branches of the brachial plexus (median, ulnar, radial) surround the axillary artery. Musculocutaneous nerve is lateral to the artery and runs through the coracobrachialis muscle and then between the biceps brachialis and coracobrachialis muscles.



Figure 16.14 Axillary block ultrasound. AA, axillary artery; MCN, musculocutaneous nerve; MN, median nerve; RN, radial nerve; UN, ulnar nerve.

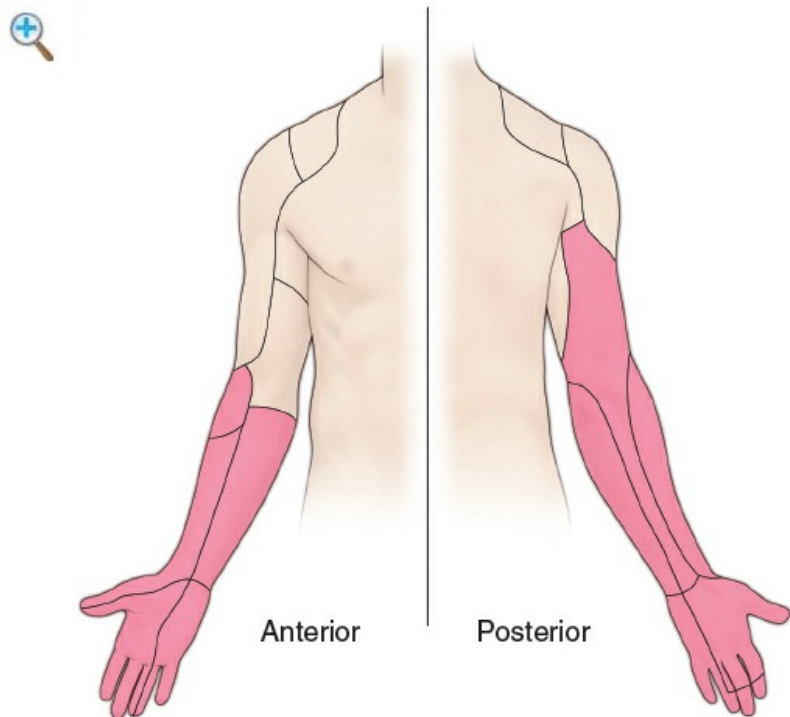


Figure 16.15 Axillary block anatomy.

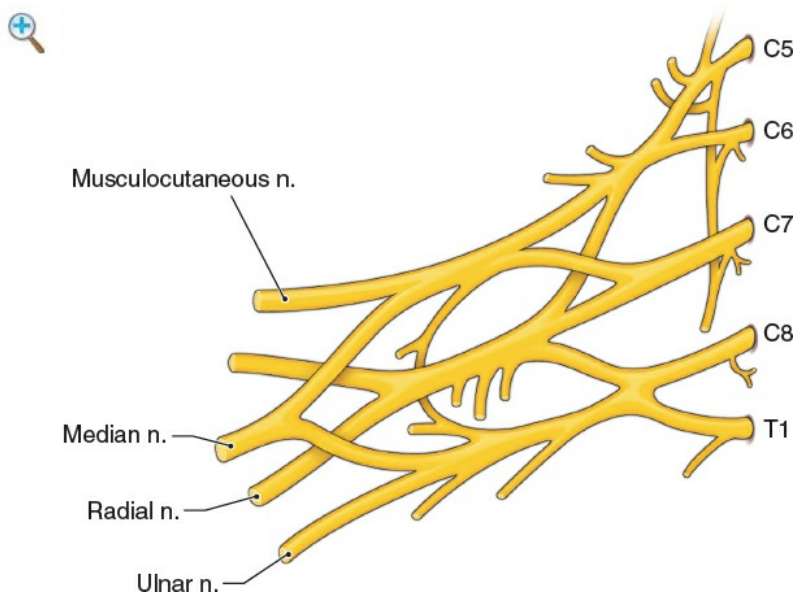


Figure 16.16 Axillary block anatomy.

	Musculocutaneous nerve is lateral to the artery and running through the coracobrachialis muscle and then between the biceps brachialis and coracobrachialis muscles
Dosage	20–40 mL (0.5% ropivacaine/bupivacaine or mepivacaine 1–1.5%)

Nerve stimulation	Distal stimulation at the hand, biceps brachialis stimulation for the musculocutaneous nerve
Side effects	Minimal
Complications	Intravascular injection/injury (<1% with ultrasound guidance)
Pearls	• Supplement with an intercostobrachial nerve block to cover the medial upper arm and improve tourniquet tolerance. • Separate injections for each nerve hasten the onset and increase success. • Catheter will only cover the adjacent nerve and usually provides only partial brachial plexus block. • Special contraindications: infected axillary lymph nodes.
Alternative	Supraclavicular block Infraclavicular block
Further reading	Nowakowski P, Bieryło A: Ultrasound guided axillary brachial plexus block. Part 1 and Part 2. Anaesthesiol Intensive Ther 2015; 47(4):409–24.

LOWER EXTREMITY BLOCKS

FEMORAL NERVE

Indications	Anterior thigh, knee, and medial ankle surgery
Anatomy	

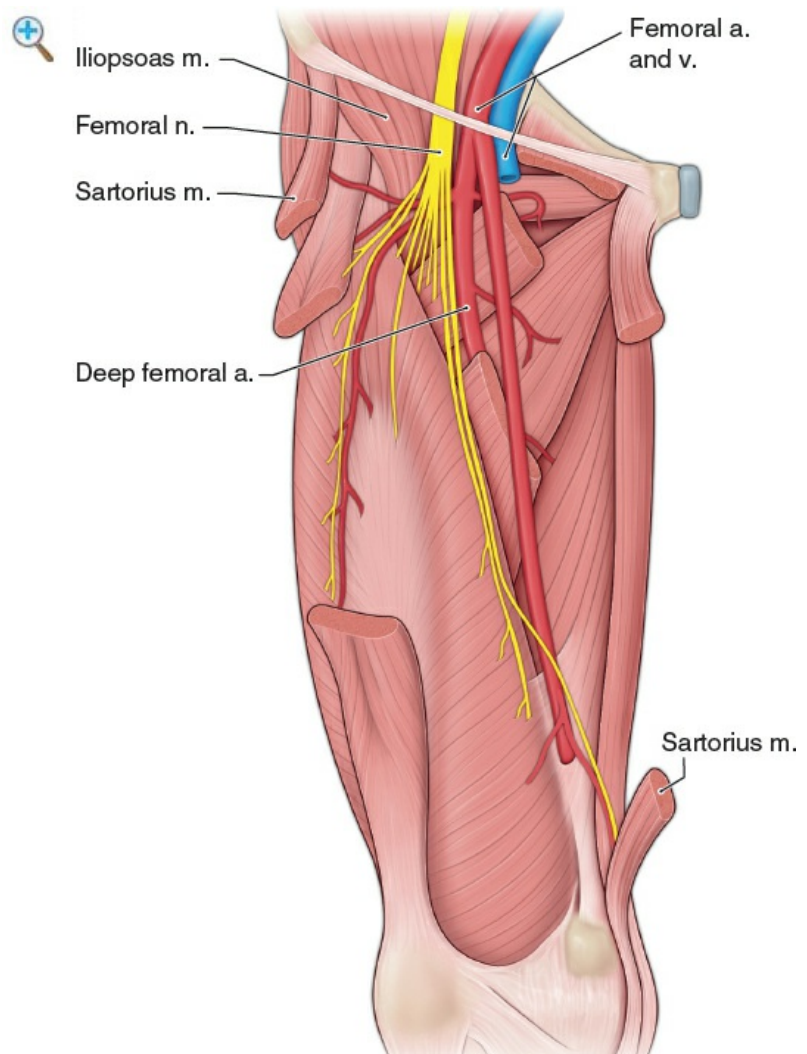


Figure 16.17 Femoral nerve anatomy. Originates from L2 to L4 nerve roots.

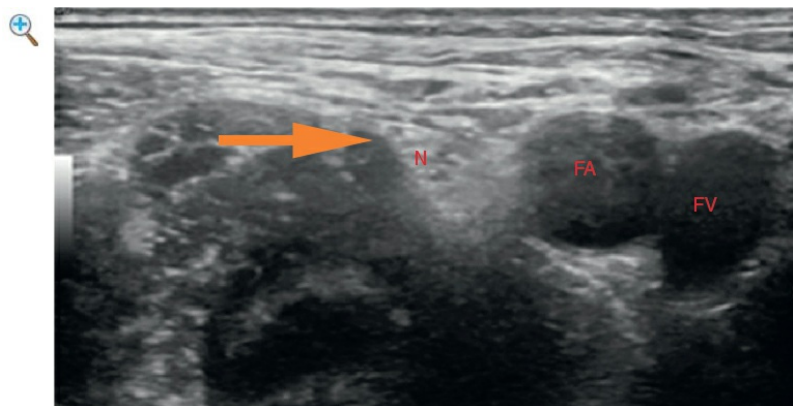


Figure 16.18 Femoral nerve ultrasound. FA, femoral artery; FV, femoral vein; N, nerve.



Figure 16.19 Femoral nerve anatomy.

Dosage	15–30 mL (0.25–0.5% ropivacaine/bupivacaine or mepivacaine 1–1.5%)
Nerve stimulation	Quadriceps femoris with patellar twitch
Side effects	Quadriceps weakness with ambulation difficulties
Complications	Vascular injury: <1% using ultrasound guidance
Pearls	• Supine with the leg extended, probe placed transverse for

	an in-plane or out-of-plane approach The nerve lies deep to the fascia iliaca while the artery is superficial; the fascia iliaca has an S shape placing the nerve in a deeper anatomical plane than the artery.
Alternative	A posterior approach of the lumbar plexus will reliably cover the femoral, obturator, and lateral femoral cutaneous nerves for surgical anesthesia. A fascia iliaca block will reliably cover the femoral, obturator, and lateral femoral cutaneous nerves for analgesia. An adductor canal block or subsartorial distal saphenous block will cover the medial ankle for surgical anesthesia and, partially, the knee for analgesia.
Further reading	Szucs S, et al: Femoral nerve blockade. Med Ultrason 2010; 12(2):139–44.

LUMBAR PLEXUS

Indications	Hip, anterior, medial and lateral thigh, and knee surgery
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Anatomy

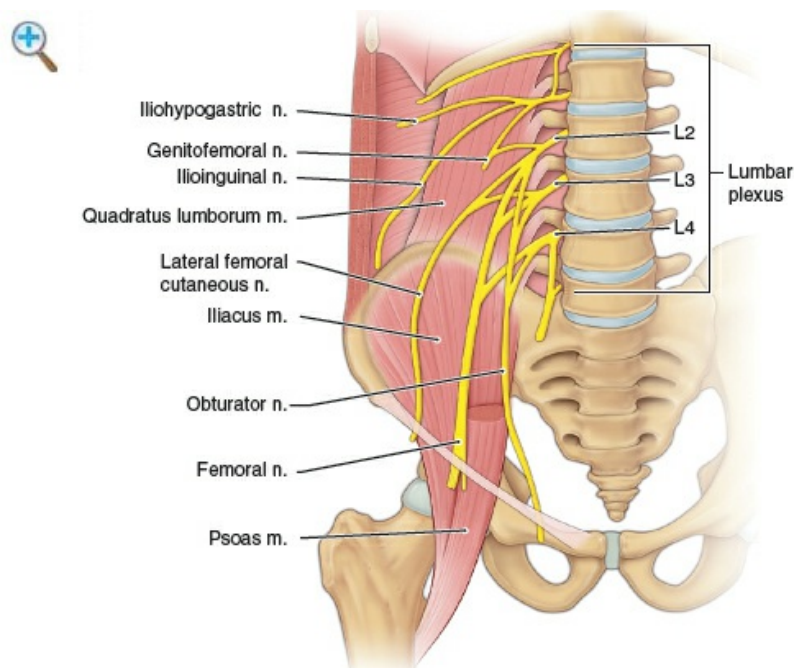


Figure 16.20 Lumbar plexus anatomy. The portion of the

lumbar plexus covering the lower extremity originates from L2 to L4 nerve roots: femoral, lateral femoral cutaneous, and obturator nerves.

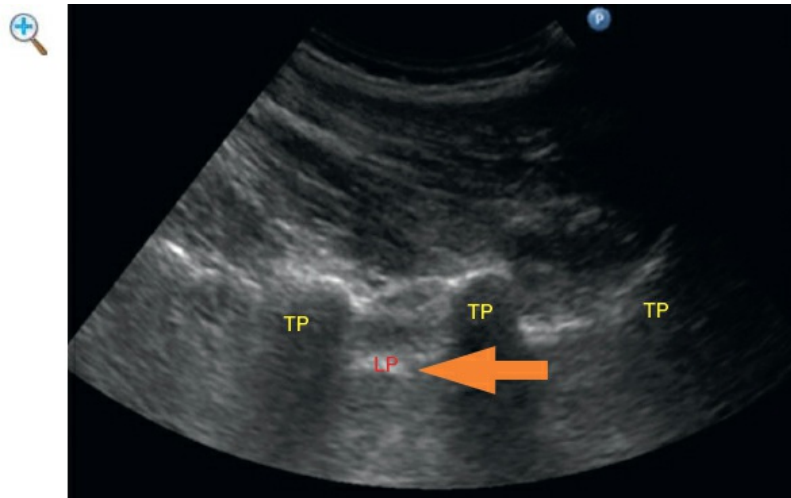


Figure 16.21 Lumbar plexus ultrasound. TP, transverse process; LP, lumbar plexus.

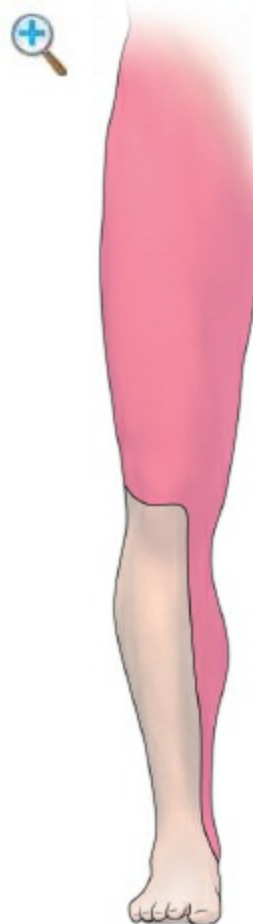


Figure 16.22 Lumbar plexus anatomy.

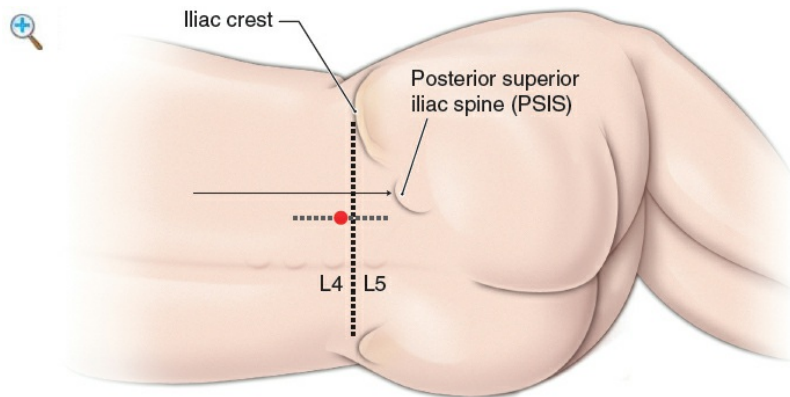


Figure 16.23 Lumbar plexus anatomy.

Dosage	10–20 mL (0.25–0.5% ropivacaine/bupivacaine)
Nerve stimulation	Quadriceps femoris with patellar twitch
Side effects	Quadriceps weakness with ambulation difficulties Epidural spread with partial sympathectomy
Complications	Epidural/bilateral spread (up to 10%) Epidural/intrathecal placement Vascular puncture: <1% Local anesthetic systemic toxicity (LAST): rare Renal puncture/retroperitoneal hematoma: rare
Pearls	• Lateral decubitus with fetal position, i.e., epidural positioning • Prescan with ultrasound to map the area with best angle of approach, angles to avoid, and to estimate depth. • Probe placed transverse and sagittal to create a 3D representation of the target • Direct ultrasound guidance is best in small and lean patients with transverse probe placement in the flank aiming toward the spine. • If hamstring twitch on nerve stimulation, move one level cephalad • Always give a test dose to diagnose intrathecal placement. • Keep volume low to prevent epidural spread.
Alternative	Fascia iliaca block Femoral block
Further reading	Chin KJ, Perlas A. Ultrasonography of the lumbar

spine for neuraxial and lumbar plexus blocks. Curr Opin Anaesthesiol 2011; 24(5):567–72.

FASCIA ILIACA AND LATERAL FEMORAL CUTANEOUS NERVE

Indications

Hip, anterior, medial and lateral thigh, and knee

Anatomy

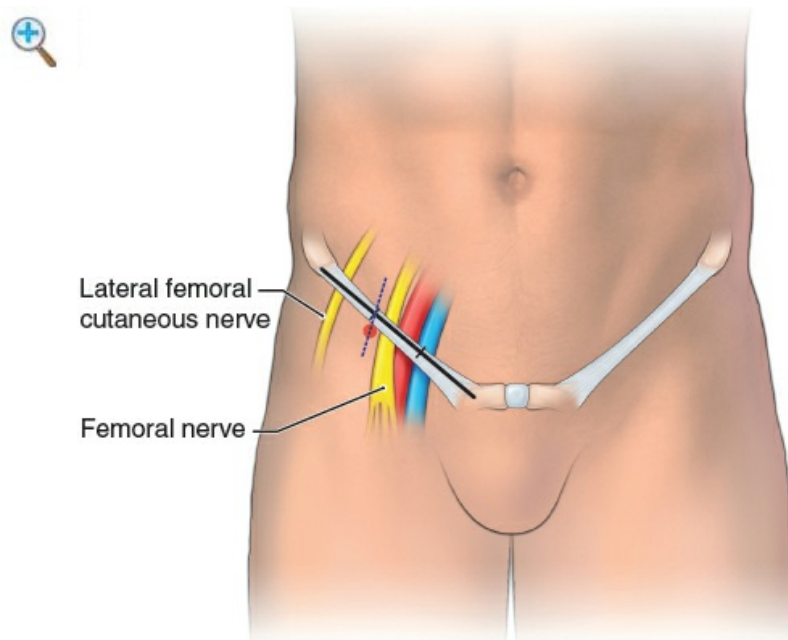


Figure 16.24 Fascia iliaca anatomy. Originates from L2 to L4 nerve roots.

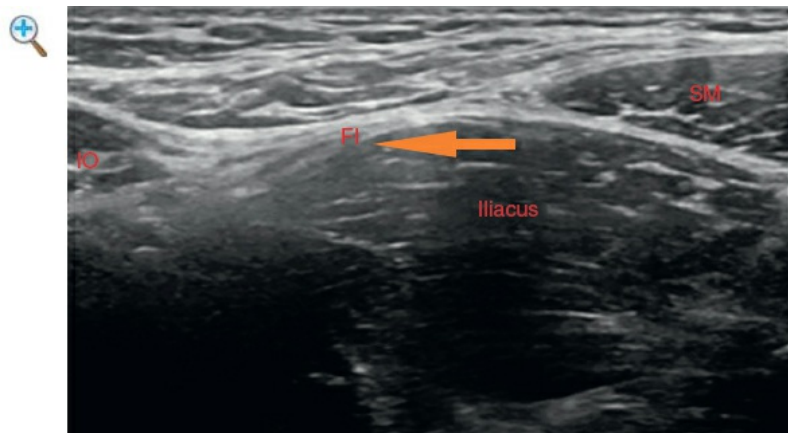


Figure 16.25 Fascia iliaca ultrasound. SM, sartorius muscle; FI, fascia iliaca; IO, internal oblique.



Figure 16.26 Fascia iliaca anatomy.

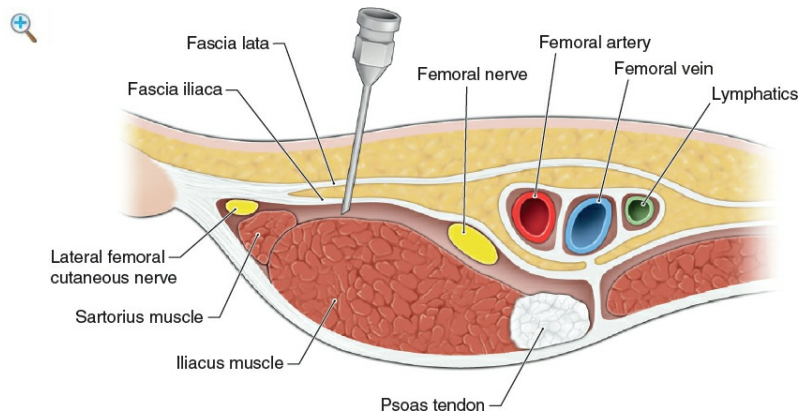


Figure 16.27 Fascia iliaca anatomy.

Dosage	20–60 mL (0.25% ropivacaine/bupivacaine)
Nerve stimulation	None
Side effects	Quadriceps weakness with ambulation difficulties
Complications	Vascular puncture: <1%

	Local anesthetic systemic toxicity (LAST): <0.1% Bowel puncture (rare)
Pearls	• Supine positioning • Position ultrasound transducer transversely near the femoral crease, superficial to the iliopsoas muscle, lateral to femoral vessels and then rotate the probe 90° to drive the needle cephalad under the inguinal ligament. • Use color to avoid the inferior epigastric vessels. • Large-volume dilute local anesthetic allows spread to block both femoral and lateral femoral cutaneous nerves, and sometimes the obturator nerve. • Nonultrasound technique describes the “double pop” as needle traverses the fascia lata and fascia iliaca, but only reliably covers the femoral nerve.
Alternative	Lumbar plexus block Femoral block
Further reading	Steenberg J, et al: Systematic review of the effects of fascia iliaca compartment block on hip fracture patients before operation. Br J Anaesth 2018; 120(6):1368–80.

ADDUCTOR CANAL (SAPHENOUS NERVE)

Indications	Knee and medial ankle surgery
Anatomy	

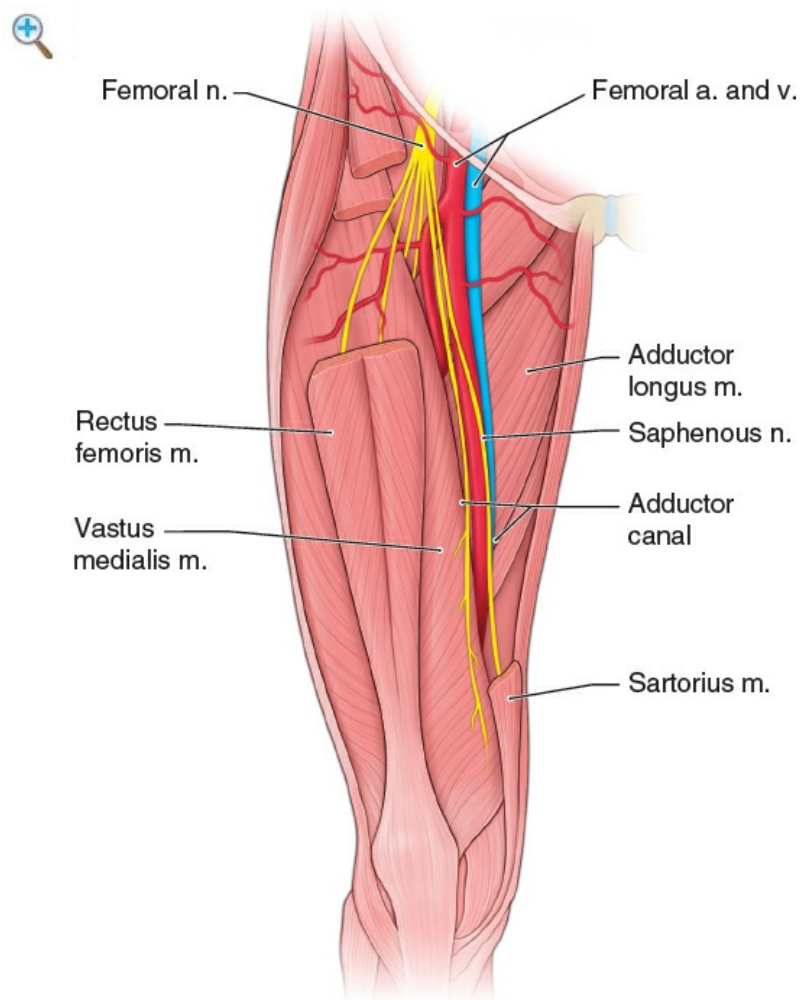


Figure 16.28 Adductor canal anatomy. At the level of the distal femoral triangle, the saphenous nerve runs with the nerve to the vastus medialis. Distal to the adductor hiatus, the nerve runs under the sartorius muscle, with the descending genicular artery.

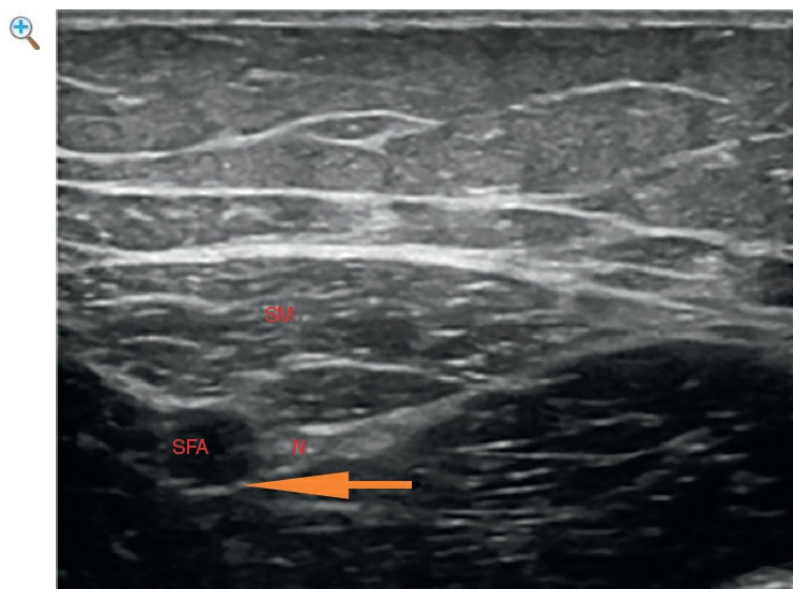


Figure 16.29 Adductor canal ultrasound. SFA, superficial femoral artery; SM, sartorius muscle; N, nerve.

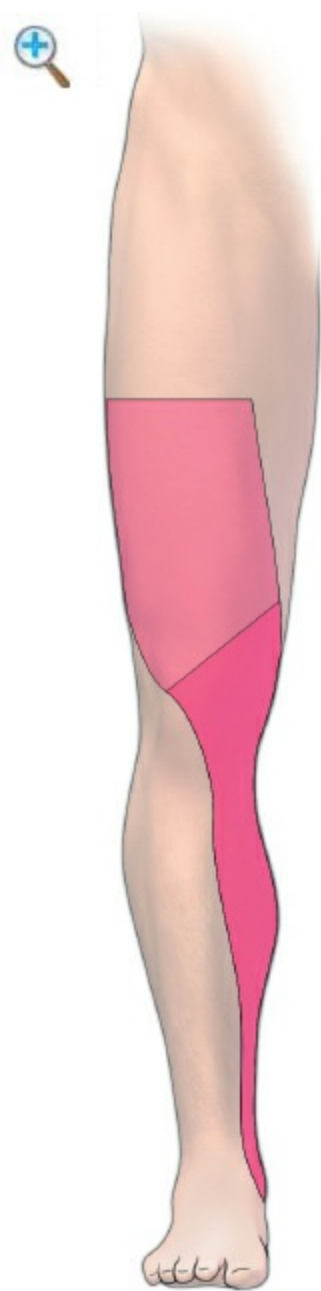


Figure 16.30 Adductor canal anatomy.

Dosage	10–30 mL (0.25–0.5% ropivacaine/bupivacaine or mepivacaine 1–1.5%)
Nerve stimulation	Sensory stimulation at ankle
Side effects	Quadriceps weakness with ambulation difficulties (rare) Infrequent spread to tibial nerve

Complications	Femoral artery injury
Pearls	• Supine with the leg externally rotated, probe placed transverse for an in-plane or out-of-plane approach • The saphenous nerve lies adjacent to the femoral vessels. • A proximal block will provide blockade of both the saphenous nerve and nerve to the vastus medialis. • A block distal to the adductor hiatus will avoid risk of injury to the femoral artery but will only provide appropriate blockade for the ankle/foot. • Despite traditional assumption, the saphenous nerve innervates deep structures of the medial ankle, talocrural joint, and sometimes the first metatarsal.
Alternatives	A posterior approach of the lumbar plexus, a fascia iliaca, or a femoral nerve block will reliably cover all branches of the femoral nerve. For foot surgery, a subcutaneous injection of the saphenous nerve at or above the ankle
Further readings	Eglitis N, et al: The importance of the saphenous nerve in ankle surgery. <i>Anesth Analg</i> 2016; 122(5):1704–6. Horn J-L, et al: Anatomical basis to the ultrasound-guided approach for saphenous nerve blockade. <i>Reg Anesth Pain Med</i> 2009; 34(5):486–9. Vora MU, et al: Adductor canal block for knee surgical procedures: review article. <i>J Clin Anesth</i> 2016; 35:295–303.

POPLITEAL SCIATIC NERVE

Indications	Posterior capsule of the knee and leg, ankle, and foot with the addition of a saphenous nerve block for the medial leg/ankle/forefoot
Anatomy	

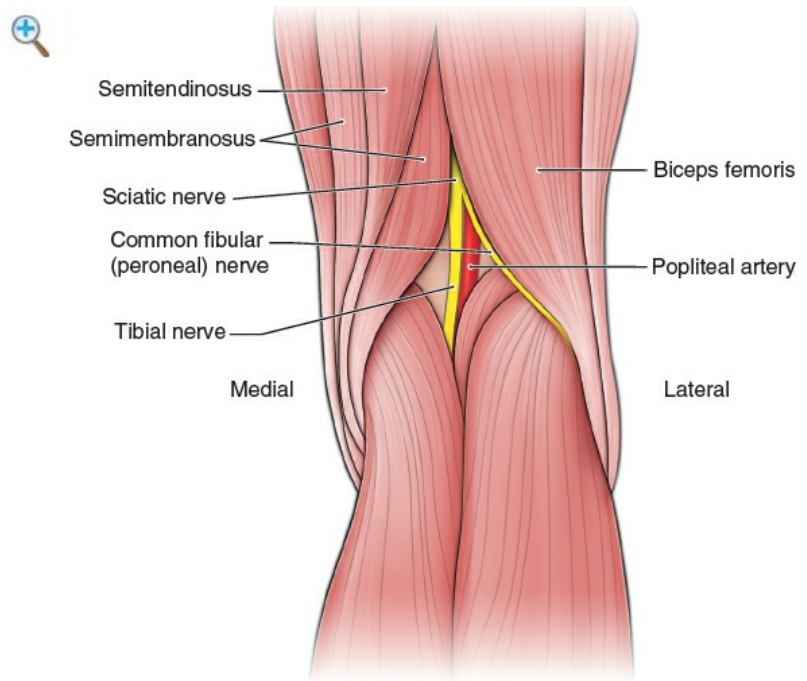


Figure 16.31 Popliteal sciatic anatomy. The sciatic nerve arises from L4 to S3 nerve roots. The level of the split into tibial and common peroneal nerves varies from knee capsule to about 10 cm more proximal.

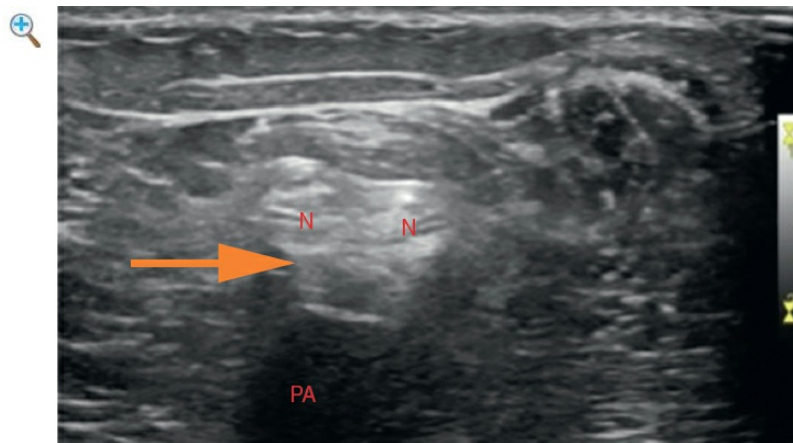


Figure 16.32 Popliteal sciatic ultrasound. PA, popliteal artery; N, nerve.

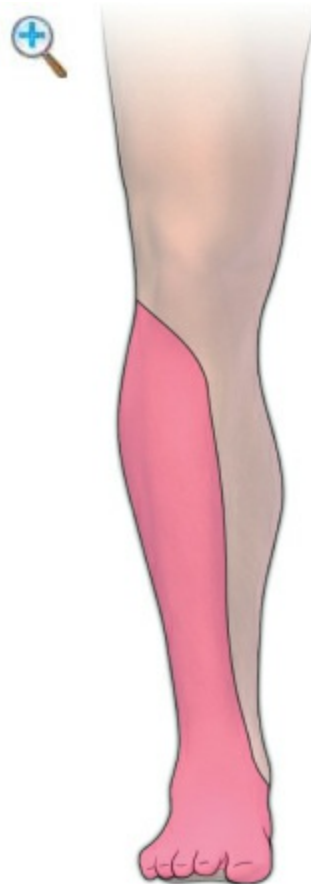


Figure 16.33 Popliteal sciatic anatomy.

	The level of the nerve split into the tibial and common peroneal nerves varies usually from the knee crease to about 10 cm more proximally
Dosage	20–30 mL (0.5% ropivacaine/bupivacaine or mepivacaine 1–1.5%)
Nerve stimulation	Dorsiflexion for common peroneal stimulation, extension for tibial stimulation
Side effects	Complete block of all muscles below the knee with ambulation difficulties
Complications	Vascular injury: <1% Nerve injury: <5%
Pearls	• Prone, lateral decubitus with probe placed transverse for an in-plane or out-of-plane approach. Supine position with elevated leg makes the needle placement more difficult but is more comfortable for the patient. • Adding a saphenous nerve block is essential when the

	medial leg, ankle, and forefoot are involved. • Single shot placed at or just distal to the nerve split speeds the onset. • Catheter placement above the split ensures more reliable results.
Alternative	Higher sciatic approach Ankle block for forefoot and toe surgery
Further reading	Dufour E, et al: Combined ultrasound and neurostimulation guidance for popliteal sciatic nerve block: a prospective, randomized comparison with neurostimulation alone. <i>Anesth Analg</i> 2008; 106:1553–8.

SCIATIC NERVE

Indications	Posterior thigh/knee and ankle/foot with the addition of a saphenous nerve block for the medial ankle/forefoot
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Anatomy

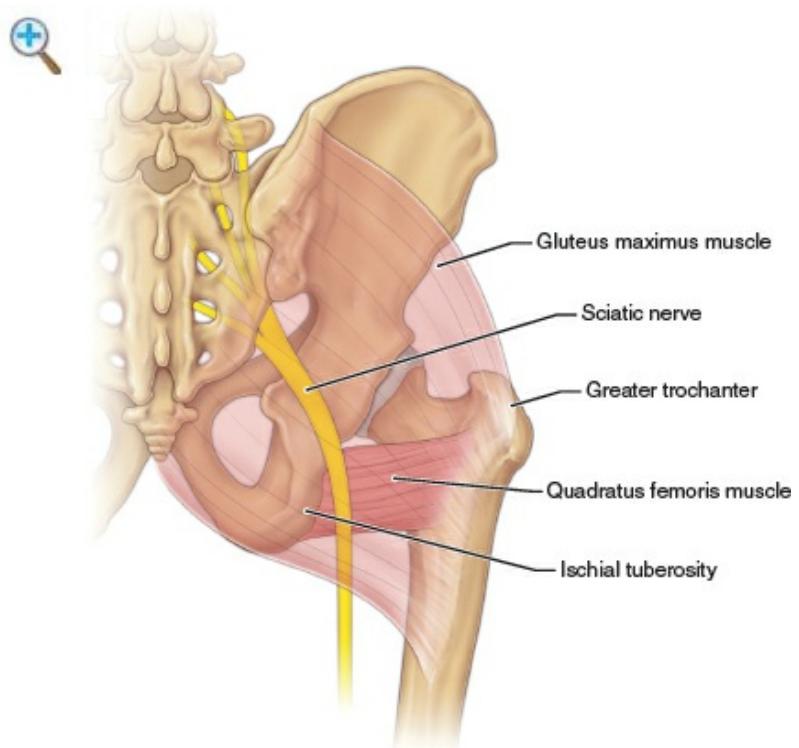


Figure 16.34 Sciatic anatomy—The sciatic nerve arises from L4 to S3 nerve roots.

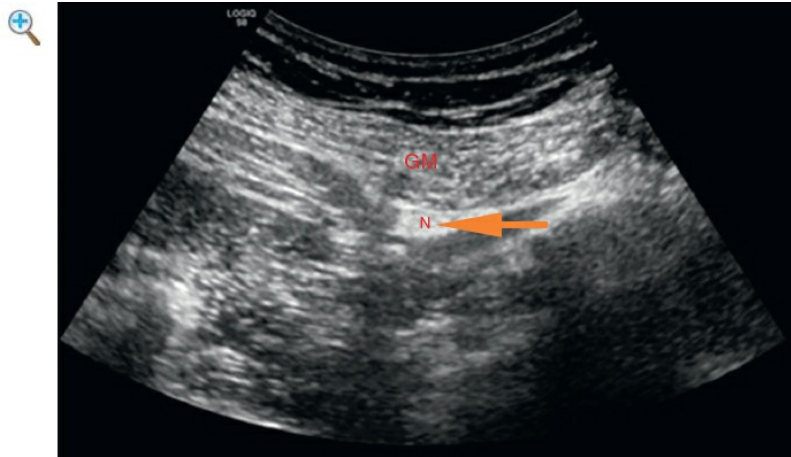


Figure 16.35 Sciatic ultrasound. GM, gluteus muscle; N, nerve.



Figure 16.36 Sciatic anatomy.

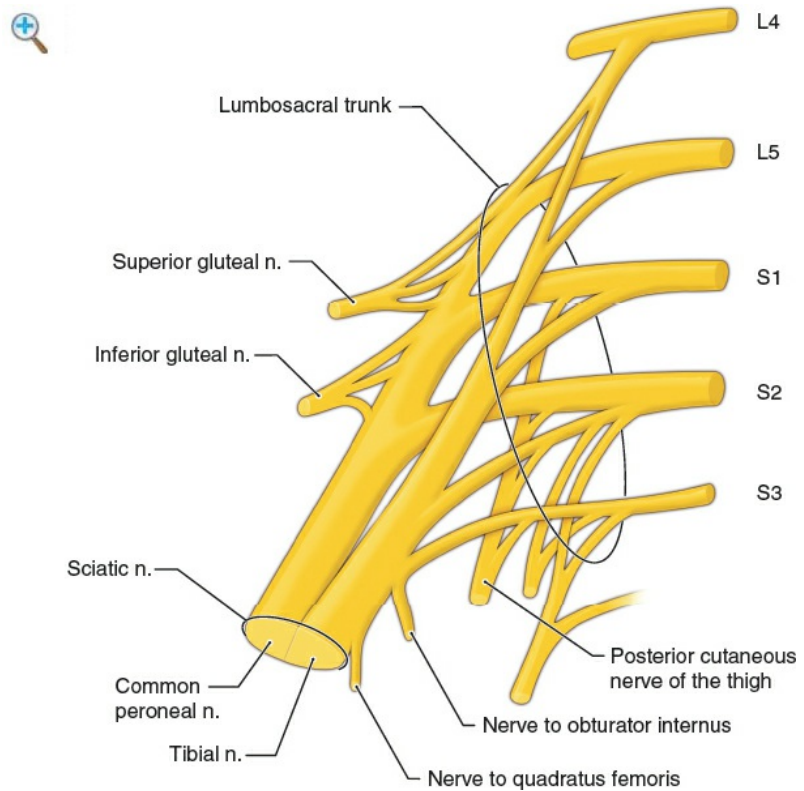


Figure 16.37 Sciatic anatomy.

Dosage	15–30 mL (0.5% ropivacaine/bupivacaine or mepivacaine 1–1.5%)
Nerve stimulation	Dorsiflexion for common peroneal stimulation, extension for tibial stimulation
Side effects	Complete block of all muscles below the knee +/- hamstring muscles with ambulation difficulties
Complications	Vascular injury: <1% Nerve injury: <5%
Pearls	• Lateral decubitus with leg flexed, probe placed transverse for an in-plane or out-of-plane approach • The parasacral approach is the most challenging due to the depth and angle of the nerve entering the sacrum. • A proximal block, parasacral, or subgluteal above the long head of the biceps is necessary to cover the posterior thigh or knee to catch the posterior femoral cutaneous nerve before it branches posterior to the biceps femoris.
Alternative	Anterior approach when the patient cannot be moved

(but can be challenging due to the depth of the nerve).

Popliteal sciatic nerve block is more superficial and works well for structures below the knee, and it may also cover the posterior capsule of the knee joint.

Ankle block for forefoot and toe surgery

Further reading

Rodziewicz TL, et al: Sciatic Nerve Block. 2017 Nov 26. StatPearls [Internet]. StatPearls Publishing, Treasure Island: 2018. Available from: <http://www.ncbi.nlm.nih.gov/books/NBK470391/>

OBTURATOR NERVE

Indications

Medial distal thigh, medial knee, preventing thigh adduction in transurethral bladder surgery, rescue after knee surgery when femoral block is insufficient

Anatomy

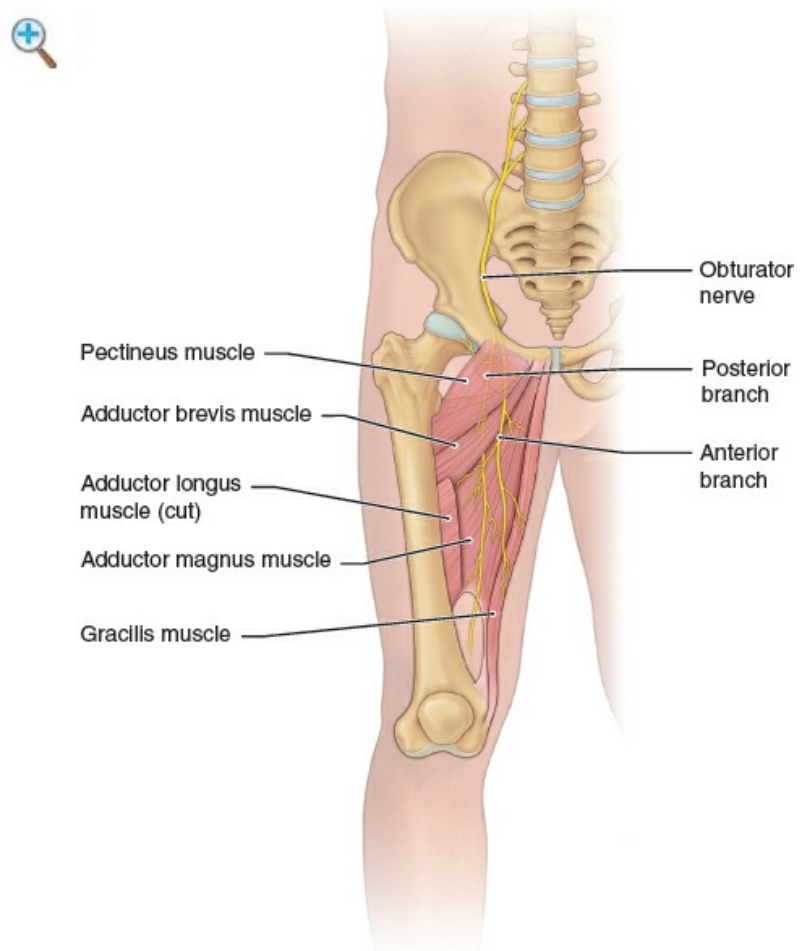


Figure 16.38 Obturator nerve anatomy. Originates from the

lumbar plexus (anterior rami of L2-L4).

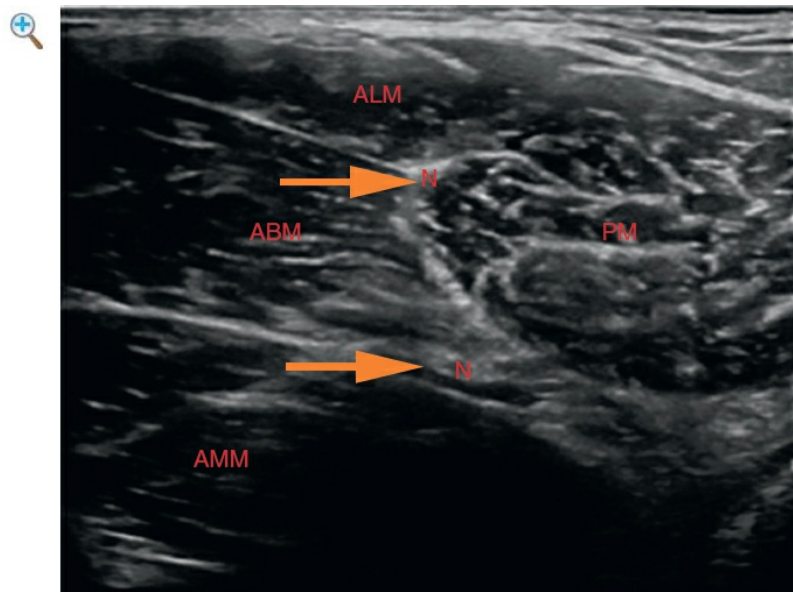


Figure 16.39 Obturator nerve ultrasound. PM, pectineus; ALM, adductor longus muscle; ABM, adductor brevis muscle; AMM, adductor magnus muscle.

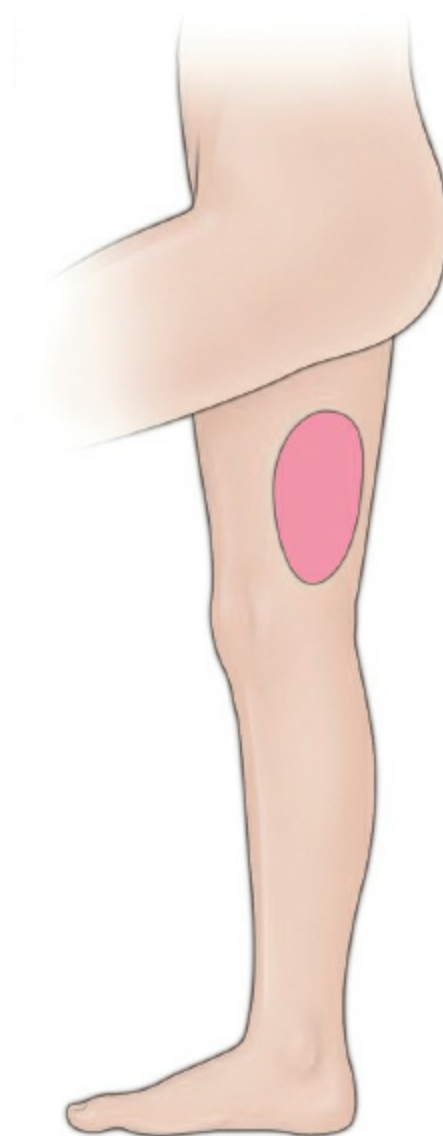


Figure 16.40 Obturator nerve anatomy.

	An interfascial plane block. The anterior branch is between the pectineus, adductor longus, and adductor brevis muscles. The posterior branch is between the adductor brevis and adductor magnus.
Dosage	10 mL (0.25–0.5% ropivacaine/bupivacaine)
Nerve stimulation	Adductor twitch (rarely used)
Side effects	Inability to adduct leg (rarely impairs mobility)
Complications	Vascular injury: rare
Pearls	• Supine with the leg extended, probe placed transversely over the femoral vessels and then slide medially over the pectineus and adductor muscles for an in-plane or out-of-

	plane approach. The anterior branch is mostly cutaneous, and posterior branch is mostly motor.
Alternative	Lumbar plexus block Fascia iliaca block
Further reading	Yoshida T, et al: Ultrasound-guided obturator nerve block: a focused review on anatomy and updated techniques. Biomed Res Int 2017; 2017:7023750.

ANKLE

Indications	Forefoot or toe surgery
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Anatomy

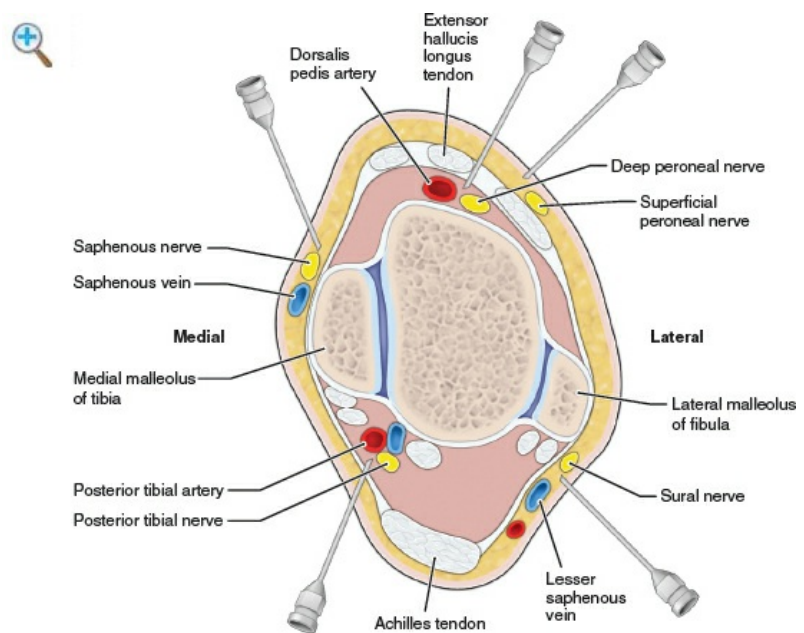


Figure 16.41 Ankle anatomy. The deep peroneal nerve is located lateral to the extensor hallucis longus tendon and dorsalis pedis artery. The posterior tibial nerve is posterior to the distal medial malleolus and posterior tibial artery.

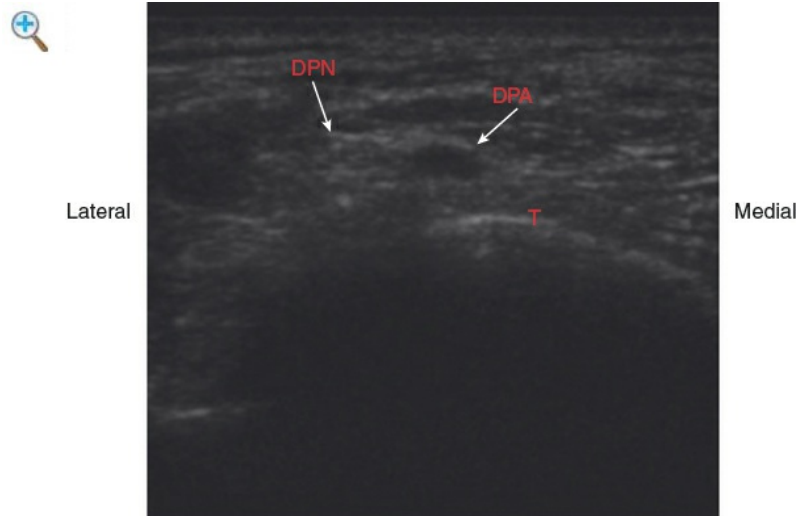


Figure 16.42 Ankle ultrasound. DPN, deep peroneal nerve; DPA, dorsalis pedis artery; T, Tibia.



Figure 16.43 Ankle anatomy.

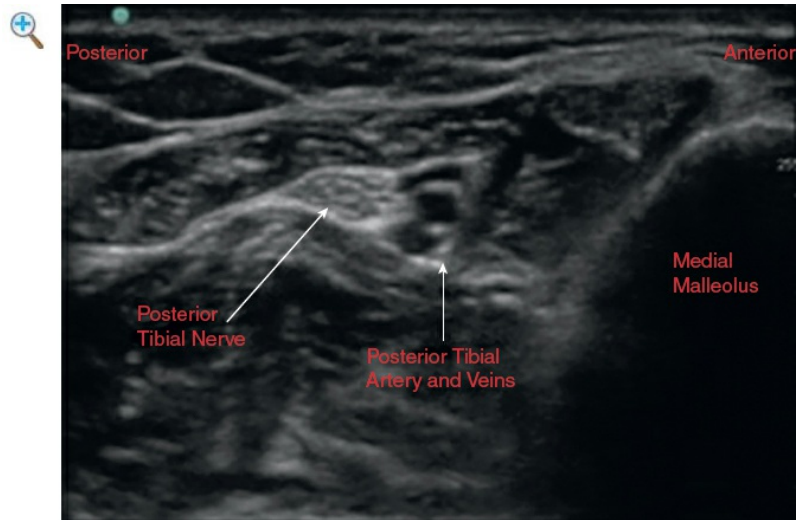


Figure 16.44 Ankle ultrasound.

	Posterior tibial and deep peroneal nerves are targeted individually, and superficial nerves (superficial peroneal, sural, and saphenous) are blocked with a subcutaneous field block. The deep peroneal nerve is located directly lateral to the extensor hallucis longus tendon and the anterior tibial artery. The posterior tibial nerve is located immediately posterior to the distal medial malleolus and the posterior tibial artery. The superficial nerves are targeted with a subcutaneous ring block at the narrowest part of the ankle proximal to the malleoli excluding the tissue overlying the Achilles tendon.
Dosage	5 mL (0.5% ropivacaine or 1.5% mepivacaine) for each nerve
Nerve stimulation	Nerve stimulation is only useful for posterior tibial nerve localization.
Side effects	Discomfort during placement Failure in acidic milieu
Complications	Vascular injury: <1%
Pearls	• Adding sodium bicarbonate (10%) to local anesthetic may correct local acidosis. • Start with the superficial nerves (S), less risk of paresthesia. • Place a circular subcutaneous wheel for the S at the

narrowest point of the ankle.

- Then block the deep peroneal nerve at the distal tibia adjacent to the anterior tibial artery or dorsalis pedis artery more distally.
- Lastly, block the posterior tibial nerve behind the medial malleolus next to the posterior tibial artery, as it is the most uncomfortable injection.

Alternative	Combined popliteal and saphenous nerve blocks
Further reading	Coe A, Ram S: Ultrasound-guided ankle block for forefoot surgery: is sural nerve block necessary? Reg Anesth Pain Med 2013; 38(3):251.

THORACOABDOMINAL BLOCKS

TRANSVERSUS ABDOMINIS PLANE (TAP)

Indications	Abdominal surgery
Anatomy	

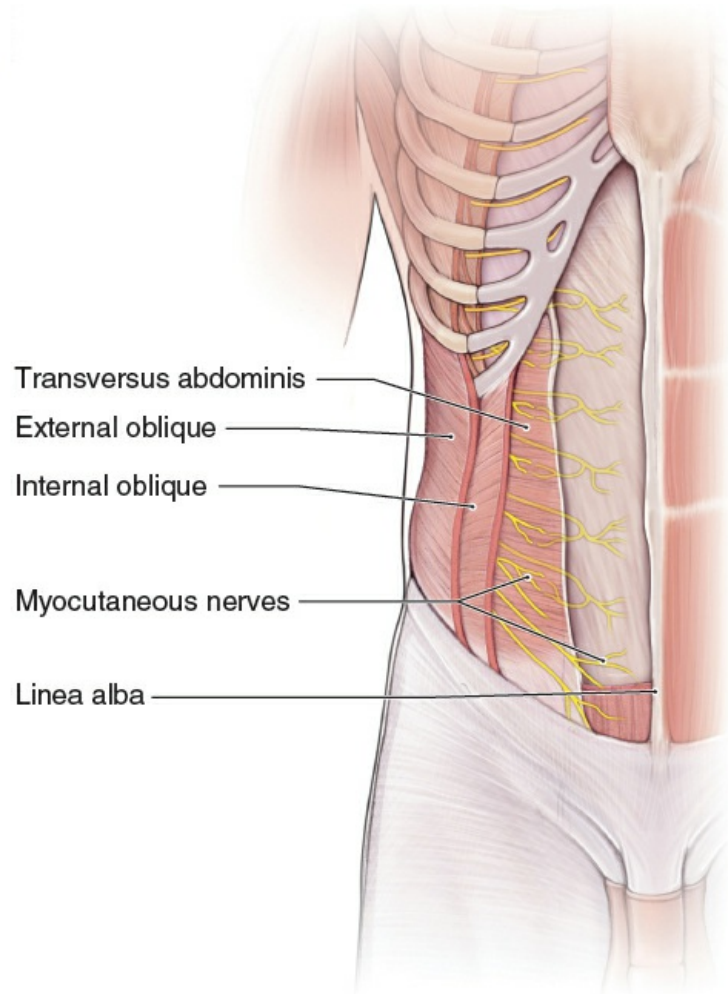


Figure 16.45 Transversus abdominis anatomy. The thoracoabdominal nerves, T6-T12, and the ilioinguinal/iliohypogastric nerves, T12-L1, run in the fascial plane deep to the internal oblique and superficial to the transversus abdominis. The subcostal approach targets higher-level dermatomes (T6-T9), while the lateral approach (mid axillary line) targets lower dermatomes (T10-L1).

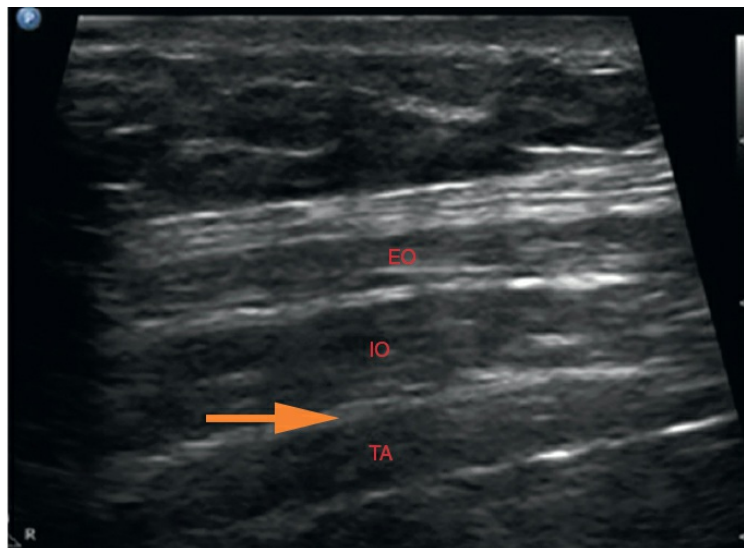


Figure 16.46 Transversus abdominis ultrasound. EO, external oblique; IO, internal oblique; TA, transversus abdominis.

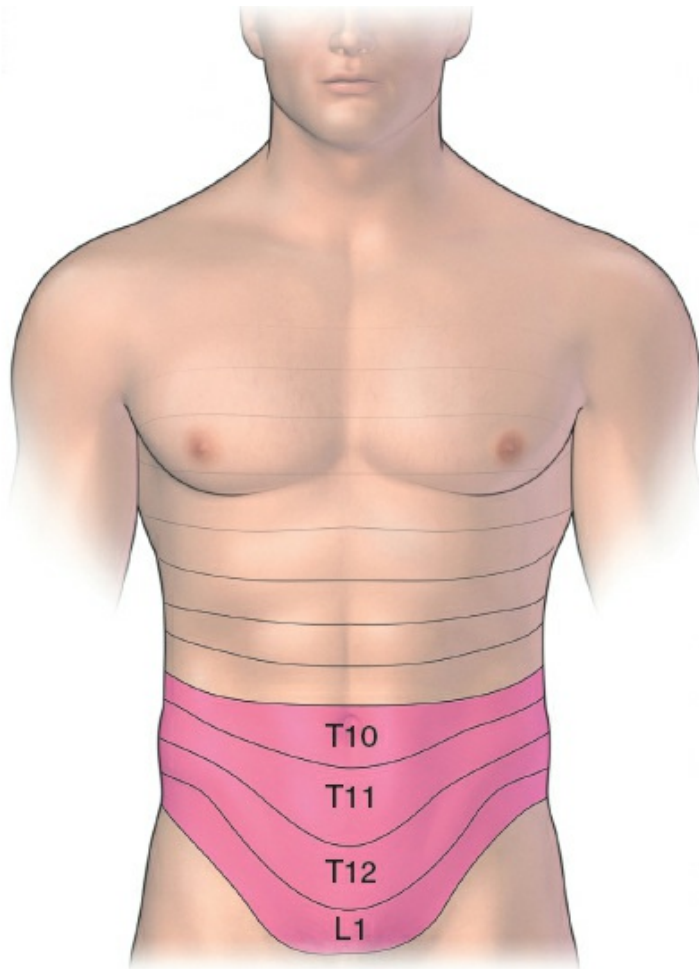


Figure 16.47 Transversus abdominis anatomy.

The thoracoabdominal nerves, T6-T12, and the ilioinguinal/iliohypogastric nerves, T12-L1, run in the fascial plane deep to the internal oblique and superficial to the transversus abdominis. The subcostal approach targets higher-level dermatomes (T6-T9), while the lateral approach (mid axillary line) targets lower dermatomes (T10-L1).

Dosage 15–30 mL (0.25% ropivacaine)

Nerve stimulation None

Side effects Minimal

Complications	Visceral puncture: <0.1% Local anesthetic systemic toxicity (LAST): <0.1% Hematoma: <1%
Pearls	• The posterior approach may extend axially like the blind angle of petit approach. • Large bolus may provide better spread/effect. • Landmarks may be altered by surgery.
Alternative	Local infiltration by the surgeon Epidural Paravertebral block Quadratus lumborum block Rectus sheath block Erector spinae plane block
Further reading	Chin KJ, et al: Essentials of our current understanding: abdominal wall blocks. Reg Anesth Pain Med 2017; 42(2):133–83.

ERECTOR SPINAE PLANE

Indications	Chest wall, abdominal wall, shoulder, and possibly hip
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Anatomy

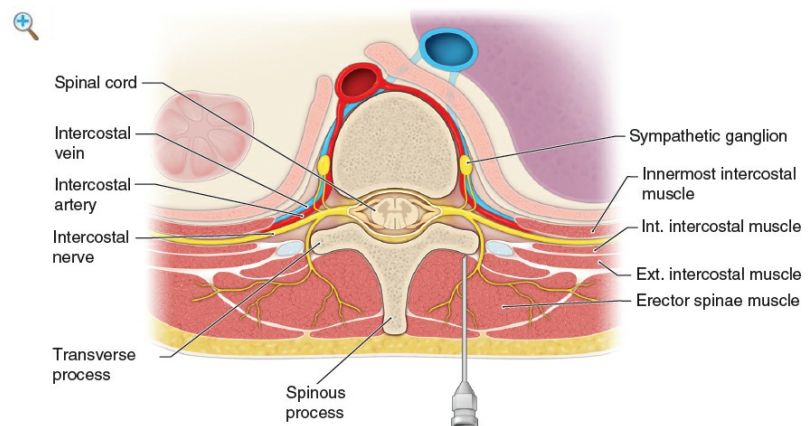


Figure 16.48 Erector spinae anatomy. Potential blockade of dorsal and ventral rami of the thoracic spinal nerves.

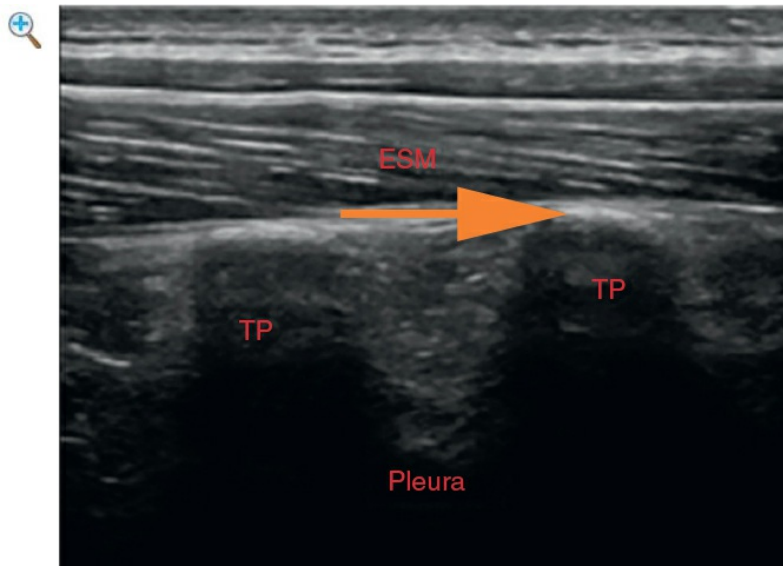


Figure 16.49 Erector spinae ultrasound. ESM, erector spinae muscle; TP, transverse process.

	Fascial plane block. Deposition of local anesthetic deep to the erector spinae muscle along the tip of the transverse processes, potentially blocking dorsal and ventral rami of the thoracic spinal nerves
Dosage	10–20 mL (0.5% ropivacaine or 1.5% mepivacaine)
Nerve stimulation	Not used
Side effects	Minimal
Complications	Paravertebral spread LAST (rare)
Pearls	• Positioning: prone, sitting, or lateral • Ultrasound guided: parasagittal or transverse probe placement with an in- or out-of-plane approach
Alternative	Epidural Intercostal block Paravertebral block
Further reading	Forero M, et al: Erector spinae plane block in the management of post thoracotomy pain syndrome: a case series. Scand J Pain 2017; 17:325–9.

PARAVERTEBRAL

Indications

Thoracic, breast, axillary, abdominal surgeries

Anatomy

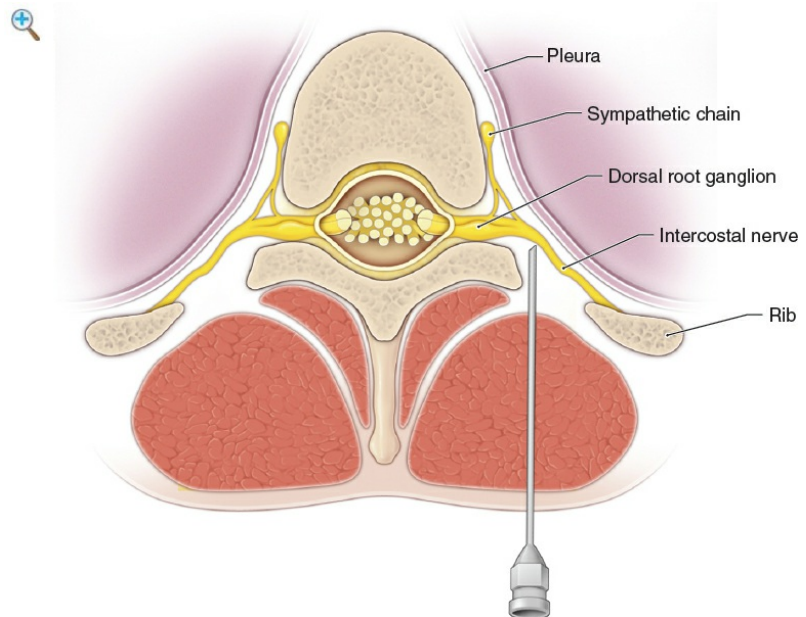


Figure 16.50 Paravertebral anatomy. Triangular spaces lateral to the vertebral column. Bordered anteriorly by the parietal pleura, posteriorly by the superior costotransverse ligament, and medially by the transverse process and foramen. Space contains fat, vessels, intercostal nerves, sympathetic chain, and dorsal rami of the spinal nerves.

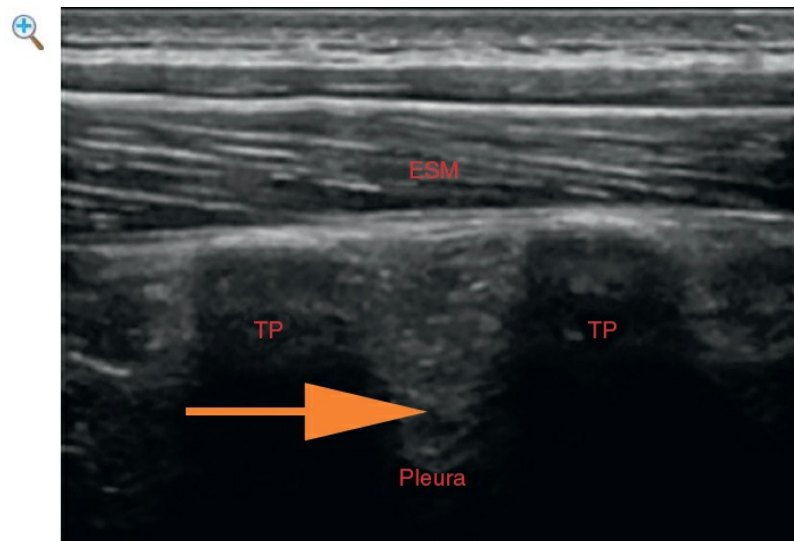


Figure 16.51 Paravertebral ultrasound. ESP, erector spinae muscle; TP, transverse process.

Target the triangular space lateral to the vertebral column, bordered anterior by the parietal pleura, posteriorly by the superior costotransverse ligament

	and transverse process, and medially by the vertebral body and foramen. The space contains fat, vessels, intercostal nerves, sympathetic chain, and dorsal rami of the spinal nerves.
Dosage	3–5 mL 0.25–0.5% ropivacaine/bupivacaine or 1.5% mepivacaine per level for multiple levels; 15–20 mL 0.25–0.5% ropivacaine/bupivacaine or 1.5% mepivacaine for single level injection
Nerve stimulation	Helpful to confirm level
Side effects	Hypotension Epidural spread
Complications	Epidural or intrathecal injection (~1%) Pneumothorax (~0.5–5%) Hematoma (2.4%)
Pearls	• Positioning: prone, sitting, or lateral • Ultrasound guided: parasagittal probe placement with in- or out-of-plane approach • For parasagittal approach, tilting the probe medial to lateral to obtain a good beam reflection on the pleura • Obtain a “loss of resistance” when passing the posterior border, with depression of the pleura with fluid spread.
Alternative	Epidural Intercostal block Erector spinae plane block
Further reading	Wardhan R: Update on paravertebral blocks. Curr Opin Anaesthesiol 2015; 28(5):588–92.

PECTORALIS PLANE (PECS 1, 2 AND SERRATUS ANTERIOR)

Indications	Breast, anterior chest surgery
Anatomy	

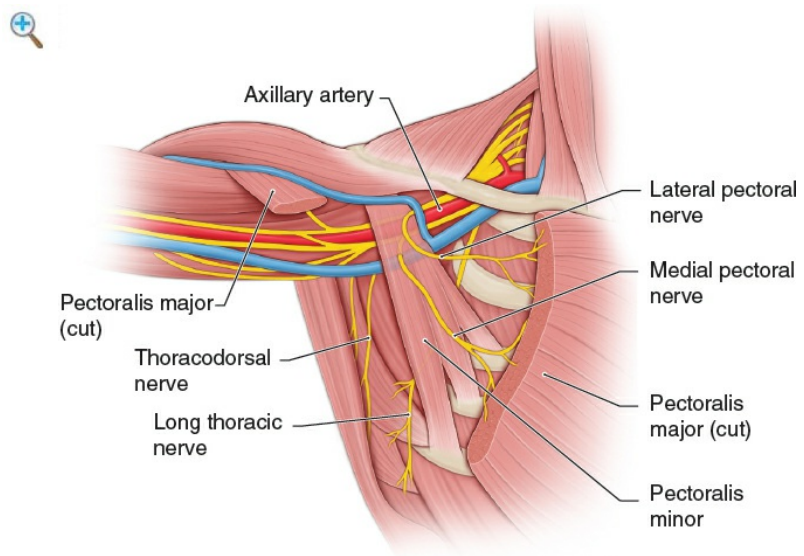


Figure 16.52 Pectoralis plane anatomy.

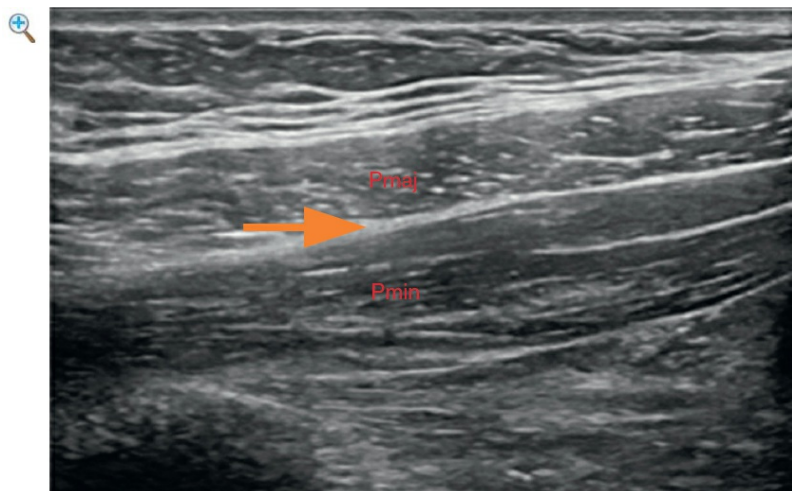


Figure 16.53 Pectoralis plane ultrasound. Pmaj, pectoralis major; Pmin, pectoralis minor.

PECS I between the pectoralis major and minor muscles targeting the lateral and medial pectoral nerves

PECS II under the pectoralis minor muscles and the ribs covering the lateral cutaneous branches of the upper intercostal nerves (T2-T4)

PECS III or serratus anterior block covering the lateral cutaneous branches of the mid intercostal nerves (T5-T7). LA is injected either superficial or deep to the serratus anterior muscle

Dosage

10–20 mL (0.25–0.5% ropivacaine or bupivacaine)

Nerve stimulation	Not used
Side effects	Long thoracic nerve block → loss of serratus anterior muscle function
Complications	Local anesthetic systemic toxicity (LAST): <0.1% Vascular puncture: thoracoacromial trunk for PECS I (6.8%) Pneumothorax (0.5%)
Pearls	• Supine positioning with arm abducted or at side • PECS I: ultrasound probe parasagittal, midclavicular line below clavicle, identify the thoracoacromial trunk adjacent to the lateral pectoralis nerve • PECS II: move the probe laterally at an angle along breast contour directing the probe to the anterior axillary line • PECS III: continue to slide the probe caudate over the serratus anterior muscle
Alternative	Epidural Intercostal block Paravertebral block Erector spinae plane block
Further reading	O'Scanail P, et al: Single-shot pectoral plane (PECS I and PECS II) blocks versus continuous local anaesthetic infusion analgesia or both after non-ambulatory breast-cancer surgery: a prospective, randomized, double-blind trial. Br J Anaesth 2018; 120(4):846–53.

RECTUS SHEATH

Indications	Abdominal midline/umbilical surgery
Anatomy	

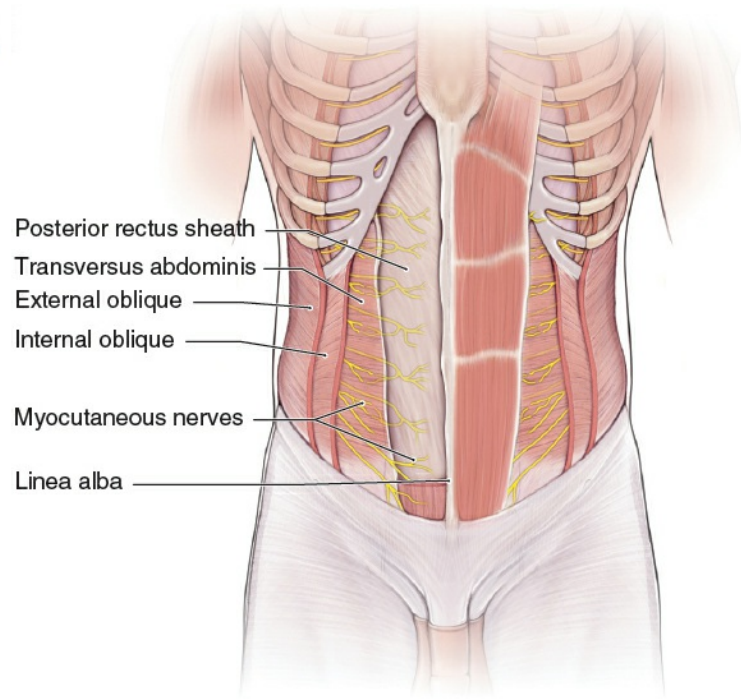


Figure 16.54 Rectus sheath anatomy. Local spread between the oval-shaped rectus muscle and its posterior fascia with blockade of T6-T11.

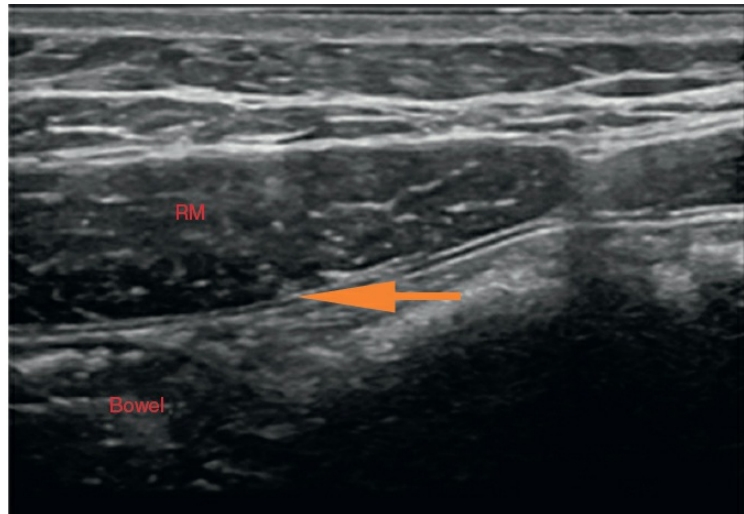


Figure 16.55 Rectus sheath ultrasound. RM, rectus muscle.

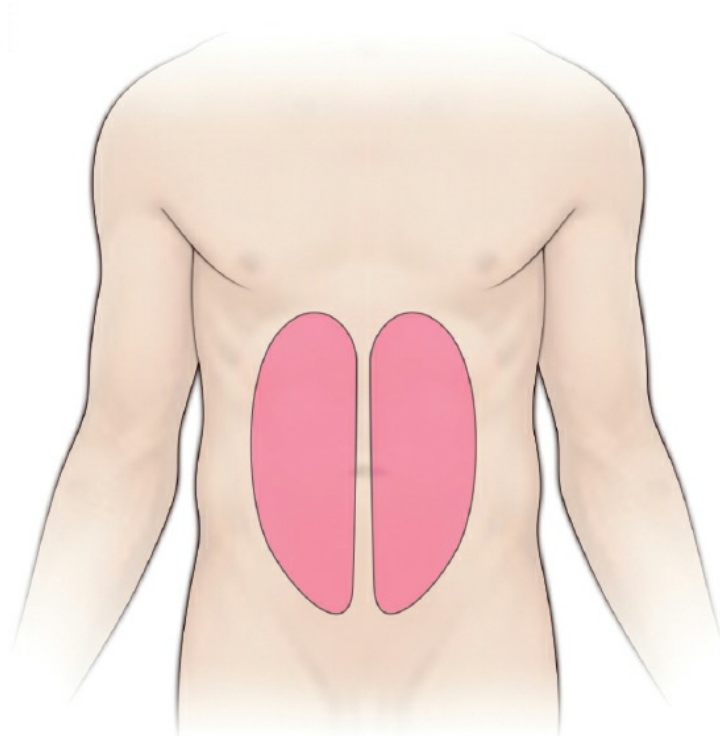


Figure 16.56 Rectus sheath anatomy.

Dosage	10–30 mL (0.25–0.5% ropivacaine or bupivacaine)
Nerve stimulation	Not used
Side effects	Minimal
Complications	Local anesthetic systemic toxicity (LAST): <0.1% Vascular puncture: epigastric artery: <1% Visceral puncture: <1%
Pearls	• Supine positioning • Ultrasound probe placed in the transverse position at the level of the desired injection site • Look for the superior epigastric at medial part of the rectus muscle, stay lateral to the artery • Injection posterior to the muscle fascicles but anterior to the posterior rectus sheath
Alternative	Epidural Transversus abdominis plane block Paravertebral block
Further reading	Chin KJ, et al: Essentials of our current understanding: abdominal wall blocks. Reg Anesth

QUADRATUS LUMBORUM (QL)

Indications

Abdominal surgery, possibly useful for hip surgery

Anatomy

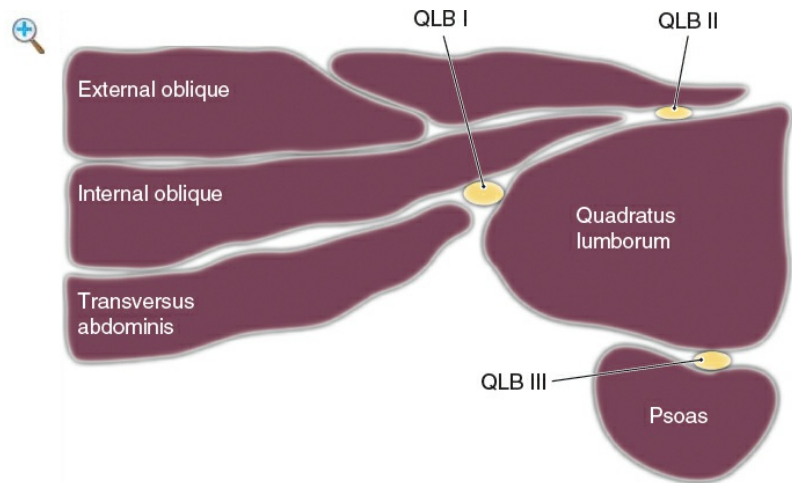


Figure 16.57 Quadratus lumborum anatomy. The quadratus lumborum lies deep and posterior to the aponeurosis of the abdominal wall musculature and is covered by the thoracoabdominal fascia.

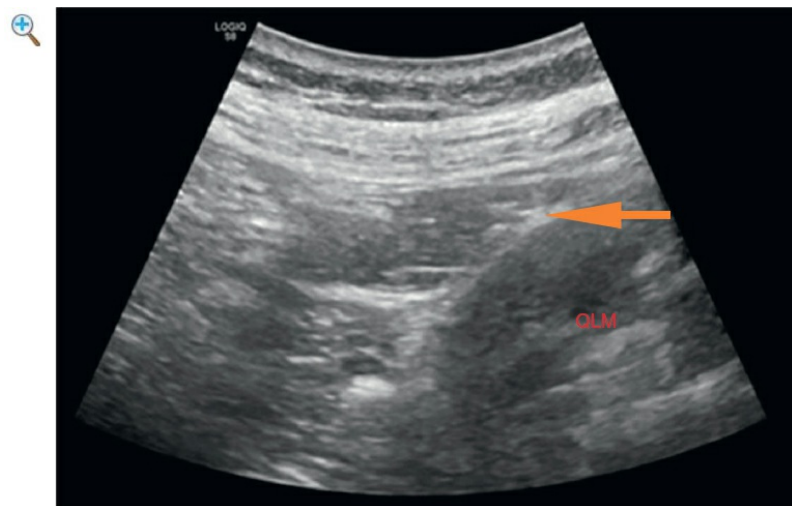


Figure 16.58 Quadratus lumborum ultrasound. QLM, quadratus lumborum muscle.

The quadratus lumborum lies deep and posterior to the aponeurosis of the abdominal wall musculature and is covered by the thoracoabdominal fascia. LA is injected either lateral, posterior, or anterior to the

	QL muscle defined as QL 1, 2, or 3 (transmuscular), respectively. All injection sites have spread of LA into the space under the transversus abdominis muscle aponeurosis with spread along the thoracoabdominal fascia with possible paravertebral space spread.
Dosage	20–30 mL (0.25% ropivacaine or bupivacaine)
Nerve stimulation	Not used
Side effects	Minimal
Complications	Local anesthetic systemic toxicity (LAST): <0.1% Vascular puncture: epigastric artery: <1% Visceral puncture: <1%
Pearls	• QL1, lateral, is close to a posterior TAP and the key landmark is a thick fascia: the lumbar interfascial triangle (LIFT). • QL2, posterior, is between the latissimus dorsi and the QL. • QL3, transmuscular, injection adjacent to the psoas muscle • Large bolus may be better.
Alternative	TAP block Epidural Erector spinae plane block Paravertebral block
Further reading	Chin KJ, et al: Essentials of our current understanding: abdominal wall blocks. Reg Anesth Pain Med 2017; 42(2):133–83.

Suggested Readings

1. Farag E, et al: Atlas of Regional Anesthesia. Elsevier, Philadelphia: 2017.
2. Grant S, Auyong D: Ultrasound Guided Regional Anesthesia Oxford University Press, Oxford, England, UK: 2017.
3. Hadzic A: Hadzic's Textbook of Regional Anesthesia and Acute Pain Management McGraw-Hill Education, New York: 2017.

APPENDICES

AUTHORS

Erin Bushell, MD
Alex J. Butwick, MD, MS, FRCA
Brendan Carvalho, MBBCh, FRCA
Rebecca E. Claire, MD
Genevieve D'Souza, MD
Andrew J. Giustini, MD, PhD
Brenda Golianu, MD
Julie Good, MD
Anita Honkanen, MD
Richard A. Jaffe, MD, PhD
Sharon Jung, PharmD
Jamie Kuo, PharmD, BCCCP
Birgit Maass, MD
Carolyn Nguyen, PharmD
Einar Ottestad, MD
C. Philip Larson Jr., MDCM
Roya Saffary, MD
Clifford A. Schmiesing, MD
Jeannie Seybold, MD
Steven L. Shafer, MD

APPENDIX A: PREOPERATIVE CONSIDERATIONS

■ PREOPERATIVE LABORATORY TESTING AND DIAGNOSTIC STUDIES

Clifford A. Schmiesing, MD

The value and utility of preop diagnostic studies have become central issues in evaluating cost-effective health care in the surgical patient. Approximately \$3 billion is spent in the United States annually on preop laboratory and diagnostic studies. Unnecessary testing is inefficient and expensive, and it requires additional technical resources. Inappropriate studies may lead to evaluation of “borderline” or false-positive laboratory abnormalities. This may result in unnecessary OR delays, cancelations, and potential patient risk through additional testing and follow-up.

Surgical patients require preop lab and diagnostic studies that are consistent with their medical conditions, the proposed operative procedures, and the potential for blood loss. Preop lab and diagnostic testing should be ordered for specific clinical indications rather than just because the patient is about to undergo a particular surgical procedure.

The following preop diagnostic guidelines provide basic recommendations. They are not absolute or standard requirements. Practice guidelines should be modified based on clinical needs and individual practice to ensure the highest quality of anesthesia and surgical patient care.

■ SUMMARY OF PREOP STUDIES

Chest x-ray (CXR)

Overview: A preop CXR should be used to assess the presence of acute, progressive, or chronic changes in cardiac/pulmonary disease. The decision to obtain a preop CXR should be individualized and based on clinical indications (see Table A-1). CXRs should not be part of a routine preop screening protocol.

Clinical indications: Pneumonia; pulmonary edema; atelectasis; aortic aneurysm; mediastinal or pulmonary masses; tracheal deviation; pulmonary HTN; cardiomegaly; advanced COPD and blebs;

dextrocardia; pulmonary embolism

Electrocardiogram (ECG)

Overview: ECGs evaluate cardiac rhythm/conduction disturbances, ischemia, myocardial infarction, hypertrophy, and metabolic and electrolyte disorders.

Clinical indications: Patients with suspected or known Hx of CAD; patient age >60; HTN; Hx of dysrhythmias, chest pain; CHF; diabetes; OSA, cerebrovascular disease, and PVD; syncope or presyncope; dizziness; SOB; DOE; PND; palpitations; leg/ankle edema; abnormal valvular murmurs

Liver function test (LFT)

Overview: LFTs establish the absence or presence of hepatic injury and the degree of hepatic reserve in disease states. LFTs consist of AST (SGOT), ALT (SGPT), GGTP, alkaline phosphatase, serum albumin, and bilirubin.

Clinical indications: Patients with suspected or known Hx of hepatitis (viral, alcohol, drugs, high BMI); infiltration (tumor, immunologic); cirrhosis; portal HTN; gallbladder or biliary tract disease; jaundice; intravascular hemolysis

Renal function testing

Overview: Renal function testing measures glomerular filtration and the magnitude of renal tubular dysfunction. These tests include serum creatinine and BUN.

Clinical indications: Renal function testing is indicated for patients with HTN; increased fluid overload (CHF/peripheral edema/ascites) associated with cardiac or hepatic disease or dehydration; diabetes; nausea, emesis, or anorexia; polyuria; nocturia; oliguria; anuria; high-risk surgery in patients with low-CO syndrome; hematuria; costovertebral angle pain; renal transplant Hx; acute and chronic kidney disease; and dialysis.

Hemoglobin (Hgb),

Overview: The decision to obtain a preop Hgb, Hct,

hematocrit (Hct), CBC	or CBC should be individualized and based on clinical indications, medical Hx, and the proposed surgical procedure. Hgb/Hct or CBC should not be part of a routine preop screening protocol. Clinical indications: Hematologic disorder; bleeding/coagulopathy Hx; malignancy; chemotherapy; radiation therapy (CBC); renal disease; anticoagulant and steroid therapy; surgical procedures with high blood loss (>1500 mL); highly invasive or trauma surgery; malabsorption/poor nutrition status; CNS disease
Pregnancy testing	Overview: The decision to obtain a preop pregnancy test may be based on clinical Hx and examination or OR policy. Several assays are available (serum hCG, urine hCG); β-hCG is detectable in maternal urine and blood 8–9 d postconception. Clinical indications: Sexually active; time of last menstrual period; presence or absence of birth control method; patient intuition. OR policy may require pregnancy testing based on specific criteria including age and ability to become pregnant (i.e., no Hx of hysterectomy).
Coagulation testing	Overview: Coagulation testing, or clotting function studies, should be considered in patients with known or suspected coagulopathies as indicated from H&P and drug therapies. Tests include PT, PTT, INR, and platelet count. Clinical indications: Bleeding disorder Hx; anticoagulants or other drugs affecting coagulation; critical-risk surgeries with significant blood loss expected; hepatic disease; malabsorption/poor nutrition
Urinalysis	Overview: Assessment of renal function, infection, intravascular volume status, and metabolic disorders Clinical indications: There are no routine anesthesia preop requirements for a urinalysis.

We gratefully acknowledge the past contributions of Stephen P. Fischer, MD.

Table A-1. Diagnosis-Based Preop Testing

Tests →	ECG	CXR	Hct/Hb	CBC	Lytes	Renal	Glucose	Coag	LFTs	Drug Levels	Ca ⁺
Preop Diagnosis ↓											
Cardiac Disease											
Chronic atrial fib	X									X*	
CHF	X	±				±					
HTN	X	±				X†	X				
MI history	X				±						
PVD	X										
Stable angina	X				±						
Valvular heart disease	X	±									
CNS Disorders											
Seizures	X			X	X		X			X	
Stroke	X			X	X		X			X	
Tumor	X			X	X						
Vascular/aneurysms	X	X									
Coagulopathies				X				X			
Endocrine Disease											
Addison disease				X	X		X				
Cushing disease				X	X		X				
Diabetes				±	X		X				X
Hyperthyroidism	X	X	X								X
Hypoparathyroidism	X		X								X
Hypothyroidism	X	X	X								X
Hematologic disorders				X							X

Hepatic Disease									
Alcohol/drug-induced								X	X
Infectious hepatitis								X	X
Tumor infiltration								X	X
Malabsorption/poor nutrition	X		X		X		X	±	
Malignancy									
Morbid obesity	X		±					X	
Pulmonary Disease									
Asthma								(PFT only if symptomatic; otherwise, no tests are required)	
Chronic bronchitis	X		±		X				
Emphysema	X		±						X [†]
Kidney Disease									
Acute kidney disease						X	X	X	
Chronic kidney disease						X	X	X	
Select Drug Therapies									
Anticoagulants						X			X
Aspirin/NSAID									X
Chemotherapy					X				
Digoxin (digitalis)	X							±	X
Dilantin									X
Diuretics							X	X	
Phenobarbital									X
Steroids					X			X	
Theophylline									X
X = OBTAIN; ± = CONSIDER. *Patients on diuretics. †Patients on digoxin. ‡Patients on theophylline.									

Suggested Readings

1. Bapojé SR, Whitaker JF, Schulz T, et al: Preoperative evaluation of the patient with pulmonary disease. *Chest* 2007; 132(5):1637–45.
2. Cohn SL, Auerbach AD: Preoperative cardiac risk stratification 2007: evolving evidence, evolving strategies. *J Hosp Med* 2007; 2(3):174–80.
3. Feely MA, Collins CS, Daniels PR, et al: Preoperative testing before noncardiac surgery: guidelines and recommendations. *Am Fam Physician* 2013; 87(6):414–8.
4. Johansson T, Fritsch G, Flamm M, et al: Effectiveness of non-cardiac preoperative testing in non-cardiac elective surgery: a systematic review. *Br J Anesth* 2013; 110(6):926–39.
5. Mooney JF, Ranasinghe I, Chow CK, et al: Preoperative estimates of glomerular filtration rate as predictors of outcome after surgery: a systematic review and meta-analysis. *Anesthesiology* 2013; 118(4):809–24.

PREOPERATIVE EVALUATION OF PATIENTS WITH A CARDIAC RHYTHM MANAGEMENT DEVICE

Preoperative evaluation should include at least the following:

Device-Specific Considerations	• Date of most recent device interrogation (preferably <6 mo) • Type of device, manufacturer, model number, implant date (information typically on “pacemaker card” carried by patient) • Indication (including underlying dysrhythmia) for pacemaker or ICD. Include whether patient is “pacemaker dependent.” • Determine underlying rate and rhythm. • Confirm adequate battery life. • Presence of freshly implanted leads (<3 mo old) → ↑ risk of dislodgement • Programming information including pacing mode, lower rate, pacer/ICD thresholds, and the response to magnet placement (asynchronous mode, ICD inactivation) • If any questions or concerns arise during evaluation, contact patient’s cardiologist. Pacemaker companies can be contacted through 24-h service centers.
Patient-Specific Considerations	• Determine if patient has developed new symptoms possibly related to pacemaker dysfunction since last interrogation, including dizziness, syncope, fatigue, dyspnea, chest pain, or palpitations. • Review status of any associated cardiac disease (CAD, cardiomyopathy, valvular lesions). • Obtain 12-lead electrocardiogram; consider rhythm strip as needed. • If patient is pacemaker-dependent, consider placing magnet for procedures above T10. • If ICD/CRT, consider placing magnet for procedures above T10.

	• Advise surgeon to use short electrocautery applications (<5 sec). • External defibrillation equipment should be immediately available. • Consider placement of external pacer pads. • Use caution when placing upper body central lines with a guidewire. • A magnet should be immediately available.
Procedure-Specific Considerations	• Anatomical location of the surgery and patient position during surgery • Is monopolar cautery necessary? • Will there be other sources of electromagnetic interference (e.g., IORT, ECT, SSEP, MEP, RF ablation, or resection). • Will cardioversion or defibrillation be used during the procedure?

A complete discussion of this topic can be found in the following consensus statement

Grossley GH, Poole JE, Rozner MA, et al: The Heart Rhythm Society (HRS)/American Society of Anesthesiologists (ASA) Expert consensus statement on the perioperative management of patients with implantable defibrillators, pacemakers, and arrhythmia monitors: facilities and patient management: this document was developed as a joint project with the ASA, and in collaboration with the American Heart Association (AHA), and the Society of Thoracic Surgeons (STS) Heart Rhythm 2011; 8(7):1114–54.

ASA PHYSICAL STATUS CLASSIFICATION SYSTEM

ASA Physical Status	Preop Health Status	Examples Include
1	A healthy patient	No Hx acute illness or chronic medical conditions, no meds, no significant substance abuse, negative ROS
2	A patient with mild	No functional limitations: mild obesity,

	systemic disease	pregnancy, smoking, well-controlled hypertension or diabetes
3	A patient with significant systemic disease	Functionally limited: morbid obesity with comorbidities, CHF, COPD, poorly controlled HTN or diabetes, stable angina or old MI, renal failure on dialysis, stable cerebrovascular disease, unruptured aneurysm
4	A patient with severe systemic disease that is a constant threat to life	Severe functional limitations: unstable angina, unstable cerebrovascular disease, symptomatic CHF or COPD, ruptured aneurysm
5	A moribund patient who is not expected to survive without the operation	Not expected to survive >24 h without surgery; severe sepsis, multiorgan failure, severe coagulopathy, or intracranial hemorrhage
6	A declared brain-dead patient whose organs are being removed for donor purposes	

PERIOPERATIVE BETA-BLOCKER THERAPY

Clifford A. Schmiesing, MD

In the absence of new data showing benefit and safety of perioperative beta-blockers (PBB), published guidelines have tempered their support for starting therapy before surgery. Although PBB decreases the risk of cardiac complications, it increases the risk of stroke. The current ACC/AHA updated report continues to give a class I recommendation (should be performed) to continue beta-blockers in patients undergoing surgery who are already receiving the drugs for management of other conditions including but not limited to HTN, afib, heart failure, and coronary artery disease. The report recommends against starting beta-blockers on the day of surgery due to the risk of bradycardia, hypotension, and stroke. Specific recommendations are listed in Table A-2 below.

When starting PBB therapy before surgery, consider the following general

recommendations. “Start low and go slow.” Initiate therapy days to weeks in advance of surgery and titrate to a heart rate of 60–80 beats per minute (assuming not hypotensive), maintain a systolic blood pressure of >100 mm Hg, and monitor vigilantly. Choose medications that are beta-1 selective (e.g., atenolol, metoprolol, and bisoprolol). Consider contraindications before initiating PBB therapy and how it will interact with other medications, medical conditions, and other surgical and anesthetic factors (Table A-3).

Table A-2. Guidelines for Perioperative Beta-Blocker (PBB) Therapy (Adapted from current ACC/AHA Guideline)

Recommendation	Recommendation Class	Level of Evidence
Continue BBs for patients taking them chronically.	Should be performed (1)	Limited (B)
Guide management of BB after surgery by clinical circumstances, independent of when the agent was started.	Reasonable to perform (2a)	Limited (B)
It may be reasonable to begin PBB Rx in patients with intermediate- or high-risk myocardial ischemia noted in preop risk stratification tests.	Reasonable to perform (2a)	Very limited (C)
It may be reasonable to begin PBB Rx in patients ≥ 3 RCRI risk factors (high-risk surgery, diabetes mellitus, heart failure, CAD, CKD, or stroke).	Reasonable to perform (2a)	Limited (B)
Uncertain benefit to begin PBB Rx in patients with a compelling long-term indication for beta-blocker therapy but no other RCRI risk factors	Reasonable to perform (2a)	Limited (B)
If beginning PBB Rx, it may be reasonable to start long enough in advance to assess effect and safety, preferably >1 d before surgery.	Reasonable to perform (2a)	Limited (B)

PBB Rx should not be started on the day of surgery.	Harmful (3)	Limited (B)
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Table A-3. Recommendations for Perioperative Beta-Blocker Therapy (Adapted from the Focused AHA/ACC Recommendations for Perioperative Beta-Blocker Therapy Based on Published Randomized Clinical Trials)

Surgery Type	≤1 Risk Factor	>1 Risk Factor	CHD or Finding of Ischemia on Preop Testing	Taking β-Blocker
Vascular	Usefulness β-blocker uncertain ^{*,†}	Reasonable to give β-blocker ^{‡,§}	Probably give β-blocker	Continue β-blocker
Intermediate risk	Usefulness β-blocker uncertain	Reasonable to give β-blocker	Reasonable to give β-blocker	Continue β-blocker
Low risk	Usefulness β-blocker uncertain	Usefulness β-blocker uncertain	Usefulness β-blocker uncertain	Continue β-blocker

*In patients who are not currently taking β-blockers.

†β-blockers should not be given to patients undergoing surgery who have absolute contraindications to β-blockade.

‡Titrated to heart rate and blood pressure.

§Routine administration of high-dose β-blockers in the absence of dose titration is not useful and may be harmful to patients not currently taking β-blockers who are undergoing noncardiac surgery.

||β-blockers should be continued in patients undergoing surgery who are receiving β-blockers for treatment of conditions with ACA/AHA class I guideline indications for the drugs.

Suggested Readings

1. Bangalore S, Wetterslev J, Pranesh S, et al: Perioperative beta blockers in patients having non-cardiac surgery: a meta-analysis. *Lancet* 2008; 372:1962–76.
2. Devereaux PJ, Yang H, Yusuf S, et al: Effects of extended-release metoprolol succinate in patients undergoing non-cardiac surgery (POISE trial): a randomized controlled trial. *Lancet* 2008; 371:1839–47.
3. Fleisher LA, Fleischmann KE, Auerbach AD, et al: 2014 ACC/AHA guideline on perioperative cardiovascular evaluation and management of patients undergoing noncardiac surgery: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *Circulation* 2014; 130:2215–45.
4. Lee TH, Marcantonio ER, Mangione CM, et al: Derivation and prospective validation of a simple index for prediction of cardiac risk of major noncardiac surgery. *Circulation* 1999; 100:1043–9.

PREOPERATIVE MEDICATION MANAGEMENT

Preop medication management is a crucial step in patient preparation for surgery. It is made challenging by several factors. Input from the surgeon and other consultants may

be needed. Patients often take a large number of medications, and many require active and deliberate management before surgery, including stopping and starting, dose adjustment, and conversion to other medications, and these adjustments often require different timing. Patients frequently find it challenging to follow these medication instructions. Many EHR systems do not provide robust support for this complex workflow. Medication management before surgery must be individualized, thorough, thoughtful, and transparent—a tall order. The following is intended as general recommendations and considerations. Specific management of preop medications must be based on a detailed assessment of the patient’s medical history including a current and accurate medication list, planned surgery, and anesthetic plan.

Preoperative Medication Management Considerations			
Medication	Day Before Surgery	Day of Surgery (DOS)	Other Management Considerations
Antihypertensives	Take*	Take*	*Hold ACE, ARBs $\geq$ 24 h. Hold diuretics DOS unless indicated for management of CHF. Note: hemodialysis (HD) may completely filter off ACE/ARB and lead to rebound HTN. Consider timing of HD and ACE/ARB cessation after HD.
Oral antidiabetics	Take	Hold	Include many antidiabetic medication classes including sulfonylureas, meglitinides, and metformin. Consider holding metformin $\geq$ 24 h inpatient with CKD and those who will receive IV contrast.
Insulin, long-acting	No change	Consider 50–100% a.m.	Consider $\downarrow$ p.m. dose day before surgery if patient is

(peakless)		dose.	prone to nighttime or morning hypoglycemia.
Insulin, intermediate-acting	No change daytime dosing 50–75% of p.m. dose	50–75% a.m. dose	Consider ↓ p.m. dose day before surgery if patient is prone to nighttime or morning hypoglycemia.
Insulin, short-acting	No change	Hold	
Antiplatelet agents (APA)	Risk dependent →	→	Consider risk of thrombosis vs. risk of hemorrhage. Need to individualize based on indication and duration for therapy and surgical urgency and risk of bleeding. Consider carefully before stopping or adjusting antiplatelet therapy, especially dual therapies (DAPT). It is often helpful to consult with the clinician who manages antiplatelet therapy and surgeon to ensure optimal periop management. Guidelines exist for a variety of antiplatelet agent regimens (see Suggested Reading). Consider other hemostatic defects when formulating optimal plan.
NSAIDs	Risk dependent →	→	Consider holding NSAID 3 d before surgery unless the anticipated benefit of pain relief outweighs the

risk of bleeding.
Ibuprofen's effect on platelet function is shorter, and it can often be held for as little as 24 h. Selective COX-2 inhibitors have limited antiplatelet effect and are often continued and even initiated preoperatively. Consider other hemostatic defects (CKD, liver disease, thrombocytopenia) when formulating optimal plan.

Anticoagulants

Risk →
dependent →

Consider risk of thrombosis vs. risk of hemorrhage. Need to individualize based on the indication for anticoagulation, surgical urgency, and risk of bleeding. Thoughtful and timely planning is crucial. It is often helpful to consult with clinician who manages anticoagulant therapy and surgeon to ensure optimal periop management that may include cessation up to a week in advance or bridging strategies. Guidelines exist for periop anticoagulant Rx. (See Suggested Reading.) Consider other hemostatic defects (CKD, liver

			disease, thrombocytopenia, etc.) when formulating optimal plan.
Pure opiate agonists	Take	Take	Consider both fixed dose and PRN use. Helpful to calculate an MME score to determine overall opiate use and the potential effect on periop dosing. Anticipate tolerance when MME is significant.
Partial and complete opiate antagonists	Individualized → →		Partial opiate agonists, especially buprenorphine, are increasingly used to manage opiate use disorder and chronic pain conditions. Preop management needs to be individualized and may include continuation with or without weaning to lower dose. See suggested readings. Full antagonists, especially naltrexone, are being increasingly used for indications including alcohol use disorder. It should usually be held 3–4 d before surgery that is painful, especially postop pain. Depot naltrexone use is on the rise and lasts up to 40 d. Recognition and management well in advance of surgery are necessary.

Antirejection medications	Take	Take	
Psychiatric medications	Take	Take if helpful	Consider drug-drug interactions, especially for MAOI inhibitors.
Acid-lowering	Take	Take	
Statins	Take	Take	
Corticosteroids	Take	Take or $\pm$ stress dose	
Dietary supplements	Consider holding supplements 7 d before, especially if constituents are hard to identify. Vitamins generally can be continued up to the day before surgery.		
Preemptive analgesia	Consider a multimodal approach that may include acetaminophen, gabapentin or pregabalin, and COX-2 inhibitors.		

Suggested Readings

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7. Levine GN, Bates ER, Bittl JA, et al: 2016 ACC/AHA guideline focused update on duration of dual antiplatelet therapy in patients with coronary artery disease: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *Circulation* 2016; 134(1):e173–187.

Clinical Practice Guidelines. J Am Coll Cardiol 2016; 68(10):1082–115.

APPENDIX B: STANDARD ADULT ANESTHETIC PROTOCOLS

Richard A. Jaffe, MD, PhD, Clifford A. Schmiesing, MD, and Steven L. Shafer, MD

■ STANDARD MONITORS (NONINVASIVE)

Blood pressure (BP)	Usually noninvasive (oscillometric) technique. Match cuff width to arm size to avoid inaccurate BP.
Capnometry/capnography	Measurement of ET CO_2 /display of waveform
Gas analyzer (e.g., IR, mass spectrometry)	Measurement of respired gases including end-tidal CO_2 and inhaled anesthetics
Electrocardiogram (ECG)	5-Lead preferred. Usual display: lead II and V_4 or V_5 . EASI lead system can be used to derive a 12-lead ECG that may improve detection of intraop ischemia.
Esophageal or precordial stethoscope	Breath and heart sounds are monitored; dysrhythmias and pulmonary system changes may be quickly detected. Seldom used today.
Nerve stimulator	Monitor status of neuromuscular blockade usually by stimulating the ulnar nerve. Automated systems to measure TOF ratio are more accurate than visual assessment but still may fail to detect residual blockade.
Oxygen analyzer	Measurement of FiO_2
Pulse oximetry	Measurement of O_2 saturation of arterial hemoglobin; also gives a pulse sound that indicates heart rate, rhythm, and change in saturation
Temperature	Nasal, esophageal, bladder, rectal, tympanic, or skin (deep esophageal best approximates core

	temperature)
Visual observation of patient	Skin color, pupils, edema, sweating, movement
Ventilator function monitors	PIP, TV, disconnect alarm, pressure-volume/flow-volume loops, etc.

STANDARD ANESTHETIC MANAGEMENT (ADULT ASA 1 AND 2)

NB: The following sections are guidelines only (for an otherwise healthy 70-kg adult). Specific drugs and drug dosages should be individualized, based on the physiologic and pharmacologic status of the patient, including factors such as age, weight, concurrent medication, and comorbidities.

PREMEDICATION FOR ANXIOLYSIS ± ANALGESIA

Light	Midazolam 1–2 mg iv - or- Lorazepam 1–2 mg - or- Hydroxyzine 25–100 mg	Before induction or transport to OR po 1 h preop po 1 h preop
Moderate	Midazolam 2–3 mg iv ± Fentanyl 25–100 mcg iv	Prior to transport to OR Monitor for respiratory depression or apnea.
Heavy	Lorazepam 1–2 mg Midazolam titrated to effect	PO the night before surgery Before transport to OR. May require >10 mg iv. NB: severe respiratory depression when combined with opiates. Flumazenil should be available.

PREINDUCTION CHECKLIST

Anesthesia machine	Complete ASA preanesthesia checkout procedures. Sample procedures for specific
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	machines/workstations are on the ASA Web site: http://www.asahq.org/For-Members/Clinical-Information/2008-ASA-Recommendations-for-PreAnesthesia-Checkout/Sample-Procedures.aspx .
Airway management	Appropriate-sized face masks Working laryngoscope and blade Working backup laryngoscope and blade Working suction ETT with nonleaking cuff + backup ETTs in other sizes ETT stylet and bougie available Oral and nasal airways available McGill forceps available Ambu® bag or equivalent self-inflating resuscitator available Unexpected difficult airway plan and equipment available
Monitoring	Standard monitors (see section above) Invasive monitoring if necessary, before induction
Patient	Working IV access Verify status of regional blocks (if in use). Positioning plans discussed with surgeon Fire risk assessment (especially if open O₂ delivery is planned)
Medication	Emergency drugs available Infusions prepared and pump function verified Anesthetic plan and drugs reviewed

INDUCTION TECHNIQUES

Preinduction	1. Administer 100% O ₂ by mask × 3 min (ideally ETO ₂ should be >90%). 2. Administer supplemental sedation/analgesia (as appropriate). e.g., fentanyl 1–3 mcg/kg iv			
Induction agents	Propofol	1.5–2.5 mg/kg iv (in increments) NB: pain on injection may be lessened by prior administration of lidocaine 1% 5–10 mL in conjunction with temporary venous occlusion.		
	Thiopental	3–5 mg/kg iv (still unavailable in the United States)		
	Etomidate	0.2–0.4 mg/kg iv NB: pain on injection; myoclonus		
Muscle relaxants for intubation	Drugs	Doses	Onset	Duration
	Vecuronium	0.1 mg/kg	2–3 min	24–30 min
		0.2 mg/kg (rapid onset)	<2 min	45–90 min
	Rocuronium	0.6 mg/kg	90–120 sec	30–45 min
		1.2 mg/kg	60–90 sec	60–120 min
	Pancuronium	0.1 mg/kg	3–4 min	40–65 min
	Gantacurium*	0.2–0.5 mg/kg	1–2 min	5–10 min
	Cisatracurium	0.2 mg/kg	2 min	40–80 min
	Succinylcholine	1.0 mg/kg	30–60 sec	4–8 min
	If given after defasciculating dose of NMR	1.5 mg/kg		
	Infusion	1 g/250–500 NS: 50–100 mcg/kg/min (titrated to effect)	+60 sec	While infusing +5–8 min (phase II block possible)
	*Gantacurium is not yet (2019) available in the United States.			

MAINTENANCE TECHNIQUES

Inhalational anesthesia	30–100% O ₂ (NB: absorption atelectasis rapid above 80% O ₂) + 0–70% N ₂ O + Sevoflurane (MAC = 2.0%) or isoflurane (MAC in 100% O ₂ = 1.2%) or titrated to effect. Desflurane (MAC = 6.6%) may be useful for rapid recovery in morbidly obese patients.
Balanced anesthesia	30–100% O ₂ (NB: absorption atelectasis rapid above 80% O ₂) + 0–70% N ₂ O + Hydromorphone 3–10 mcg/kg/3–4 h (intermittent bolus) or fentanyl 1–10 mcg/kg prn response to surgical stimulation or remifentanyl 0.05–2 mcg/kg/min prn response to surgical stimulation (no residual analgesia)

	+ Isoflurane or sevoflurane titrated to effect or propofol 50–200 mcg/kg/min (titrated to effect)		
Total intravenous anesthesia (TIVA)*	Oxygen usually mixed with air to a final concentration of 30–50%		
	Remifentanyl [†]	Induction infusion	@ 0.5–1 mcg/kg/min × 1–2 min
		Maintenance infusion	@ 0.1–0.35 mcg/kg/min
		10 min before end of surgery, reduce infusion rate	@ 0.05–0.1 mcg/kg/min
		5 min before end of surgery	Infusion off
	Propofol	Induction bolus	1–1.5 mg/kg
		Maintenance infusion	@ 60–90 mcg/kg/min
		10 min before end of surgery, reduce infusion rate	@ 20–40 mcg/kg/min
		3 min before end of surgery	Infusion off
	*Steven Shafer, MD.		
	†No residual analgesia: Postop pain management depends on the type of surgery, and analgesic requirements may be substantial. Recommend having fentanyl or meperidine for quick pain relief followed by hydromorphone as a component of multimodal analgesia for prolonged pain relief.		

If **continued muscle relaxation** is required during the above maintenance techniques, several options are available. Always use a nerve stimulator to assess block before bolus redosing.

Short-acting	Gantacurium*	0.2 mg/kg/10–20 min
	Rocuronium	Rocuronium can be rapidly reversed shortly (3 min) after administration (1.2 mg/kg bolus) by using

		sugammadex (16 mg/kg iv).
Intermediate	Vecuronium	0.025 mg/kg/30 min; 1–2 mcg/kg/min infusion
	Rocuronium	0.6 mg/kg/30 min; 5–12 mcg/kg/min infusion
	Cisatracurium	0.2 mg/kg/40 min; 1–2 mcg/kg/min infusion
Long-acting	Pancuronium	0.02 mg/kg/60–90 min

*Not yet (2019) available in the United States.

EMERGENCE

1. Reversal of muscle relaxant	As surgical conditions permit, reverse residual muscle relaxant (when at least one twitch is present in train-of-four) with one of the following: Sugammadex 2 mg/kg actual body weight if twitch 2 in TOF has reappeared, otherwise 4 mg/kg Neostigmine 0.05–0.07 (maximum dose) mg/kg iv + glycopyrrolate 0.01 mg/kg iv Edrophonium 0.5–1.0 (maximum dose) mg/kg iv + atropine 0.015 mg/kg iv
2. Analgesia	If remifentanyl was used during surgery, supplemental analgesics will be necessary and should be given promptly after emergence.
3. Nausea prophylaxis	Ondansetron (4 mg iv before emergence) or granisetron (1 mg iv for a longer duration effect). Also, consider adding dexamethasone 4 mg at induction, metoclopramide (20 mg iv before emergence), or aprepitant (40 mg po 1–3 h before induction). The use of droperidol (0.625 mg iv) is unfortunately controversial because of QT prolongation at higher doses. Consider OG tube suction to empty stomach. See Prevention of PONV (below).
4. O₂	d/c N ₂ O/volatile agents and administer 100% O ₂ .

5. Suction	Suction oropharynx thoroughly.
6. Extubation	Extubate after protective airway reflexes have returned and the patient is breathing spontaneously and ideally (with adequate analgesia) can follow commands.

LUNG PROTECTIVE VENTILATION

The traditional use of high tidal volume (10–15 mL/kg) mechanical ventilation has been associated with alveolar overdistension → ventilator-associated lung injury → release of inflammatory mediators → pneumonia and multiorgan dysfunction in ICU patients. Recent studies have suggested that a strategy of lung-protective ventilation using a pressure-controlled, volume-directed ventilator mode may improve postop outcomes in a variety of surgical patient populations. The types of surgery most commonly associated with postop pulmonary dysfunction include thoracic abdominal aortic aneurysm, neurosurgery, upper abdominal surgery, and emergency surgery.

Recommendations for lung-protective ventilation in hemodynamically stable patients (without ARDS) at risk for adverse postop pulmonary complications include:

- FiO₂ titrated to keep SpO₂ > 92% (optimal PaO₂ is patient dependent)
- Tidal volume = 6–8 mL/kg predicted body weight (see Table B-1)
- PEEP = 4–8 cm H₂O (less if venous return is impaired)
- Recruitment maneuvers (CPAP of 30 cm H₂O × 30 sec, prn atelectasis)

In the IMPROVE trial, recruitment maneuvers started immediately after intubation and were standardized as follows:

1. Pressure support ventilation mode
2. PEEP set to 30 cm of water
3. Inspiratory gas flow set to the highest value
4. Duration of the maneuver = 30 sec

Table B-1. Duration of the maneuver = 30 sec

Height (inches)	Female Predicted Body Weight (kg)	Female Tidal Volume (mL) (@6–8 mL/kg)	Male Predicted Body Weight (kg)	Male Tidal Volume (mL) (@6–8 mL/kg)
5' 0"	45.5	273–364	50	300–400
5' 1"	47.8	287–382	52.3	314–418
5' 2"	50.1	301–401	54.6	328–437
5' 3"	52.4	314–419	56.9	341–455
5' 4"	54.7	328–438	59.2	355–474
5' 5"	57	342–456	61.5	369–492
5' 6"	59.3	356–474	63.8	383–510
5' 7"	61.6	370–493	66.1	397–529
5' 8"	63.9	383–511	68.4	410–547
5' 9"	66.2	397–530	70.7	424–566
5' 10"	68.5	411–548	73	438–584
5' 11"	70.8	425–566	75.3	452–602
6' 0"	73.1	439–585	77.6	466–621

Formula: males, PBW (kg) = 50 + 2.3 (height [inches] – 60); females, PBW (kg) = 45.5 + 2.3 (height [inches] – 60)

Suggested Readings

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2. Mills GH: Respiratory complications of anaesthesia. *Anaesthesia* 2018; 73(Suppl 1):25–33.

MONITORED ANESTHESIA CARE (MAC)

1. Standard monitoring with regular verbal contact.
2. Nasal O₂ (qualitative measurement of respiratory rate, and ETCO₂ can be obtained by attaching a sampling catheter to the nasal cannula). The open administration of O₂ should be limited to a maximum concentration of 30% O₂ for procedures above T4 to minimize fire risk.
3. Light-to-moderate levels of sedation (±analgesia) can be maintained using a propofol (25–100 mcg/kg/min) or dexmedetomidine (see below) infusions, with PRN intermittent bolus injections of midazolam (1 mg) or propofol (20–30 mg), or with a remifentanyl infusion (0.025–0.07 mcg/kg/min), titrated to effect. Monitor closely for respiratory depression.

- l. Alternatively, dexmedetomidine (a α -2 agonist) can produce excellent sedation and analgesia without respiratory depression. The loading dose is typically 1 mcg/kg iv over ~10 min ($\downarrow$ BP may occur) followed by an infusion of 0.2–0.7 mcg/kg/h.
5. If the initial local anesthetic injection is painful (e.g., retrobulbar block), then a brief period of analgesia, sedation, and amnesia can be induced with

	Advantages	Disadvantages
A. Midazolam (0.5–2 mg) $\pm$ ketamine (10–20 mg) $\pm$ alfentanil 3–7 mcg/kg or remifentanil 0.5 mcg/kg all injected 2–3 min before acute pain	Profound amnesia and analgesia. Usually no apnea, and airway reflexes are maintained. The patient is usually able to cooperate.	Patient not “asleep” Possible $\uparrow$ BP and HR
B. Propofol (0.5–1 mg/kg) $\pm$ fentanyl (25–50 mcg/kg)	Patient “asleep”	Possible apnea with loss of airway. $\downarrow$ BP may occur; patient may become unresponsive.

RAPID-SEQUENCE INDUCTION OF ANESTHESIA (FULL-STOMACH PRECAUTIONS)¹

1. $\downarrow$ Gastric volume/acidity	Ranitidine 50 mg iv at least 30–60 min before induction if possible Metoclopramide 10 mg iv 30–60 min before induction if possible 0.3 M sodium citrate 30 mL po within 15 min of induction
2. Induction	Preoxygenation $\times$ 3 min (or until $\text{ETO}_2 > 90\%$). Suction must be readily available. Stylet ETT. $\pm$ Defasciculate, e.g., rocuronium 3–4 mg iv 1.5–3 min before using succinylcholine. Cricoid pressure (Sellick maneuver) by an assistant (8–10 lb; beware of force transfer to C6 vertebra). Effective cricoid pressure makes laryngoscopy, mask ventilation, and LMA insertion difficult and will not

	eliminate the risk of aspiration. Choice of induction agent depends on patient condition including hemodynamic stability. Options include Etomidate 0.1–0.4 mg/kg (minimal hemodynamic effects) or Ketamine 1 mg/kg (usually ↑BP and HR) For patients better able to tolerate ↓ BP: Propofol 1–2.5 mg/kg + succinylcholine 1.5 mg/kg or rocuronium 1.2 mg/kg (slightly slower but <2 min) for intubation (sugammadex 16 mg/kg can be used for rapid reversal) Lungs may be ventilated while waiting for NMB—avoid airway pressure >20 cm H₂O (to reduce stomach insufflation).
3. Intubation	Intubate only when the patient is fully relaxed. After intubation, watch chest movement and auscultate for equal BBS. ✓ expired CO₂ on monitor. Listen over the stomach. Secure ETT and release cricoid pressure. Pass NG/OG tube and suction stomach contents.
4. Failed intubation protocol	[See Anesthetic Considerations for Cesarean Section, Obstetric Surgery, p. 297]
5. Maintenance	As indicated by patient's condition and type of surgery
6. Extubation	Repeating NG/OG suction is often useful before emergence and extubation. Extubate when patient is awake and has active laryngeal protective reflexes. Postop ICU care may be necessary until safe extubation can be assured.

1SPECIAL PEDIATRIC CONSIDERATIONS

1. The same principles apply in children requiring surgery and in those who may have

full stomachs. Consider emptying the stomach with an OG tube before induction in patients with pyloric stenosis or with high-grade intestinal obstruction following po barium. If iv is placed, continue as indicated previously. If iv access is difficult, O₂/sevoflurane induction with cricoid pressure, and succinylcholine (2–4 mg/kg im) will permit intubation and minimize risks of gastric aspiration.

2. Awake intubation in neonates and sick infants may be the safest method.

PREVENTION OF PONV (ADAPTED FROM GAN ET AL., 2014)

Clifford A. Schmiesing, MD

PONV Risk Factors	Patient-related Anesthetic-related Surgical-related	Female gender Nonsmoking status Hx PONV or motion sickness Volatile anesthetics Nitrous oxide Intra-/postop opiates High-dose neostigmine Duration of surgery Type of surgery	Simple risk score for PONV in adults: assign 1 point for each of the following risk factors—female gender, nonsmoker, hx of PONV, and postop opioids. Risk of PONV: point sum 0 = 10%; 1 = 20%; 2 = 40%; 3 = 60%, and 4 = 80%.
Antiemetics	Pharmacologic Agents Serotonin blockers Dexamethasone 4–8 mg iv Droperidol 0.625–1.25 mg iv or Haloperidol 1 mg iv Metoclopramide 10–25 mg iv	Timing of Administration End of surgery Postinduction 30 min before end of surgery 30 min before end of surgery 30 min before end of surgery 2–4 h before surgery	Ondansetron 4–8 mg iv or po, granisetron 0.35–1.5 mg iv, dolasetron 12.5 mg iv, palonosetron 0.075 mg iv (long-acting) Adverse effects not seen with single-bolus dose.

	Promethazine 6.25–25 mg iv Scopolamine transdermal patch Neurokinin-1 receptor antagonists	Before surgery	May cause perianal pruritus when given iv to awake patient Minimal risk of QT prolongation, cardiac dysrhythmias, or extrapyramidal reactions at these low doses Can cause anxiety, confusion, or visual changes. 20–25 mg more effective than lower doses Sedating and may cause local tissue damage if iv catheter infiltration occurs May cause sedation, dry mouth, dilated pupils, visual changes, and confusion, especially in the elderly Aprepitant 40–80 mg po. Long-acting
Nonpharmacologic PONV prevention	Acupuncture, acupressure	Before or after surgery	
PONV prevention strategies	↓ Anesthesia-related RFs	Consider regional anesthesia where	

		appropriate. Propofol for induction and maintenance where appropriate ↓ N ₂ O and volatile agents ↓ Intraop and postop opioids ↓ Neostigmine	
PONV risk stratification and management	Low risk Intermediate risk High risk Very high risk	No prophylaxis → treat PONV when it occurs Choose 1–2 interventions Choose 2 interventions Consider 3–4 interventions	Consider including both short- and long-lasting antiemetics that will be effective post PACU discharge.

Adapted from Gan TJ, Diemunsch P, Habib AS, et al. Consensus guidelines for the management of postoperative nausea and vomiting. *Anesth Analg* 2014; 118(1):85–113. Copyright © 2017 Elsevier. With permission.

Suggested Readings

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2. Apfel CC, Philip BK, Cakmakkaya OS, et al: Who is at risk for postdischarge nausea and vomiting after ambulatory surgery? *Anesthesiology* 2012; 117(3):475–86.
3. Apfelbaum JL, Silverstein JH, Chung FF, et al: Practice guidelines for postanesthetic care: an updated report by the American Society of Anesthesiologists Task Force on Postanesthetic Care. *Anesthesiology* 2013; 118(2):291–307.
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VENOUS THROMBOEMBOLISM (VTE) PROPHYLAXIS

Clifford A. Schmiesing, MD

Common patient risk factors for VTE	• Increasing age • Previous VTE or family h/o VTE • Pregnancy and the postpartum period • Obesity • Inherited or acquired thrombophilia • Immobility, lower extremity paresis
Common disease-related risk factors for VTE	• Estrogen-containing oral contraceptives or hormone replacement therapy • Major surgery trauma (major trauma or lower extremity injury) • Cancer (known or occult) • Inflammatory bowel disease • Nephrotic syndrome • Myeloproliferative disorders • Paroxysmal nocturnal hemoglobinuria

VTE Risk Factor Assessment: Caprini Score Worksheet

Venous Thromboembolism Risk Factor Assessment Please Choose All That Apply				
Each Risk Factor Represents 1 Point	Each Risk Factor Represents 2 Points	Each Risk Factor Represents 3 Points	Each Risk Factor Represents 5 Points	Women Only: Each Risk Factor Represents 1 Point
☐ Ages 41–60 y ☐ Minor surgery planned (<45 min) ☐ History of prior major surgery (>1 mo) ☐ Varicose veins ☐ History of inflammatory bowel disease ☐ Swollen legs (current) ☐ Obesity (BMI > 25) ☐ Acute myocardial infarction (<1 mo) ☐ Congestive heart failure (<1 mo) ☐ Sepsis (<1 mo) ☐ Serious lung disease including pneumonia (<1 mo) ☐ Abnormal pulmonary function (COPD) ☐ Medical patient currently at bed rest ☐ Other risk factors: _____	☐ Ages 61–74 y ☐ Major surgery (>45 min) ☐ Arthroscopic surgery (>45 min) ☐ Laparoscopic surgery (>45 min) ☐ Previous or present malignancy ☐ Central venous access ☐ Immobilizing leg plaster cast (<1 mo) ☐ Patient confined to bed (>72 h)	☐ Age 75 y or more ☐ History of DVT/PE ☐ Family history of DVT/PE ☐ Positive factor V Leiden ☐ Positive prothrombin 20210A ☐ Elevated serum homocysteine ☐ Positive lupus anticoagulant ☐ Elevated anticardiolipin antibodies ☐ Heparin-induced thrombocytopenia (HIT) ☐ Other congenital or acquired thrombophilia ☐ If yes: Type_____	☐ Elective major lower extremity arthroplasty ☐ Hip, pelvis, or leg fracture (<1 mo) ☐ Stroke (<1 mo) ☐ Multiple trauma (<1 mo) ☐ Acute spinal cord injury (paralysis) (<1 mo)	☐ Oral contraceptives or hormone replacement therapy ☐ Pregnancy or postpartum (<1 mo) ☐ History of unexplained stillborn infant, recurrent spontaneous abortion (≥3), premature birth with toxemia, or growth-restricted infant
Total risk factor score =				

Venous Thromboembolism Prophylaxis Recommendations for Caprin Risk Score

Caprini Score	Risk Level	Prophylaxis Recommendation
0	Very low	Early ambulation
1–2	Low	Mechanical prophylaxis (sequential compression)
3–4	Moderate	Chemoprophylaxis (mechanical prophylaxis optional): heparin, 5000 units sc tid; low-

		molecular-weight heparin, 30 units sc bid; low-molecular-weight heparin, 40 units sc QD
≥5	High	Chemoprophylaxis and mechanical prophylaxis: heparin, 5000 units sc tid; low-molecular-weight heparin, 30 units sc bid; low-molecular-weight heparin, 40 units sc QD

Adapted from Jacobs B, Henke PK. Evidence-Based Therapies for Pharmacologic Prevention and Treatment of Acute Deep Vein Thrombosis and Pulmonary Embolism. Surg Clin North Am 2018; 98(2):239–53. Copyright © 2017 Elsevier. With permission.

Suggested Readings

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POSTOPERATIVE DELIRIUM: PRACTICAL GUIDELINES

Postoperative delirium (POD) can occur in any patient. DSM-5 criteria define delirium as “a disturbance in attention (i.e., reduced ability to direct, focus, sustain, and shift attention) and awareness (reduced orientation to the environment)” that develops over a short period of time and includes impaired cognition not explained by preexisting conditions. POD may be characterized as hyperactive or hypoactive, the latter being more difficult to diagnose. POD has been associated with long-term morbidity including POCD (postoperative cognitive dysfunction) and PTSD (posttraumatic stress disorder). POD usually starts in the recovery room and may persist for several days postop. The complex and often unpredictable interactions of both predisposing and precipitating factors make an accurate preop risk assessment for POD difficult.

RISK FACTORS ASSOCIATED WITH POD INCLUDE:

Preop

- Cognitive impairment
- Patient self-assessment of health if fair, poor, or very poor (compared to good or very good)
- Functional age greater than 65 years

	• Functional impairments • Dehydration/malnutrition • Significant comorbidities (e.g., CVD, CAD, PVD, DM, PD, anemia, depression/anxiety disorders) • Vision or hearing impairments • Polypharmacy or h/o substance abuse • Emergency surgery
Intraop	• Surgery of long duration • General anesthesia or deep sedation • Hypo or hyperglycemia • Hypotension (SBP < 90 mm Hg)
Postop	• Surgical complications • Hypothermia on arrival in recovery • Poorly controlled pain

Prevention should include when possible the use of fast-track (ERAS) protocols, preop Midazolam for significant anxiety, intraop dexmedetomidine, perioperative multimodal analgesia, and avoidance of deep anesthesia/sedation.

Treatment should include the return of any hearing or vision aids, minimal restraint, frequent reassurance and reorientation, supplemented (if necessary) with low-dose dexmedetomidine (e.g., 0.2 mcg/kg over 2 minutes followed by 0.2 mcg/kg/h infusion) or haloperidol (e.g., 0.25 mg titrated up to 2 mg iv at 30-minute intervals).

In recovery, patients should be screened for POD and if confirmed, treated before discharge. The Nursing Delirium Screening Scale (Nu-DESC) is often used for POD diagnosis (see Fig. B-1 below), but like similar screening tests, its reported sensitivity (32–95%) is quite variable. As usual, other potentially treatable explanations for POD-like Sx (e.g., electrolytes, ventilation, temperature, Sz, stroke, central cholinergic syndrome) should be ruled out.



Features and descriptions		Symptoms Rating (0-2)		
Symptom	Time Period	Midnight - 8 AM	8 AM - 4 PM	4 PM - Midnight
I. Disorientation Verbal or behavioural manifestation of not being oriented to time or place or misperceiving persons in the environment				
II. Inappropriate behaviour Behaviour inappropriate to place and/or for the person; e.g., pulling at tubes or dressings, attempting to get out of bed when that is contraindicated, and the like.				
III. Inappropriate communication Communication inappropriate to place and/or for the person; e.g., incoherence, noncommunicativeness, nonsensical or unintelligible speech.				
IV. Illusions/Hallucinations Seeing or hearing things that are not there; distortions of visual objects.				
V. Psychomotor retardation Delayed responsiveness, few or no spontaneous actions/words; e.g., when the patient is prodded, reaction is deferred and/or the patient is unarousable.				
Total score				

Fig. B-1. The Nursing Delirium Screening Scale (Nu-DESC). Symptoms are rated from 0 to 2 based on the presence and intensity of each symptom, and individual ratings are added to obtain a total score per shift. A score of 2 or more is strongly correlated with the presence of postop delirium.

Suggested Readings

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APPENDIX C: MULTIMODAL PERIOPERATIVE PAIN MANAGEMENT

Roya Saffary, MD and Einar Ottestad, MD

INTRODUCTION

Periop pain management continues to be a challenging aspect of anesthesiology with significant effects on patient satisfaction and clinical outcomes. According to the Centers for Disease Control and Prevention (CDC), about 48 million surgical and nonsurgical procedures were performed in the United States in 2010. A large survey of periop pain management revealed that ~80% of patients experienced acute pain postop, with 86% of those patients characterizing their pain as ranging from moderately to extremely painful. Furthermore, postop pain management was the most common concern of the patients surveyed. Expertise in neuraxial and regional anesthesia, as well as analgesic pharmacology, makes the anesthesiologist the ideal advocate for improving periop pain management.

The Joint Commission has recognized the importance of pain management in reducing periop pain while facilitating improvements in functionality. Poor pain control leads to adverse clinical outcomes, including decreased ability to ambulate with increased risk for thromboembolic events and fatal pulmonary embolism. Inadequate pain control following abdominal and thoracic surgeries may → splinting, atelectasis, and pneumonia.

The neuroendocrine stress response to surgery involves the release of stress hormones and catecholamines, which lead to many deleterious clinical effects and outcomes. These include weight loss, fatigue, immunosuppression, thromboembolism, hypercoagulability, dysrhythmias, urinary retention, and impaired pulmonary function. Furthermore, ongoing, uncontrolled pain in the postop period is a risk factor for chronic postsurgical pain. The continuous nociceptive barrage to the spinal cord and brain can lead to central sensitization, or “windup,” which is thought to result in persistent pain beyond the acute recovery period. As we learn more about periop pain management, we can minimize pain while reducing morbidity and mortality in our surgical patients.

Opioids remain the primary treatment for postop pain. However, the recent opioid epidemic and the clinical application of enhanced recovery after surgery (ERAS) protocols have emphasized the potential risks and weaknesses of opioid-dominated

treatment. In addition to respiratory depression, opioids can lead to dependence, opioid-induced hyperalgesia, decreased testosterone levels, depressed immune function, lack of functionality, slowed recovery, and morphologic brain changes, particularly in patients with a preexisting chronic pain condition and/or opioid tolerance. In contrast to traditional pain management, ERAS protocols utilize an opioid-sparing approach by incorporating other medication families. Initially introduced to standardize key aspects of periop management including pain management for colorectal surgery, ERAS has since been applied to other surgical subspecialties, including gynecologic and orthopedic surgeries. Studies have shown that this approach significantly improves patient comfort and overall recovery and may even decrease hospital stay.

MULTIMODAL ANALGESIA

The concept of multimodal analgesia is based on the understanding that no single analgesic medication targeted to one receptor system allows for sufficient postop pain control. Therefore, rather than relying solely on opioids, ERAS protocols utilize agents that act on several receptor systems in the nociceptive pathway often combined with neuraxial or regional techniques to provide more effective postop analgesia. A number of medication classes have been effectively used for this purpose throughout the periop period. Please refer to Table C-1 for an outline of medication classes and examples.

Table C-1. Pain Medication Classes	
Class	Examples
Mu opioid receptor agonists	IV or po morphine; sublingual sufentanil
Mixed COX inhibitors	IV or po ketorolac
COX-2 inhibitors	PO celecoxib
Central COX inhibitors	IV or po acetaminophen
Sodium channel blockers	IV lidocaine; po mexiletine
Multisite effects including Ca ⁺⁺ channel modulation	PO gabapentin or pregabalin
SNRIs	PO duloxetine or venlafaxine
TCAs	PO desipramine or nortriptyline

Alpha-2 agonists	Patch clonidine or IV dexmedetomidine
NMDA antagonists	IV ketamine; magnesium
Inhaled low-dose anesthetic	Methoxyflurane (experimental)

Postop pain management and patient recovery are most successful when an integrated, multispecialty, rehabilitation-orientated approach is utilized. Input from surgeons, anesthesiologists, rehabilitation specialists, nursing, pharmacy, and other key health care providers is necessary for this to occur. Best outcomes require that the patient be given appropriate preop education, excellent periop nociceptive blockade, attenuation of the neuroendocrine stress response, postop exercise, and early enteral nutrition.

PREEMPTIVE ANALGESIA

Preemptive analgesia refers to the administration of antinociceptive agents especially during the pre- and intraop periods to prevent the formation of central sensitization caused by incisional and inflammatory injuries. The benefit of preventing sensitization is a reduction in the patient's risk of developing chronic pain after surgery. It has been demonstrated clinically that preincisional analgesics help improve postop pain control. A number of agents have been studied in this context with promising results, including lidocaine, ketamine, neuraxial anesthesia, peripheral nerve blockade, topical anesthetics, peripheral opioid, gabapentin, and central opioid agonists. ERAS protocols often recommend oral administration of nonopioid analgesics in the preop area, including NSAIDs, gabapentin, and acetaminophen.

THE NEW ANALGESIA PARADIGM

This new paradigm, with ERAS at the forefront, emphasizes the preemptive and multimodal administration of a variety of analgesic classes through various routes of administration, including iv, po, neuraxial, peripheral nerve exposure, and topical agents. The traditional approach using mainly short-duration opiates has been shown to be suboptimal and inferior to providing the patient with a more comprehensive periop pain management regimen. In addition to improved pain control, this approach leads to less reliance on opioids and their associated side effects, as well as reduced risk of sensitization and development of chronic postop pain. When used in conjunction with a structured postop rehabilitation program, these techniques can lead to decreased patient morbidity and mortality, increased patient satisfaction, decreased recovery time, and shorter hospitalization (Table C-2).

Table C-2. An Example of Multimodal Analgesia for a 70-kg Patient Without Specific Drug Contraindications, Undergoing Major Surgery

Preinduction	Systemic	Ibuprofen 800 mg or celecoxib 200 mg (decreased bleeding risk) po 30–60 min before induction and/or acetaminophen 1000 mg po 30–60 min before induction and/or gabapentin* 600–1200 mg or pregabalin* 150–300 mg po 60 min before induction
	Regional	Neuraxial analgesia (e.g., epidural or spinal opiates) or regional/peripheral nerve blocks
Preincision	Systemic	Fentanyl 100–250 mcg iv and/or ketamine 0.5 mg/kg iv bolus + 0.25 mg/kg/h infusion during surgery (especially useful in opioid-dependent patients)
	Local	Incision-site local anesthetic infiltration
Intraop	Systemic	IV opiates (e.g., fentanyl, hydromorphone)
		Lidocaine infusion 1–3 mg/kg/h
		Acetaminophen 1000 mg IV if not given preop
		Magnesium sulfate infusion 1 g/h
		Continuous regional anesthetic
Postop	Systemic	IV opiates (e.g., fentanyl, hydromorphone) bolus and PCA
		Ketorolac 15 mg iv q6h or ibuprofen 200–800 mg po q8h or celecoxib 100 mg bid
		Acetaminophen 650–1000 mg q6–8h

*Recent meta-analyses suggest that gabapentinoids may be associated with an increased incidence of adverse events (e.g. sedation, respiratory depression) while providing minimal opioid-sparing effects.

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■ STANDARD ADULT POSTOP PCA ORDER

Typical orders

1. Call anesthesiologist with any questions about PCA.
2. Check respiratory rate q1h while PCA is in use.
3. Call anesthesiologist if respiratory rate <10/min. If rate <6/min, treat with naloxone 0.4–0.8 mg iv push (may repeat q2–3min to 2 mg), and assist ventilation while waiting for anesthesiologist.
4. Encourage patient to ambulate 4–6 h after surgery (unless contraindicated).

5. After PCA d/c'd, start po pain medications (per surgeon).
6. If patient required a large loading dose, the PCA lockout dose may need to be increased. Expect patient to need about 1/3 loading dose q h. Change PCA dose or lockout time to permit this dosing.

Analgesics	Morphine	1–2 mg q10–15min
	Hydromorphone	0.2–0.4 mg q10–12min
	Fentanyl	10–25 µg q8–10min*

*Very short-acting might need small basal infusion dose.

■ Epidural Analgesia*

Lumbar Epidural		Thoracic Epidural	
Morphine Bolus	Hydromorphone Bolus	Morphine Bolus	Hydromorphone Bolus
2–4 mg	0.4–0.8 mg	1–2 mg	0.1–0.4 mg
Morphine infusion	Hydromorphone infusion	Morphine infusion	Hydromorphone infusion
0.5–1 mg/h	0.05–0.2 mg/h	0.25–0.5 mg/h	0.1–0.2 mg/h

*Age <65 consider half dosing.

Special considerations

1. Concentrations of opioids used for epidural infusions (in preservative-free solution):
 - Morphine, 0.15 mg/mL
 - Hydromorphone, 0.05 mg/mL

■ Epidural Anesthesia/Postop Analgesia*

Surgical Site	Epidural Catheter Location	Initial Bolus of 0.5% Bupivacaine	Infusion Rate of 0.125% Bupivacaine
Thoracic or upper abdomen	T6-T8	4–6 mL	4–8 mL/h

Lower abdomen	T10	10 mL	6–10 mL/h
Hip or knee	L2-L3	8 mL	6–10 mL/h
*Divided doses.			

Special considerations

1. Give initial bolus dose before incision. Then, if patient is hemodynamically stable, give 1/2 bolus dose 30 min before end of surgery.
2. In recovery room, ✓ sensory level. If no sensory block, ✓ whether catheter is functioning with 5–10 mL 2% lidocaine bolus. (✓ vital signs.) If thoracic epidural starts with 2–3 mL, redose with 2 mL q5–10min, up to 10 mL.
3. Start infusions: If catheter is functional, as evidenced by loss of sensation, start local anesthetic + opioid infusions (see table, above).
4. Best results: Local anesthetics and opioids are mixed in line using two separate infusion pumps. Thus, if either causes side effects, one can be stopped without affecting the other.

■ PATIENT-CONTROLLED EPIDURAL ANALGESIA (PCEA)

Bupivacaine	0.125–0.25% with 3–5 mL bolus q12–20min
Fentanyl	2–4 mcg/mL

Special considerations

1. For thoracic epidural, decrease all doses by one-third; if high thoracic, decrease by one-half.

TYPICAL ORDERS FOR POSTOP EPIDURAL/SPINAL ANALGESIA

I. EPIDURAL MEDICATIONS

_____ 1. Epidural Infusion—Opiate

- ☐ Hydromorphone (Dilaudid) 0.05 mg/mL concentration @ _____mg/h = _____mL/h
- ☐ Morphine sulfate 0.15 mg/mL concentration @ _____mg/h = _____mL/h
- ☐ Fentanyl (Sublimaze) 10 mcg/mL concentration @ _____mcg/h = _____mL/h
- ☐ Range: May vary infusion between _____/h and _____/h prn pain.

_____ 2. Epidural Infusion—Anesthetic

- ☐ Bupivacaine 0.125% concentration @ _____mL/h
- ☐ Bupivacaine 0.25% concentration @ _____mL/h
- ☐ Turn off bupivacaine on POD # 1 @ 5 a.m.

_____ 3 Epidural Bolus—Opiate

- ☐ Hydromorphone (Dilaudid) _____mg q _____h in 10 mL preservative-free NS (PFNS)
- ☐ Morphine sulfate _____mg q _____h in 10 mL PFNS

_____ 4. Breakthrough Pain

- ☐ Fentanyl PCEA 10–20 mcg q10min or Bupi 0.125% with fentanyl 4 mcg/mL 3 mL bolus q10min
- ☐ Call MD if > 2 doses in [2] h.
- ☐ Ketorolac (Toradol) [15] mg iv q6h × [3] d (for a maximum of 12 doses)

II. NURSING CARE

_____ 1. Vital Signs

- ☐ For opiates: T, P, BP q4h; RR q2h; then q8h, if stable.
Continuous O₂ sats × 24 h; then d/c if O₂ sats >90% on room air.
- ☐ For anesthetics: BP & P q2h; then q4h, if stable.
Orthostatic BP before ambulation × 24 h.

_____ 2. Label tubing, pump, infusion as “epidural.” Place sign over bed; cover “Y” ports on tubing.

_____ 3. Place naloxone (Narcan) and syringe at bedside.

_____ 4. Administration of any additional opiates, sedatives, or antiemetics must be approved by anesthesiologist.

_____ 5. Assess patient’s pain (0–10 scale: 0 = no pain; 10 = worst pain imaginable) and level of sedation (awake, drowsy, asleep, unresponsive), along with VS.

_____ 6. d/c Epidural Protocol when epidural d/c’d.

III. SIDE EFFECTS MANAGEMENT

_____ 1. **Respiratory depression:** For RR < 6 or O₂ sat <86% (on two separate occasions <5 min apart):

Administer naloxone (Narcan) 0.1–0.2 mg iv STAT; may repeat, to total of 0.6 mg.

Administer O₂ 10 L/min via nonrebreathing mask. TURN OFF INFUSION. CALL ON-CALL MD STAT.

_____ 2. **Excessive somnolence/sedation:**

Administer naloxone (Narcan) 0.1–0.2 mg iv STAT; may repeat, to total of 0.6 mg.

Administer O₂ 10 L/min via nonrebreathing mask. TURN OFF INFUSION. CALL ON-CALL MD STAT.

_____ 3. **Nausea/Vomiting:**

- ☐ Metoclopramide (Reglan) [10 mg] iv q4–6h prn
- ☐ Ondansetron [4 mg] q6h prn
- ☐ Nalbuphine (Nubain) [2.5–5.0 mg] iv q4h prn

_____ 4. **Pruritus:**

- ☐ Nalbuphine (Nubain) [2.5–5.0 mg] iv q2–4h prn
- ☐ Diphenhydramine (Benadryl) [10–25 mg] iv q4h prn

Suggested dosage shown in brackets [–]

APPENDIX D: STANDARD PEDIATRIC ANESTHETIC MANAGEMENT

Birgit Maass, MD, and Brenda Golianu, MD

Standard Pediatric Monitors (Noninvasive)	
Blood pressure (BP)	
Capnometry/capnography	Measurement of ET CO_2 display of waveform
Gas analyzer (e.g., Raman, IR, or mass spectroscopy)	Measurement of respired gases and anesthetics
Electrocardiogram (ECG)	5-lead preferred
Esophageal or precordial stethoscope	Breath and heart sounds monitored; dysrhythmias detected
Nerve stimulator	Monitor status of neuromuscular blockade.
Oxygen analyzer	Measurement of FiO_2
Pulse oximetry	Measurement of O_2 saturation; two pulse oximeters in neonates—one preductal; one postductal
Temperature	Nasal, esophageal, rectal, skin, or Foley
Visual observation of patient	Skin color, pupils, temperature, edema, sweating, movement
Ventilator function monitors	PIP, TV, disconnect alarm, etc.

Standard Preop Fasting (NPO) Guidelines
• All solid foods should be withheld after midnight or 6–8 h before procedure. Fatty (fried) solids, 8 h • Nonclear liquids (e.g., milk, infant formula, orange juice) should be withheld 6 h prior to surgery. • Breast milk has a relatively short transit time through the stomach and may be given up to 4 h prior to surgery.

- All clear liquids should be discontinued 2 h prior to scheduled surgery.

***NB: the following sections are guidelines only. Specific drugs and drug dosages should be individualized, based on the physiologic and pharmacologic status of the patient, including factors such as age, weight, medication, and concurrent diseases.**

ASSESSMENT OF FLUID STATUS

Preop fluid management is essential and includes assessment of preexisting dehydration before fasting, minimization of NPO status, and estimation of any ongoing fluid or blood losses before surgery.

In addition to an historical assessment of recent oral intake, enteral nutrition, iv or other source of fluid administration, and recent urine output, a physical examination will include evaluation of vital signs, level of alertness, skin turgor, and the fontanelle status in infants. Laboratory analysis (e.g. electrolytes, Hb) may be useful.

Dehydration can be classified as mild (50 ml/kg, 5% body weight loss), moderate (100 ml/kg, 10% body weight loss) and severe (150 ml/kg, 15% body weight loss). For severe dehydration, it may be difficult to obtain iv access, and intraosseous access may be needed. In emergency surgery, optimizing fluid status will need to occur concurrently with surgery. For neonates and some infants, concurrent administration of dextrose may be needed.

Intraop volume management is guided by several parameters: preop fluid status, maintenance requirements during surgery, “third-space” fluid loss, urine output, and blood loss. Alterations in body temperature, patient positioning, and the use of vasopressors, vasodilators or other agents will affect assessment of volume status.

Positioning of the patient.

Surgical requirements (e.g. volume loading before reperfusion of transplanted organ) must also be considered

Postop volume management requires assessment of any ongoing bleeding, extracellular fluid shifts, and urine output. Volume assessment can be affected by administration of opioids, regional anesthesia, and patient comfort.

PREMEDICATION

- In general, patients <8 mo of age do not need sedative premedication. Infants <1 mo

and premature infants may require atropine 10–20 mcg/kg iv or 20 mcg/kg im or glycopyrrolate 5–10 mcg/kg iv before intubation. Older children (8 mo to 10 yr) can be premedicated successfully by using po midazolam (0.5–0.75 mg/kg with maximum of 20 mg) in syrup, grape Kool-Aid, or cherry-flavored acetaminophen elixir (10–15 mg/kg po) 20–30 min before surgery.

- Another option for premedication is intranasal dexmedetomidine 1–2 mg/kg or intranasal midazolam 0.2–0.4 mg/kg via atomizer.
- For patients >35 kg, po lorazepam (0.03–0.05 mg/kg) or diazepam (0.1–0.15 mg/kg) may be given 60 min before surgery with a sip of water if needed. Distraction with computer games, movies, and interaction with child life personnel has been used successfully.

Parental presence may not decrease the need for premedication but has been shown to increase parental satisfaction.

INDUCTION TECHNIQUES

Preinduction	1. ✓ anesthesia machine, suction, airway equipment, drugs 2. Attach monitors and verify function. 3. Premedication: <6 mo—consider atropine 0.01–0.02 mg/kg iv or glycopyrrolate 5–10 mcg/kg iv before laryngoscopy to prevent vagally mediated bradycardia.
IV induction	Indicated in cases with a difficult airway, in full-stomach considerations, in complex medical conditions, and in older children (>10). If iv induction is planned, topical anesthesia, such as Synera® (lidocaine/tetracaine) patch (30 min), EMLA® (lidocaine/prilocaine, 1 h), and ELA-Max® (liposomal lidocaine, 20 min), may be applied prior to iv placement.
Induction	Routes of administration: 1. IM: ketamine hydrochloride 3–5 mg/kg (with atropine 0.02 mg/kg or glycopyrrolate 0.01/mg/kg and 0.05–0.1 mg/kg midazolam) 2. IV: propofol 2–3 mg/kg 3. Inhalational: sevoflurane (MAC = 3.3% for neonates and younger infants, 2.5% for older infants and children) in N₂O (up to 70%)/O₂. Increase inspired concentration of

	sevoflurane incrementally every three breaths (up to 8%). Monitor BP and HR closely.	
Muscle relaxation	1. Rocuronium: 0.6–1 mg/kg iv; 1 mg/kg recommended for rapid-sequence induction. If airway concerns exist, consider awake intubation and use of succinylcholine. 2. Succinylcholine: 1–2 mg/kg iv. May be useful for rapid-sequence intubations, especially if airway concerns exist. Succinylcholine 2–4 mg/kg im should always be available for urgent administration if needed for treatment of laryngospasm during induction that has not responded to CPAP and deepening of inhalational anesthesia. It may cause bradycardia; consider concomitant atropine (20 mcg/kg). 3. Vecuronium: 0.1 mg/kg iv 4. Deep sevoflurane anesthesia	
Laryngoscope	Blade Miller 0 Miller 1 Wis-Hipple 1.5 Macintosh 2 Macintosh 3 or Miller 2	Age Neonate 6–9 mo 9 mo–3 yr 1–5 yr >5 yr
	*Succinylcholine may trigger MH or hyperkalemia with cardiac arrest in susceptible patients; therefore, many pediatric anesthesiologists avoid the use of succinylcholine especially in male patients under 6 yr unless airway concerns exist. Sugammadex is able to reverse paralysis with rocuronium at any time; rocuronium is becoming more popular as an agent for a rapid-sequence induction in pediatric patients.	

TYPICAL ETT SIZE AT DIFFERENT AGES

Age	Wt	Uncuffed ETT Size	Cuffed ETT Size	LMA Size
Premature Newborn	1–3 kg 3–4 kg	2.5 or 3.0 3.5	n/a 3.0	1 (<5 kg)
≤6–8 mo	6–8 kg	3.5 or 4.0	3.0 or 3.5	1.5 (5–10 kg)
≤8–16 mo	10–12 kg	4.0 or 4.5	3.5 or 4.0	2 (10–20 kg)
2–3 yr	13–15 kg	4.5	4.0 or 4.5	2 (10–20 kg)
6 yr	20	Formula (uncuffed tubes): $4 + (\text{age}/4) = \text{ETT size}$ (to allow for a slight leak when positive pressure is applied). For cuffed tubes, use 0.5 size smaller and inflate as necessary to attain leak of 20–30 cm H ₂ O.		2.5 (20–30 kg)
9 yr	30 kg			3 (30–50 kg)
12 yr	40 kg			3 (30–50 kg)

NB: these ETT sizes are guidance only. ✓ ET placement of tube by auscultation of breath sounds bilaterally. ✓ depth of carina by auscultation over left axilla as ETT is slowly advanced. Withdraw and secure ETT 2 cm from carina. Positive pressure leak between 20 and 30 cm H₂O is desirable to prevent airway tissue necrosis due to inadequate perfusion. Leaks <20 cm may result in volume loss and difficulty in providing appropriate ventilation during critical phases intraop or postop. Conversely, leaks >30 cm H₂O may carry a higher risk of subglottic edema and/or stenosis. Consider length of case, difficulty of placing ETT, and adequacy of ventilation. A throat pack may assist in decreasing the leak if tube exchange is not desired.

In the past few years, mostly cuffed ETT are used in the OR in order to minimize need for tube exchange and optimize ventilation. This has not been associated with an increased risk of stridor; however, care should be exercised if N₂O is utilized in long cases, as it will cause an increase in cuff inflation pressure.

MAINTENANCE TECHNIQUES

Inhalational anesthesia only

30–100% O₂
+ 0–70% N₂O. In preemies and for cases where N₂O is contraindicated, air should be used to lower FiO₂.
+ Sevoflurane or isoflurane, titrated to effect
Consider warming and humidifying all gases for long cases. Warm room to 75–80°F for infants, 70–75°F

	for children.	
Balanced anesthesia	30–100% O₂ + 0–70% N₂O + ~0.5% isoflurane or propofol (50–250 mcg/kg/min) TIVA—propofol 100–250 mcg/kg/min, + remifentanyl 0.05–0.2 mcg/kg/min. Consider including amnestic agents: midazolam and low-dose inhalational agents (e.g., sevoflurane, isoflurane). Dexmedetomidine 0.5–2 mcg/kg/h + remifentanyl 0.05–0.1 mcg/kg/min Ketamine 0.2–1.5 mg/kg/h + Morphine (0.05 mg/kg/h) or fentanyl (1–3 mcg/kg/h) + hydromorphone 2 mcg/kg/h + Consider acetaminophen PR 30–40 mg/kg or iv 10–15 mg/kg for pain control. Temperature monitoring; forced-air warming; fluid warming if volume/blood/resuscitation is expected Fluid requirements LR/NS at maintenance: 0–10 kg = 4 mL/kg/h +11–20 kg = 2 mL/kg/h +>20 kg = 1 mL/kg/h (e.g., 25 kg = 65 mL/h) If continued muscle relaxation is required during the above maintenance techniques, several options are available. Always use a nerve stimulator to assess block before redosing. Neonates may demonstrate delayed return of neuromuscular function.	
Intermediate	Rocuronium (most commonly used)	0.6 mg/kg → 30 min 1 mg/kg for rapid-sequence induction → onset in 1 min, lasts 45–60 min
Vecuronium	0.1 mg/kg → 25–30 min	

EMERGENCE

1. Reverse muscle relaxant	As surgical conditions permit, reverse residual muscle relaxant (when at least one twitch is present in train-of-four) with one of the following: • Neostigmine 0.05–0.07 (maximum dose) mg/kg iv + glycopyrrolate 0.01 mg/kg iv or sugammadex: for rocuronium and vecuronium • A dose of 4 mg/kg is recommended if spontaneous recovery of the twitch response has reached 1–2 posttetanic counts (PTC) and there are no twitch responses to train-of-four (TOF) stimulation. • A dose of 2 mg/kg is recommended if spontaneous recovery has reached the reappearance of the second twitch (T2) in response to TOF stimulation for rocuronium only. • A dose of 16 mg/kg is recommended if there is a clinical need to reverse neuromuscular blockade soon (~3 min) after administration of a single dose of 1.2 mg/kg of rocuronium. • The efficacy of the 16-mg/kg dose following administration of vecuronium has not been studied. • NB: Sugammadex may decrease the effectiveness of hormonal contraceptives for up to 7 d.
2. Nausea prophylaxis	Ondansetron 0.1 mg/kg iv and/or metoclopramide 0.1 mg/kg iv (~1 h before emergence). Dexamethasone 0.1 mg/kg can be considered (avoid in oncology patients due to special protocols). TIVA. A scopolamine patch (e.g., one patch if >15 kg) may be useful in selected patients with h/o severe PONV although no safe/effective dose has been established by the manufacturer.
3. O₂	d/c N₂O/volatile agents and administer 100% O₂.
4. Suction	Suction oropharynx thoroughly.
5. Extubation	Laryngeal spasm is common in children; therefore, it is common practice to extubate them when they are awake, moving all limbs, and breathing adequately. Infants and children with full stomachs or difficult airways must be extubated when they are fully

awake. Deep extubation on 1.5–2 MAC sevoflurane or on propofol infusion is also common, except in patients <1 yr, moderate to severe sleep apnea, full stomach, or difficult airway. The pharynx and stomach should be suctioned thoroughly prior to extubation. If laryngeal spasm occurs, Rx with 100% O₂ and CPAP or PPV. If spasm fails to resolve and hypoxemia occurs, give succinylcholine 0.1–0.5 mg/kg and administer PPV. Consider reintubation if hypoxemia fails to resolve quickly. Consider atropine 0.01–0.02 mg/kg before succinylcholine, to preempt bradycardia.

PEDIATRIC EPIDURAL ANESTHESIA

Epidural anesthesia may be combined with GA for infants and children undergoing surgery involving the lower extremities, abdomen, chest, or spine. Single-dose (“single-shot”) techniques may be used, or epidural catheters may be placed for longer procedures and to facilitate postop epidural analgesia (see below). Ropivacaine 0.1–0.2% ± 1:200K epinephrine or bupivacaine 0.25% ± 1:200K epinephrine is most commonly used intraop. For patients admitted following surgery, opioids (e.g., hydromorphone) are generally added, together with ropivacaine or bupivacaine 0.1% (see [Appendix E](#)). Use saline-filled syringe for loss of resistance to minimize chances of VAE.

TECHNIQUES AND DOSAGES

1. Caudal

- Single shot: 22-ga iv catheter is inserted via sacrococcygeal membrane. Usually only performed for age <6 yr.
- A test dose with 0.1 mL/kg (max 3 mL) lidocaine or bupivacaine + epinephrine 1:100K or 1:200K is given. (If epidural catheter is used, add 0.8 mL to test dose volume due to dead space volume of catheter.)
- Initial dose: 0.5 mL/kg for lower extremity/perineal/genital procedures, 1.0 mL/kg for abdominal procedures (max 15 mL).
- Catheter technique: 18-ga iv catheter is used; in patients <10 kg, first dilate the epidural space with 5–10 mL NS; then, a 20-ga epidural catheter may be inserted through 18-ga iv catheter and advanced so that the tip is located near the level of incision (e.g., ~17 cm from the skin to T4 in infant). If resistance is met, it may be necessary to pull the catheter back slightly, together with the needle (to avoid

shearing catheter), or repeat procedure. The tip of the catheter should be visualized with ultrasound or via x-ray with 0.5–1 mL of contrast. (If epidural catheter is used, add 0.8 mL to test dose volume due to dead space of catheter.)

2. LUMBAR

- 18-ga epidural needle is inserted via L3-L4 or L4-L5 interspace for single-shot injection or placement of 20-ga epidural catheter.
- Use loss-of-resistance technique with fluid-filled syringe (air may cause VAE).
- Good estimate of depth to epidural space is 1 mm/kg, up to ~30 kg.
- Catheter should be threaded 3–4 cm beyond the tip of needle.
- A test dose with 0.1 mL/kg lidocaine or bupivacaine + epinephrine 1:200K is given, max 3 mL. (If epidural catheter is used, add 0.8 mL to test dose volume due to dead space of catheter.)
- Initial dose: 0.5 mL/kg for lower abdominal procedures, 1.0 mL/kg for upper abdominal and thoracic procedures

3. THORACIC

- The technique described for lumbar epidural catheter placement may be used between T6 and T12 in children.
- A test dose with 0.1 mL/kg lidocaine or bupivacaine + epinephrine 1:100K or 1:200K is given, max 3 mL. (If epidural catheter is used, add 0.8 mL to test dose volume due to dead space of catheter.)
- Initial dose: 0.5 mL/kg
- Local anesthetic–dosing guidelines same as above (note that volume will be ~1/3 less than with caudal approach).
- Erector spinae plane block:

Most recently, erector spinae catheters have emerged as a safe alternative to central neuraxial blockade for major pediatric surgeries. Though still described primarily via case studies, these catheters hold promise as a simpler and safer alternative to thoracic epidural or paravertebral blocks, as the site can be easily visualized with sonography and is distant from the neuraxis, pleura, and major vascular structures. Catheter placement may be bilateral if warranted by the surgical incision. Dosing is accomplished by alternating hourly boluses via R and L catheters.

For indwelling catheter techniques, hourly maintenance doses of half the initial dose may be given. For continuous infusion, maximum rate for ropivacaine and bupivacaine is 0.5 mg/kg/h (0.3 mg/kg/h for neonates).

PEDIATRIC SPINAL ANESTHESIA AND ANALGESIA

Spinal anesthesia is used primarily for procedures such as inguinal herniorrhaphy in former preterm infants at risk for postop apnea following GA. By avoiding GA, the incidence of postop apnea is reduced, but not eliminated.

After standard monitors are applied, the infant is placed in a supine or lateral decubitus position. Care is taken to avoid neck flexion, which may cause airway obstruction. The skin is infiltrated with 1% lidocaine using a 27- or 30-ga needle. Lumbar puncture is performed with a 22-ga 1.5-inch spinal needle to an average depth of 1.5 cm from the skin. The most commonly used local anesthetic for spinal anesthesia in infants is tetracaine 1.0%, mixed with an equal volume of 10% dextrose in a dose of 0.8–1.0 mg/kg. Epinephrine 1:1000 0.01 mL/kg is added. This dose usually provides adequate anesthesia for 90–120 min for inguinal herniorrhaphy. Little change in BP or HR occurs in infants <6 mo of age.

Complications include high spinal anesthesia requiring tracheal intubation. PDPH is very uncommon in children <12 yr of age and probably rare in infants.

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APPENDIX E: MULTIMODAL PEDIATRIC POSTOPERATIVE PAIN MANAGEMENT

Genevieve D'Souza, MD, Julie Good, MD, and Brenda Golianu, MD

INTRODUCTION

Traditionally, children have been undermedicated for their pain because of difficulty with assessment, concerns for safety, and lack of understanding of the physiologic consequences of untreated pain. There is now strong evidence that it is not only safe but also beneficial to treat children's procedural pain. For example, infants have adverse behavioral cardiorespiratory and neuroendocrine responses to pain, which improve when appropriate analgesia is administered.

Pain assessment	Unlike most adults, infants and children <7 yr have difficulty understanding and using a Visual Analog Scale (VAS). A frequently used tool for assessing pain in children 3–7 yr old is the Wong-Baker Faces Scale (Fig. E.1). For infants and nonverbal children, observational scales that rely on behavioral and/or physiologic parameters are often used (e.g., the FLACC Pain Scale, Table E.1, PIPP, NIPS, CHEOPS, CRIES, as well as appropriate translations). The “gold standard” of pain measurement is to solicit a direct subjective report from the child whenever possible.
Oral medications	For simple outpatient procedures, such as hernia repair, circumcision, or closed reduction of a fracture, a weak oral opioid, in combination with acetaminophen, is appropriate. Care must be exercised in the use of opioids for management of pain in outpatients following tonsillectomy due to risk of hypoventilation and airway obstruction, especially in patients with severe sleep apnea.
Intravenous medications	Patient-controlled analgesia (PCA) can be considered in patients >5 yr who are expected to

remain hospitalized overnight, especially in those who are unlikely to tolerate oral intake during the initial hours after surgery. For younger children, a **nurse-controlled analgesia (NCA)** can be employed. When the patient begins oral medications, d/c the basal rate but continue to provide the lockout dose for several more hours to be sure that the child is tolerating the oral medication. For opioid-related side effects such as nausea and vomiting, ondansetron can be added (0.1 mg/kg q6h to maximum of 4 mg q6h), and for pruritus, diphenhydramine (0.5 mg/kg q6h max 50 mg), or nalbuphine (0.05 mg/kg q6h max 20 mg), especially effective for epidural opioid-related pruritus. For respiratory depression, naloxone can be carefully titrated beginning at 0.001 mg/kg q1–2min as needed to restore adequate respiratory effort and wakefulness. Care must be taken, as naloxone will have a short half-life (10–15 min), and respiratory depression may recur. Reversal of analgesia may also occur if naloxone is administered for treatment of pruritus or respiratory depression.

Adjuvant medications

In addition to opioids, several adjuvant medications are useful in the periop period. **Lorazepam** (Ativan) 0.001 mg/kg iv q6h is useful in preventing spasms and lessening fear in an unfamiliar environment. It is important to remember that lorazepam has a half-life of up to 16 h, and doses can be additive.

Another useful adjuvant is **ketorolac** 0.5 mg/kg, up to a maximum of 30 mg loading dose, followed by 0.5 mg/kg (up to 15 mg) q6h, up to 72 h. Ketorolac is not recommended in patients with poor renal function or following surgeries where there is a large bleeding surface or complex bone repair. Ketorolac has been shown to cause delay in bony fusion in animal models, but conflicting information has been presented in humans. It also has been reported to cause acute renal failure with prolonged use and is not recommended for use in bariatric patients due to

its potential adverse impact on renal function. Given these concerns, it is helpful to observe hydration status, plt count, and Cr, and discuss the use of this medication with the surgical team.

Other adjuvants to consider include

acetaminophen iv or po (10–15 mg/kg/dose q6h, maximum 650 mg). Medications to manage side effects, such as pruritus, N/V, constipation, and respiratory depression, are presented in detail on the PCA order form. **Gabapentin** (15 mg/kg po, maximum 300 mg po tid) or pregabalin (2 mg/kg po) preop, followed by gabapentin 5 mg/kg po postop 3×/day for 5 days or **pregabalin** 1 mg/kg po, maximum 150 mg po bid postop for 5 days has been shown to decrease opioid requirement postop and should be considered for major surgeries. **Celecoxib** (2 mg/kg/dose, maximum 100 mg po bid twice a day for 5 days) can also be considered instead of ketorolac if the surgeon allows. Ketamine (0.1–0.3 mg/kg/h iv) may also be used as an adjuvant, though studies have not found a postop opioid-sparing effect.



Figure E.1 Wong-Baker Faces Scale used in pain assessment in children (and other patients developmentally from 3 to 7 years old). The higher the score, the greater the child's pain. (Courtesy of the Wong-Baker FACES Foundation (2019). Wong-Baker FACES® Pain Rating Scale. Retrieved Aug 7, 2019 with permission from <http://www.WongBakerFACES.org>. Originally published in Whaley & Wong's Nursing Care of Infants and Children. © Elsevier Inc.)

Epidural pain management

In children, it is generally considered safer to place epidurals after induction of GA to avoid movement

during placement. For children 0–12 mo, a **caudal technique** can be used. The patient is placed in the lateral decubitus position, the caudal anatomy is identified, and an 18-ga iv catheter is used to enter the caudal space. The catheter is advanced and aspirated. The space is then dilated with 5–8 mL of preservative-free NS. A 20-ga epidural catheter is then advanced to the desired location or until obstruction is felt (~10–15 cm). The catheter tip may be visualized using ultrasound or nerve stimulation, if available. The catheter is taped securely with a moisture-resistant dressing, as it can easily become dislodged with regular activity. A test dose is given (lidocaine 1.5% with 1:200,000 epinephrine 0.1 mL/kg + 0.8 mL for the dead space of the catheter).

Lumbar and thoracic epidurals can be placed in children >10–12 mo or earlier in experienced hands.

Contraindications for epidural catheters include infection at the local site, coagulopathy, low plt, sepsis, progressive neurologic deficit, and refusal of patient or parent.

Epidural placement should be checked by syringe aspiration before starting the infusion. Return of any bloody fluid or >0.5 mL clear aspirant necessitates further evaluation before use of the catheter for pain control. Patients with indwelling epidural catheters may receive postop analgesia with either continuous infusion or automatic programmed intermittent epidural bolus (PIEB), which delivers the full hourly rate at once each hour on schedule.

Continuous infusion can be used alone or with intermittent bolus dosing (**patient-controlled epidural analgesia** [PCEA]). Similar to PCA, PCEA can be used in ages = 5–7 yr. Typical starting infusions are described in the table below.

Medication dosage will depend on patient's age, location of surgical pain, catheter insertion level, and catheter tip location. If the patient has an extra-wide

incision or other areas of discomfort, or if tolerant to opioids, the anesthesiologist may choose to run the epidural infusion at a fixed rate with local anesthetic alone (ropivacaine 0.1% or bupivacaine 0.1%) and provide the patient with an IV PCA with bolus (lockout) dose only (i.e., no continuous or basal rate).

A bolus dose equivalent to the hourly volume of infusion may be given when a patient is uncomfortable. Increase the infusion by 10% to maintain the new level of analgesia. Infusion should not exceed 0.3 mg/kg/h in infants under 6 mo or 0.5 mg/kg/h in older patients. If inadequate analgesia persists after two dosing increases, it may be appropriate to abandon this form of treatment in favor of systemic analgesia. Do not leave a patient in pain for an extended period of time trying to “fix” an epidural catheter. If in doubt, check catheter placement, then bolus catheter with 2% lidocaine w/ 1:200,000 epi 0.1 mL/kg to determine if a level is able to be obtained.

Erector spinae catheters

When epidural infusions are not desired, or if there are concerns related to increased risk of bleeding, such as thrombocytopenia, coagulopathy, or need for intraop anticoagulation, erector spinae catheters may be an effective alternative to epidural catheter insertion. Catheters may be placed unilaterally or bilaterally, depending on surgical intervention. If unilateral, an hourly bolus dose can be administered. If bilateral, a bolus dose can be administered alternately to each side every 2 h, taking care not to exceed the maximum hourly dose of local anesthetic noted above.

Peripheral nerve catheters

Increasingly, peripheral nerve catheters are being employed for periop pain control. Local anesthetic infusions (ropivacaine 0.1% or bupivacaine 0.1%) are continued for 3–5 d postop.

Table E-1. FLACC Pain Scale for Pain Assessment in Nonverbal Patients

	FLACC Pain Scale (For Nonverbal Patients)		
	Scoring		
Categories	0	1	2
Face	No particular expression or smile	Occasional grimace or frown, withdrawn, disinterested	Frequent-to-constant quivering chin, clenched jaw
Legs	Normal position or relaxed	Uneasy, restless, tense	Kicking, or legs drawn up
Activity	Lying quietly, normal position, moves easily	Squirming, shifting back and forth, tense	Arched, rigid, or jerking
Cry	No cry (awake or asleep)	Moans and whimpers; occasional complaint	Crying steadily, screams or sobs, frequent complaints
Consolability	Content, relaxed	Reassured by occasional touching, hugging, or being talked to; distractible	Difficult to console or comfort

Each of the five categories (F) Face, (L) Legs, (A) Activity, (C) Cry, and Consolability is scored from 0 to 2, which results in a total score between 0 and 10.

From Merkel SI, Voepel-Lewis T, Shayevitz JR, Malviya S. The FLACC: a behavioral scale for scoring postoperative pain in young children. *Pediatr Nurs*. 1997 May-Jun;23(3):293–7. The FLACC scale was developed by Sandra Merkel, MS, RN; Terri Voepel-Lewis, MS, RN; and Shobha Malviya, MD at C.S. Mott Children's Hospital, University of Michigan Health System, Ann Arbor, MI.

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STANDARD PEDIATRIC INTRAOP EPIDURAL/CAUDAL DOSING

	Ropivacaine 0.2% ± epi	Bupivacaine 0.25% ± epi	Lidocaine
Caudal	0.25 mL/kg	0.25 mL/kg	0.25 mL/kg
Lumbar	0.5 mL/kg	0.5 mL/kg	0.5 mL/kg
Thoracic	0.3 mL/kg	0.3 mL/kg	0.3 mL/kg

Maximum LA dosage: bupivacaine/ropivacaine/lidocaine 3 mg/kg.

Common additives for single dose administration

1. Preservative-free morphine: 30–50 mcg/kg
2. Hydromorphone 3–5 mcg/kg (commonly 3 mcg/kg in younger children, 5 mcg/kg in older children and teenagers)
3. Fentanyl 0.25–0.5 mcg/kg

1. Clonidine 0.5–2 mcg/kg (commonly 1 mcg/kg)

Lumbar epidural space depth (mm): $\approx 18 + 1.5 \times \text{age (yr)}$

STANDARD PEDIATRIC POSTOP ANALGESICS AND ANTIEMETICS

Analgesics	Acetaminophen	10 mg/kg po iv q6h 20 mg/kg po pr q6h	Max: 30–45 mg/kg/24 h (neo); 60 mg/kg/24 h (infants); max 80 mg/kg/24h pr (children>1)
	Hydrocodone/Acetaminophen Liquid or Tablet	0.15 mg/kg po iv q6h	Dose based on maximum acetaminophen allowed for weight (up to 10 mg/kg/dose) q4–6h, but not to exceed 60 mg/kg/d
	Oxycodone/Acetaminophen	0.1 mg/kg po q6h Oxycodone may also be prescribed without acetaminophen, and doses may be increased as needed, commonly 0.15 mg/kg po q6h	Dose should also be based on maximum acetaminophen allowed for weight (up to 10 mg/kg/dose) q4–6h, but not to exceed 60 mg/kg/d)
	* NB: Codeine is not recommended in pediatrics. Patients with CYP2D6 polymorphism may be ultrarapid metabolizers of		

	codeine, leading to oversedation and in some cases to fatalities in patients with OSA following tonsillectomy. In addition, 16% of people cannot metabolize codeine and thus may have suboptimal analgesia.		
Antiemetics	Metoclopramide* Ondansetron	0.1 mg/kg, may repeat × 1 iv 0.1 mg/kg iv	May increase risk of extrapyramidal reactions

Notes: To prevent irreversible liver damage, it is important to d/c or supervise all other use of acetaminophen when prescribing the above medications, so as not to exceed the maximum dose of 60 mg/kg/d (40 mg/kg/d in neonates) of acetaminophen.

Hydro, hydrocodone; APAP, acetaminophen; Cod, codeine.

PATIENT-CONTROLLED IV ANALGESIA (PCA)

In a setting with trained nursing supervision, PCA can be used safely by children <5 yr old (about the age they are able to play video games). The lockout time usually is set at 10 min but can be as short as 5 min for fentanyl. Common PCA medications and recommended starting doses in opiate-naïve patients are listed below, followed by typical order for iv PCA.

Common PCA Medications			
Medication	Loading Dose	Basal Rate	Patient-Controlled Bolus
Morphine (1 or 5 mg/mL)	0.03 mg/kg	0.01 mg/kg/h	0.02–0.03 mg/kg
Hydromorphone (100 mcg/mL)	3–5 mcg/kg	1 mcg/kg/h	2 mcg/kg
Fentanyl (50 mcg/mL)	0.3 mcg/kg	0.1 mcg/kg/h	0.2–0.3 mcg/kg

Continuous iv infusion

When PCA is not practical (e.g., in children unable to understand PCA), continuous iv infusion of opiates may be used. Morphine infusion of 10–30 mcg/kg/h results in serum concentrations of 10–22 ng/mL and provides adequate analgesia. A common technique is to initiate iv morphine infusion with 1 mg/kg of morphine in 100 mL of D5W at 1 mL/h (the effective infusion rate is 10 mcg/kg/h). The infusion rate is slowly increased

to provide adequate pain relief.

TYPICAL ORDERS FOR PATIENT-CONTROLLED IV ANALGESIA (PCA)

Patient's Name: _____ Medical Record No.: _____

Weight (kg): _____ Allergies: _____

Drug	Recommended Concentration	Loading Dose	Mode (select one)			PCA Dose	Lockout Interval	Basal Rate
			PCA Only	Continuous	PCA + Continuous			
Morphine	1 mg/mL	30 mcg/kg*				20–30 mcg/kg*	8–10 min	10 mg/kg/h*
Hydromorphone	100 mcg/mL	5 mcg/kg*				2 mcg/kg*	8–10 min	1 mcg/kg/h*
Fentanyl	50 mcg/mL	0.3 mcg/kg*				0.2–0.3 mcg/kg*	8–10 min	0.1 mcg/kg/h*

Physician: Check all orders that apply:

**Recommended starting dose for opiate-naïve patients.*

Nursing Care:

- ☐ 1. For renewed or changed PCA orders only, continue PCA nursing as previously ordered.
- ☐ 2. Do not administer any other opioids, benzodiazepines, sedatives, or antiemetics unless approved by:
Service Pager No.:
- ☐ 3. Continuous O₂ sat monitor. For SaO₂ < 94%, place order for and administer O₂ per nasal cannula to maintain SaO₂ ≥ 94% and re-evaluate PCA doses.
- ☐ 4. Assess and record at least q2h: RR, O₂ sat, and level of pain, using appropriate tool.
(0–10 Scale, Wong-Baker Faces Scale, FLACC Scale, or PIPP Scale)
- ☐ 5. Assess and record at least q4h: HR, BP, and T. Spot-check SaO₂ with vitals.
- ☐ 6. Bag and mask at bedside.
- ☐ 7. Assess for bladder retention. If no void for _____ h (since surgery or last void), may straight cath prn × 2.

Medications:

- ☐ 1. For renewed or changed PCA orders only, continue prn medications for PCA as previously ordered.
- ☐ 2. Metoclopramide (Reglan) _____ mg iv q6h prn N/V (0.1 mg/kg/dose; max 15 mg).
- ☐ 3. Ondansetron (Zofran) _____ mg iv q8h prn N/V (0.1 mg/kg/dose; max 4 mg).
- ☐ 4. Diphenhydramine (Benadryl) _____ mg iv q6h prn pruritus (0.5 mg/kg/dose; max 50 mg).
- ☐ 5. Nalbuphine (Nubain) _____ mg iv q4 prn pruritus (0.05 mg/kg/dose; max 20 mg).
- ☐ 6. Polyethylene glycol (Miralax) T packet po bid.
- ☐ 7. If no stool by _____ (POD 3), administer:
- ☐ 1 pediatric glycerin suppository pr 1 or 2 × daily prn (ages 2–6 yr).
 - ☐ Bisacodyl 5 mg (½ suppository) pr daily prn (ages 6–11 yr).
 - ☐ Bisacodyl 10 mg (1 suppository) pr daily prn (ages >11 yr).
 - ☐ Senna (Senokot) 8.6 mg tablet po bid prn.
 - ☐ If no stool within 24 h of beginning treatment, notify physician listed above.
- ☐ 8. If unable to arouse or SaO₂ < 90%, turn off PCA pump, administer O₂ and/or ambu-bag, stimulate patient, administer naloxone _____ mg (0.001 mg/kg) q1–2min (obtain naloxone from floor stock) as needed to restore consciousness, and STAT page physician.
- ☐ 9. Other medications: _____

Date:	Time:	Physician's Signature:	Pager:	Noted by:	Date/Time:
Orders signed				RN:	Date/Time:

POSTOPERATIVE EPIDURAL ANALGESIA

Patients with indwelling epidural catheters may receive postop analgesia with either continuous infusion alone or continuous infusion with intermittent bolus dosing (PCEA or PIEB). The infusate may be either local anesthetic with an opioid, local anesthetic alone, or opioid alone. At Stanford, the most commonly used epidural infusate for continuous infusion or PIEB is ropivacaine 0.1% with hydromorphone 3 mcg/mL. In patients receiving PCEA, ropivacaine 0.1% with hydromorphone 25 mcg/mL is often used.

At Stanford, patients receiving epidural analgesia are managed by the Pediatric Pain Service.

Hydromorphone (Dilaudid): May bolus dose 5–10 mcg/kg or just add to continuous infusion. Reduced dose for thoracic catheters.

When opioid epidural analgesia is indicated, hydromorphone is most commonly used because

1. It causes less itching and nausea, compared with morphine.
2. It is more water soluble than fentanyl (decreased systemic absorption, spread over

more dermatomes).

- It is less water soluble than morphine (thereby minimizing late respiratory depression).

STARTING DOSES FOR EPIDURAL INFUSION

Age	Starting Dose
0–6 mo	Hydromorphone 3 mcg/mL + 0.1% ropivacaine @ 0.1–0.15 mL/kg/h (max ropi 0.3 mg/kg/h); for neonates, consider 0.1% ropivacaine infusion only
6 mo to 3 yr	Hydromorphone 3 mcg/mL + 0.1% ropivacaine @ 0.1–0.15 mL/kg/h (max ropi 0.5 mg/kg/h)
3–7 yr	Hydromorphone 3–5 mcg/mL + 0.1% ropivacaine @ 0.1–0.15 mL/kg/h (max ropi 0.5 mg/kg/h)
>7 yr	Hydromorphone 5–10 mcg/mL + 0.1% ropivacaine @ 0.1–0.15 mL/kg/h + 0.05 mL/kg/h PCEA dose q30min lockout (max ropi 0.5 mg/kg/h)

Notes: In general, use a lower concentration of opiate infusion for neonates. When spread is needed (for surgeries with incisions that cross multiple dermatomes or when catheter placement is below the level of anticipated pain), a lower concentration and higher volume infusion may be desired.

Suggested Reading

1. Ivani G, Mosetti V: Regional anesthesia for post-operative pain control in children: focus on continuous central and perineural infusions. *Paediatr Drugs* 2008; 10(2):107–14.

TYPICAL ORDERS FOR CONTINUOUS EPIDURAL ANALGESIA

(may also be utilized for Peripheral Nerve Blockade)

Patient's Name: _____ Medical Record No.: _____

Weight(kg): _____ Allergies: _____

Drug and Concentration (Check one)	Basal Rate
	(0.15 mL/kg/h) (maximum 0.3 mL/kg/h for 0–6 mo)
☐ Hydromorphone 3 mcg/mL with bupivacaine 0.1% →	Epidural infusion rate _____ mL/h
☐ Hydromorphone 3 mcg/mL with bupivacaine 0.1% → and Clonidine 0.2 mcg/mL	Epidural infusion rate _____ mL/h
☐ Hydromorphone 3 mcg/mL with levobupivacaine 0.1% →	Epidural infusion rate _____ mL/h

Physician: Check all orders that apply:

Nursing Care:

- ☐ 1. Do not administer any other opioids, benzodiazepines, sedatives, or antiemetics unless first approved by Pain Management Service.
- ☐ 2. Maintain iv access until epidural catheter d/c'd.
- ☐ 3. Check epidural site qd and notify Pain Management Service if site is soiled, red, or tender.
- ☐ 4. Check lower extremities q8h while awake, and notify Pain Management if numbness or weakness exist.
- ☐ 5. Continuous O₂ sat monitor. For SaO₂ < 94%, administer O₂ per nasal cannula to maintain SaO₂ ≥ 94%.
- ☐ 6. Assess and record at least q4h: RR, O₂ sat, and level of pain, using appropriate tool.
(0–10/Numeric Scale, Wong-Baker Faces Scale, FLACC Scale, or PIPP Scale)
- ☐ 7. Assess and record at least q4h: HR, BP. Confirm absence of postural hypotension before ambulating for the first time.
- ☐ 8. Assess for bladder retention. If no void for 8 h (since surgery or last void), may straight cath prn × 2.
- ☐ 9. Bag and mask at bedside.
- ☐ 10. Check catheter entry site q8h and notify Pain Management Service if soiled, red, or tender

Medications:

- ☐ 1. Metoclopramide (Reglan)_____mg iv q6h prn N/V (0.1 mg/kg/dose; max 15 mg).
- ☐ 2. Ondansetron (Zofran)_____mg iv q6h prn N/V (0.1 mg/kg/dose; max 4 mg).
- ☐ 3. Diphenhydramine (Benadryl)_____mg iv q6h prn pruritus (0.5 mg/kg/dose; max 50 mg).
- ☐ 4. Nalbuphine (Nubain)_____mg iv q4 prn pruritus (0.05 mg/kg/dose; max 20 mg).
- ☐ 5. Polyethylene glycol (Miralax) 1 packet po bid
- ☐ 6. If no stool by _____(POD 3), administer:
- ☐ 1 pediatric glycerin suppository pr 1–2 × daily prn (2–6 yr).
 - ☐ Bisacodyl 5 mg (½ suppository) pr daily prn (6–11 yr).
 - ☐ Bisacodyl 10 mg (1 suppository) pr daily prn (> 11 yr).
 - ☐ Senna (Senokot) 8.6 mg tablet T po bid prn
 - ☐ If no stool within 24 h of beginning treatment, notify physician listed above.
- ☐ 6. If unable to arouse or SaO₂ is <85%, turn off PCA pump, administer O₂ and/or ambu-bag, stimulate patient, administer naloxone (Narcan)_____mg (0.001 mg/kg) q1–2min (obtain naloxone from floor stock) as needed to restore LOC, and STAT page pain Management Service.
- ☐ 7. Other medications:_____

Date:	Time:	Physician's Signature:	Pager:	RN Signature:	Date/Time:
Orders signed					

Electronic Order Sets for Epidural and Regional Block Infusions in Pediatrics

Pump type : _____ CADD—On-Q—Nimbus—Other

Local anesthetic Infusion:

Bupivacaine _____% 0.05%; 0.0625%; 0.08%; 0.1%; 0.125% 0.2% 0.25% **(Usually 0.1%)**

Lidocaine _____% 0.25%; 0.5%; 1%

Ropivacaine _____% 0.05%; 0.08%; 0.1%; 0.2%; 0.25%(Usually 0.1% or 0.2%)

Adjuvants

Clonidine _____ mcg/mL 0.1 mcg/mL; 0.2 mcg/mL; 0.5 mcg/mL; 1 mcg/mL **(Usually 0.2 mcg/mL)—
decrease for sedation—higher doses may be used if not using opioid)**

Fentanyl _____ mcg/mL 0.5 mcg/mL; 1 mcg/mL; 2 mcg/mL; 5 mcg/mL

Hydromorphone _____ mcg/mL 3 mcg/mL; 5 mcg/mL; 10 mcg/mL; 20 mcg/mL **(Usually 3 mcg/mL)**

Morphine (Preservative free) _____ mcg/mL

Example of frequently utilized infusions:

Immediately postoperatively: Ropivacaine 0.2% (or bupivacaine 0.1%) with clonidine 0.1 or 0.2 mcg/mL (depending on possible concerns of postoperative hypotension), hydromorphone 3 mcg/mL

Run infusion at 0.2–0.3mL/kg; check anesthetic level.

For thoracic infusions, reduce dose to 0.15–0.2 mL/kg. Check anesthetic level and assess for hypotension.

It can be particularly effective to administer the hourly dose of anesthetic as an hourly bolus—e.g., 8 cc delivered in one bolus, and the subsequent bolus is delivered in 1 h. This can be accomplished with the CADD and Nimbus devices but not with On-Q.

Maximum local anesthetic dose infusion: neonates–9 months 0.3 mg/kg/h; > 9 mo 0.5 mg/kg/h

When ambulation is being considered, reduce ropivacaine concentration to 0.1%. Verify orthostatic vital signs, lower extremity strength. Patient should always ambulate with assistance.

PCEA (patients >12 yr old): Local anesthetic and adjuvant mixture as above.

Begin Basal infusions at 0.1 – 0.15 mL/kg/h, and epidural bolus at 0.05 mL/kg/h q30min. Thus to total volume in the epidural will not exceed 0.25 mL/kg/h. This can be adjusted as needed for patient comfort to not exceed the above noted 0.3 mL/kg in epidural volume, or 0.5 mg/kg/h of local anesthetic.

Vital sign parameters for regional block infusions in pediatrics:

Age Group	1–9 mo	9 mo–3 yr	3–6 yr	6–15 yr	>15 yr
Heart Rate Minimum (pulse/min)	100	85	75	65	60
Heart Rate Maximum (pulse/min)	170	155	150	130	115
Resp Rate Minimum (breaths/min)	10	10	10	10	10
Resp Rate Maximum (breaths/min)	55	50	50	50	50
Systolic BP minimum (mm Hg)	60	75	75	90	90
Systolic BP maximum (mm Hg)	120	130	130	135	140
Diastolic BP minimum (mm Hg)	40	40	45	50	50
Diastolic BP maximum (mm Hg)	70	80	80	80	90
Temp Minimum (°C)	36.5	36.5	36.5	36.5	36.5
Temp Maximum (°C)	38	38	38	38	38

APPENDIX F: TABLES OF DRUG INTERACTIONS AND RECOMMENDATIONS FOR PREOP MANAGEMENT

Carolyn Nguyen, PharmD

These tables are intended only as an advisory overview of potential interactions between various drug classes or herbals that patients may be taking preop and the drugs used in anesthetic practice. They are not intended to be comprehensive lists. Drug interactions are dynamic and depend on a variety of circumstances including dosage, duration of administration, as well as the patient’s genetics and current physiologic state. An excellent source for the predictive values of drug interactions due to altered metabolism can also be found in references for inhibitors, inducers, and substrates of the cytochrome P450 enzymes. In view of the constant flow of new drug information, the reader is strongly encouraged to check the primary literature of each drug or online resources, such as DRUG-REAX, Lexi-Interact, and Facts and Comparisons for drug interactions, and then tailor drug usage to the patient-specific clinical situation.

TABLE F-2 HERBAL AGENTS

It may be difficult to accurately predict the potential for drug interactions, because the majority of people neglect to inform health care providers of their consumption of herbal agents, natural remedies, alternative or complementary medicines, nutritional supplements, and illicit substances. The significance of the potential interaction is also difficult to assess due to variations of botanical species, the different parts of plants that are used, assay of active ingredient(s), and product formulation. Herbs that alter hemostasis should be discontinued 14 d before surgical, dental, or invasive procedures.

■ TABLE F-1 DRUG-DRUG INTERACTIONS			
Preop Drug or Drug Class	Anesthetic Drug or Drug Class	Interaction	Clinical Management
Acetylcholinesterase inhibitors (neostigmine, pyridostigmine, donepezil, rivastigmine)	Nondepolarizing neuromuscular blockers (NMBs)	↓ blockade	Titrate

	(atracurium, rocuronium, vecuronium)		
Acetylcholinesterase inhibitors	Succinylcholine	↑ blockade	Mon supp funct
Adrenergics/sympathomimetics (ephedrine, epinephrine, norepinephrine, isoproterenol)	Anesthetic gases (desflurane, enflurane, isoflurane, sevoflurane)	↑ risk of dysrhythmias	Limit epine local
Alfentanil	Propofol, fospropofol	Opisthotonos, seizure	Mon seda hypoc
Aliskiren	Any cardiovascular depressant or vasodilator	↓↓ BP	d/c 5 surg refra with
Aminoglycosides (gentamicin, tobramycin, amikacin)	Anesthetic gases	↑ potential for nephrotoxicity secondary to fluoride	Mon funct
Aminoglycosides	Botulinum toxins	↑ neuromuscular blocking	Mon block
Aminoglycosides	NMB	↑ respiratory depression	Mon supp funct
Amiodarone	Anesthetic gases	↑ myocardial depression and conduction defects	Mon rhyth
Amiodarone	Phenylpiperidone derivative opiate agonists (alfentanil, fentanyl,	↓ HR and ↓ BP, sinus arrest	Mon hemo funct

	sufentanil)		
Amphetamines	Cocaine (topical)	↑↑ BP, ↑ HR	Mon BP.
Amphetamines	Opioids	↑ analgesic effects	Mon supp resp
Anthracyclines (daunorubicin, doxorubicin, idarubicin)	Isoflurane	Prolonged QT interval	Mon rhyth
Aprepitant, fosaprepitant	Opiate agonists (alfentanil, fentanyl, sufentanil) Meperidine	↑ respiratory or CNS depression ↑ opioid effects	Mon supp resp
Aprepitant, fosaprepitant	Midazolam	↑ midazolam levels	Mon supp resp
Aripiprazole	Midazolam	Genotype-dependent ↑ aripiprazole concentrations	Mon Cons dose aripip
Armodafinil	Midazolam	↓ midazolam effect	Titra effec
Arsenic trioxide	Halothane	Prolonged QT interval	Mon rhyth
Atorvastatin	Midazolam	↑ midazolam effect	Titra effec
Atracurium	Ketamine	↑ neuromuscular blocking	Adju moni prolc resp depr
Atracurium	Propofol, fospropofol	Bronchospasm	Anap reac

Azole antifungals (fluconazole, isavuconazonium, itraconazole, ketoconazole, posaconazole, voriconazole)	Alfentanil	↑ opioid effect	Consider alternative Monitor and support respiration
Azole antifungals	Midazolam	↑ midazolam effect	Dose reduction needed midazolam Monitor and support respiration
Barbiturates	Midazolam	Potential of CNS depression	Monitor and support respiration
Benzodiazepines	Nonpolarizing NMBs	May prolong or antagonize blockade	Monitor and support respiration
Beta-blockers	Alfentanil, fentanyl, remifentanyl, sufentanil	↑ risk of hypotension and bradycardia	Monitor BP.
Beta-blockers	Alpha/beta-agonists (ephedrine, epinephrine, norepinephrine, phenylephrine)	Enhanced vasopressor effect	Use cautiously Reactive decompensation agent
Boceprevir	Midazolam	↑ midazolam effect	Consider alternative
Botulinum toxins	NMBs	↑ blockade	Toxicity from causative weakness
Calcium channel blockers— nondihydropyridines (diltiazem,	Alfentanil, fentanyl,	↑ risk of hypotension and bradycardia	Monitor BP.

verapamil)	remifentanyl, sufentanyl		
Calcium channel blockers— nondihydropyridines	Midazolam	Prolonged sedation	Avoid midazolam dose
Calcium channel blockers— nondihydropyridines	Nondepolarizing NMBs	↑ prolonged blockade	Monitor support resp
Carbamazepine	Nondepolarizing NMBs	↓ blockade	Titrate therapeutic
Carbamazepine	Midazolam	Possible ↓ midazolam effect	Titrate effect
Catecholamines, sympathomimetics	Halothane	↑ risk of catecholamine-induced dysrhythmias	Use inhalation
Clindamycin	Nondepolarizing NMBs	↑ intensity or prolongation of blockade	Monitor support resp Anticholinergic or cardiac benefit
Clindamycin	Succinylcholine	Prolonged blockade	Monitor support resp
Clonidine	Naloxone	Possible hypertension	Monitor
Conivaptan	Alfentanil, fentanyl, sufentanyl	↑ analgesic effect	Avoid
Conivaptan	Midazolam	↑ midazolam effect	Avoid
Corticosteroids	Nondepolarizing NMBs	Muscle weakness possibly progressing to polyneuropathies and myopathies	Titrate therapeutic
Crizotinib	Alfentanil	↑ analgesic effect	Avoid

Cyclophosphamide	Succinylcholine	↑ blockade	Cons alter rece cycle in the dose succ Mon supp resp
Cyclosporine	Midazolam	Decreases clearance of midazolam. Prolonged sedation	Mon supp resp
Cyclosporine	Nondepolarizing NMBs	↑ blockade	Mon supp resp
Dasatinib	Alfentanil, fentanyl, sufentanil	↑ analgesic effect	Mon supp resp
Digoxin	Succinylcholine	Precipitate new dysrhythmias or potential existing dysrhythmias.	Mon
Disulfiram	Anesthetic gases	Reduction in gas metabolism	Mon of int
Dobutamine	Anesthetic gases	Ventricular dysrhythmias	Mon rhyth
Dofetilide	Sevoflurane	Prolonged QT interval	Mon rhyth
Dopamine	Anesthetic gases	↑ risk of dysrhythmia	Avoid
Echothiophate iodide (ophthalmic)	Succinylcholine	Prolonged blockade	Cons redu alter

Enzalutamide	Alfentanil	↓ analgesic effect	Avoid
Gabapentin	Anesthetic gases	Case report of myotonia and dystonia	Monitor for adverse reactions
Imatinib	Fentanyl	Increases fentanyl concentration and duration	Use lower doses for pain relief
Isoniazid (INH)	Enflurane	Fast acetylators of INH facilitate defluorination of enflurane—high-output renal failure	Monitor renal function
Isoniazid (INH)	Halothane	↑ risk of hepatotoxicity	Avoid rifampin after anesthesia
Ivacaftor	Alfentanil, fentanyl, sufentanil	↑ opioid effect	Monitor for respiratory depression
Ivacaftor	Midazolam	↑ midazolam effect	Monitor for respiratory depression
Ketamine	Halothane	↓ BP, CO	Monitor
Ketamine	Nondepolarizing NMBs	↑ blockade	Monitor for respiratory depression
Ketorolac	Nondepolarizing NMBs	Case reports of apnea	Monitor respiratory
Labetalol	Anesthetic gases	↓ BP	Monitor
Linezolid	Alpha/beta-agonists	Enhanced vasopressor effect	Use

Linezolid	Metoclopramide	Serotonin syndrome and NMS	Monosymptomatic syndrome possible
Lithium	Nondepolarizing NMBs	Possible ↑ blockade	Monosymptomatic respiratory depression
Local anesthetic (large doses)	Nondepolarizing NMBs	↑ blockade	Monosymptomatic respiratory depression
Loop diuretics (bumetanide, ethacrynic acid, furosemide, torsemide)	Nondepolarizing NMBs	Low dose—enhance blockade; high dose—diminish blockade	Titratable
Loop diuretics	Succinylcholine	Dose-dependent prolongation (low doses) or antagonism (high doses) of blockade	Titratable
Macrolides (clarithromycin, erythromycin, telithromycin)	Alfentanil	↑ analgesic activity	Monosymptomatic respiratory depression after surgery. Safe to use
Macrolides	Midazolam	Prolonged sedation	Monosymptomatic respiratory depression
Magnesium	NMBs	Enhanced blockade	Monosymptomatic respiratory depression
MAO inhibitors (nonselective = phenelzine, tranylcypromine) (type B = rasagiline, selegiline)	Alfentanil, fentanyl, remifentanil, sufentanil	Serotonin syndrome, orthostatic hypotension	Monosymptomatic and

MAO inhibitors	Alpha/beta-agonists	Enhanced vasopressor effect	Use reduced doses
MAO inhibitors	Metoclopramide	Serotonin syndrome	Monitor symptoms, avoid if possible
MAO inhibitors	Succinylcholine	↑ blockade	Monitor and support respiration
Mercaptopurine	Nondepolarizing NMBs	May decrease or reverse blockade	Titrate to effect
Methadone	Alfentanil, fentanyl, remifentanyl, sufentanil	↑ CNS depression	Avoid
Methotrexate	Nitrous oxide	Potential of cytotoxic effects	Avoid during methotrexate treatment
Methylphenidate	Anesthetic gases	Enhanced hypertensive effect	Hold day or avoid
Midazolam	Propofol, fospropofol	↓ clearance of midazolam	Monitor and support respiration
Mifepristone	Alfentanil, fentanyl, sufentanil	↑ analgesic effect	Monitor and support respiration
Mirtazapine	Alfentanil, fentanyl, remifentanyl, sufentanil	↑ CNS depression	Avoid

Nitrates	Pancuronium	↑ blockade	Mon supp resp
NMBs	Anesthetic gases	↑ blockade	Titra ager
NNRTIs (note, these drugs are also in combination medications)	Midazolam	↑ midazolam effect	Cont
Octreotide	Opioids	↑ analgesic effects	Mon supp resp
Olanzapine	Midazolam	Possible excessive sedation and cardiorespiratory depression	Avoid olanz pare benz
Opioids	Anesthetic gases	↑ potential for opiate toxicity	Mon supp resp
Opioids	Propofol, fospropofol	↓ BP	Titra each
Opioids	Succinylcholine	↓ HR	Mon brad hear
Phenothiazines (chlorpromazine, fluphenazine, prochlorperazine, thioridazine)	Metoclopramide	Extrapyramidal symptoms	Mon symp
Phenothiazines	Ondansetron, granisetron, dolasetron	Prolonged QTc interval	Mon rhyth
Phenothiazines	Opioids	↑ respiratory depression	Mon supp resp

Phenytoin, fosphenytoin	Nondepolarizing NMBs	Acute (<14 d—↑ blockade). Long term (>14 d—↓ blockade) QTc prolongation	Mon supp resp Avoid
Pimozide	Propofol, fospropofol		
Piperacillin/tazobactam	Nondepolarizing NMBs	↑ blockade	Mon supp resp
Polypeptide antibiotics (bacitracin, colistimethate, polymyxin B)	Nondepolarizing NMBs	↑ blockade. Colistimethate and polymyxin B may have independent NMB activity.	Supp resp Cons gluc reve blocl colis
Polypeptide antibiotics	Succinylcholine	↑ blockade. Colistimethate and polymyxin B may have independent NMB activity.	Supp resp Cons gluc reve blocl colis
Procaine, procainamide	Nondepolarizing NMBs	↑ blockade	Mon supp resp
Procaine, procainamide	Propofol, fospropofol	↑ blockade secondary to competition for pseudocholinesterases	Mon supp resp
Protease inhibitors (note, these drugs are also in combination medications)	Fentanyl	CYP3A4 interaction	Cons litera spec contr
Protease inhibitors	Midazolam	↑ midazolam effect	Cont with mida

			midazolam containing with and midazolam be used cautiously
Quinidine	NMBs	↑ blockade	Monitor support resp
Quinine	NMBs	↑ blockade	Avoid use.
Quinolones (ciprofloxacin, levofloxacin, moxifloxacin, norfloxacin)	Ondansetron, granisetron, dolasetron	Prolong QTc interval	Monitor rhythm
Rifamycin derivatives (rifampin, rifabutin, rifapentine)	Alfentanil	↓ analgesic effect	Increase titration appropriate relief
Rifamycin derivatives	Halothane	↑ risk of hepatotoxicity	Avoid rifamycin after anesthesia
Ropivacaine	Propofol, fospropofol	↑ ropivacaine toxicity	Consider reducing ropivacaine
Sotalol	Sevoflurane	Prolonged QTc interval	Monitor rhythm
Spironolactone	Nondepolarizing NMBs	↑ blockade	Monitor support function
SSRIs (citalopram, escitalopram, fluoxetine, paroxetine, sertraline)	Metoclopramide	NMS	Monitor signs Consider dantrolene

			NMS
SSRIs	Ondansetron, granisetron, dolasetron	Prolonged QTc interval	Mon rhyth
SSRIs	Opioids	Serotonin syndrome	Mon and
Succinylcholine	Nondepolarizing NMBs	↑ blockade	Mon supp resp
Succinylcholine	Propofol, fospropofol	↓ HR	Cons prem prop succ
TCA's (amitriptyline, imipramine, nortriptyline)	Metoclopramide	Extrapyramidal symptoms and NMS	Mon signs
Telaprevir	Midazolam	Increases midazolam concentrations	Oral cont mida be u cauti redu
Tetracyclines	NMBs	↑ blockade	Mon supp resp
Theophylline	Halothane	↑ risk of catecholamine-induced dysrhythmias	Use inhal
Theophylline	Midazolam	↓ midazolam effects	Titra
Thiazide diuretics	Nondepolarizing NMBs	↑ blockade may be secondary to hypokalemia	Corr hypoc
Thiazide diuretics	Opioids	Orthostatic hypotension	Mon BP.

Thiopental	Nondepolarizing NMBs	↑ blockade	Mon supp resp
Tranexamic acid	Estrogens, HRT	↑ VTE risk	Mon
Vancomycin	NMBs	↑ blockade	Mon supp resp
Vecuronium	Propofol, fospropofol	↑ blockade	Mon supp resp
Vemurafenib	Midazolam	↓ midazolam concentration	Titra to ef

■ TABLE F-2 HERBAL AGENTS

Plant	Precautions for Anesthesia and Surgery
Aloe vera	May impair hemostasis
Bilberry (<i>Vaccinium myrtillus</i>)	May impair hemostasis
Black cohosh (<i>Cimicifuga racemosa</i>)	Potential for hypotension and bradycardia
Bladderwrack (<i>Fucus vesiculosus</i>)	May impair hemostasis
Cat's claw (<i>Uncaria tomentosa</i>)	May impair hemostasis (inhibit platelet aggregation)
Cayenne (<i>Capsicum annum</i>)	May impair hemostasis (inhibit platelet aggregation). Has biologic effect of increased catecholamine secretion. Overdose may cause hypothermia.
Chamomile. German (<i>Matricaria chamomilla</i>)	May enhance CNS depression
Coleus (<i>Coleus forskohlii</i>)	May impair hemostasis. Has antithrombotic effects and potential for hypotension
Devil's claw (<i>Harpagophytum</i>)	May have chronotropic and inotropic effects.

procumbens) Dong quai (<i>Angelica sinensis</i>)	May impair hemostasis Has anticoagulant effects. May impair hemostasis. May cause vasodilation
Echinacea	May increase the sedative effect of midazolam
Ephedra (ma huang) (<i>Ephedra sinica</i>)	Potent sympathomimetic may cause cardiac arrhythmias, heart failure, and stroke.
Evening primrose (<i>Oenothera biennis</i>)	May impair hemostasis (inhibit platelet aggregation)
Fenugreek (<i>Trigonella foenum-graecum</i> L.)	May impair hemostasis. Contains coumarin
Feverfew (<i>Tanacetum parthenium</i>)	May impair hemostasis (inhibit platelet aggregation). Increase risk of serotonin syndrome
Fish oils	May impair hemostasis (inhibit platelet aggregation). Potential prolonging bleeding during and after surgery
Garlic (<i>Allium sativum</i> L.)	May impair hemostasis. May inhibit platelet aggregation; potential for enhanced ↓ BP. May ↓ blood glucose
Ginger (<i>Zingiber officinale</i>)	May → prolonged bleeding time (irreversible platelet inhibition); possible ↓ catecholamine secretion; cardioactive in large and prolonged doses
Ginkgo (<i>Ginkgo biloba</i>)	Selective antagonist of platelet aggregation; may cause vasodilation. Antithrombotic effect
Ginseng, American (<i>Panax quinquefolius</i>)	May impair hemostasis (possibly irreversible platelet inhibitor)
Ginseng, Panax (<i>Panax ginseng</i>) root	Dose-dependent effects on BP. May cause tachycardia. May impair hemostasis
Ginseng, Siberian (<i>Eleutherococcus senticosus</i>)	May impair hemostasis. Use barbiturates with caution. May affect BP.

Goldenseal (<i>Hydrastis canadensis</i>)	May impair hemostasis. Potential for hypotension and bradycardia. May alter liver enzymes
Grapefruit	Cytochrome P450 (CYP3A4) inhibition. Onset of midazolam may be delayed and action increased.
Grape seed (<i>Vitis vinifera</i>)	May impair hemostasis
Green tea (<i>Camellia sinensis</i>)	May impair hemostasis
Guggul (<i>Commiphora mukul</i>)	May impair hemostasis
Hawthorn (<i>Crataegus oxyacantha</i>)	High doses may cause hypotension + CNS depression. May ↑ digoxin and ACE inhibition.
Horse chestnut (<i>Aesculus hippocastanum</i>)	May impair hemostasis (contains coumarin). Has cholinergic properties
Kava kava (<i>Piper methysticum</i>)	Synergy with midazolam and anesthetic drugs. Chronic use may ↑ anesthetic requirements. Antagonizes dopamine
Licorice (<i>Glycyrrhiza glabra</i>)	May impair hemostasis. Mineralocorticoid effect ↓ K ⁺
Melatonin	May enhance CNS depressants
Passion flower (<i>Passiflora</i> spp.)	Synergy with CNS depressants. Avoid combination with MAOI.
Red clover (<i>Trifolium pratense</i>)	May impair hemostasis. Contains coumarins
Reishi (<i>Ganoderma lucidum</i>)	May impair hemostasis
Saw palmetto (<i>Serenoa repens</i>)	May cause ↑ BP. Antiestrogen effect
Schisandra (<i>Schisandra chinensis</i>)	Inducer of cytochrome P450 enzyme system
St. John's wort (<i>Hypericum perforatum</i>)	May have some MAOI activity. May reduce midazolam levels due to enzyme induction. Delayed emergence from anesthesia. Avoid meperidine.

Turmeric (<i>Curcuma longa</i>) Valerian (<i>Valeriana officinalis</i>)	May impair hemostasis Potentially synergistic with opiates and CNS depressants, including anesthetics
White willow (<i>Salix alba</i>)	Salicylate, may impair hemostasis
Yohimbe (<i>Corynanthe yohimbe</i>) (<i>Pausinystalia yohimbe</i>)	May cause CNS stimulation. May have cardiovascular effects

■ TABLE F-3 PREOPERATIVE MANAGEMENT OF PATIENT MEDICATIONS

Drug/Drug Class	Continue to Time of Surgery?	Otherwise d/c Before Surgery
Aliskiren (renin inhibitor)	Do not continue unless essential for BP control (consult prescribing MD).	At least 2 d and preferably 7–10 d before lengthy or high blood loss surgery in patients taking other renin/angiotensin system drugs.
Alpha- and alpha-beta-blockers	Continue except for tamsulosin in cataract surgery.	Stop tamsulosin several days prior to cataract surgery to prevent floppy iris syndrome. Consider use of labetalol for intraop blood pressure control.
Amiodarone	Continue (long elimination half-life).	
Angiotensin-converting enzyme (ACE) inhibitors		May intensify hypotension in anesthesia—ensure adequate intravascular volume d/c 24 h prior to surgery.
Angiotensin receptor blockers (ARB)		May intensify hypotension in anesthesia. Hypotension may be less responsive to vasopressors—ensure

		adequate intravascular volume d/c 24 h prior to surgery.
Anticytokines in general	Consider consult.	
Anticytokines, adalimumab		d/c 1–2 wk prior to surgery. Restart 1–2 wk after surgery.
Anticytokines, anakinra		d/c 1–2 wk prior to surgery. Restart 1–2 wk after surgery.
Anticytokines, etanercept		d/c 1–2 wk prior to surgery. Restart 1–2 wk after surgery.
Anticytokines, infliximab	May continue if indication is inflammatory bowel disease	
Anticytokines, rituximab		d/c 1–2 wk prior to surgery. Restart 1–2 wk after surgery.
Antigout medications in general	Continue.	Monitor for fluid and electrolyte shift leading to changes in uric acid levels. Treat flares with NSAIDs or corticosteroids.
Antiparkinsonian medications	Continue prior to surgery as prescribed to prevent relapse.	
Antiparkinsonian medications, levodopa/carbidopa	Give dose evening before surgery.	
Antiparkinsonian medications, entacapone		Hold 24 h before surgery.

Antiparkinsonian medications, tolcapone		Hold 24 h before surgery. Resume when the patient is taking po and stable.
Antiparkinsonian medications, bromocriptine		Hold 24 h before surgery. Resume when the patient is taking po and stable.
Antiparkinsonian medications, pergolide		Hold 24 h before surgery. Resume when patient is taking po and stable.
Antiparkinsonian medications, ropinirole	Administer morning of surgery to avoid precipitating neuroleptic malignant syndrome.	Resume when the patient is taking po.
Antiparkinsonian medications, pramipexole	Administer morning of surgery to avoid precipitating neuroleptic malignant syndrome (NMS).	Resume when the patient is taking po.
Antiparkinsonian medications, selegiline		d/c 3 d before surgery because it is a MAOI type B.
Antiparkinsonian medications, anticholinergic agents		Parenteral diphenhydramine and benztropine can be used to control symptoms in patients who cannot take po medications.
Antipsychotic medications	Continue.	Haloperidol, olanzapine, and ziprasidone are available parenterally.
Antiseizure medications (AEDs)	Continue. Anesthesia, metabolic changes, ethanol withdrawal, and neurosurgical procedures may increase risk of seizures.	Check drug levels. Substitute iv alternatives.

Antithyroid medications	Continue antithyroid medications and beta-blockers.	
Apixaban		d/c 24 h before minor surgery and 48 h before any surgery with a moderate or higher bleeding risk.
Aspirin	Continue in vascular surgery or stents. Secondary prophylaxis in patients at high risk	d/c 5–7 d (up to 10 d) before surgery if risk or consequences of bleed are high or if taken only for primary prophylaxis.
Benzodiazepines	Continue to prevent development of withdrawal symptoms.	Support with iv preparations if necessary.
Beta-adrenergic blockers	Continue because abrupt discontinuation is associated with rebound ischemia, hypertension, and tachycardia.	If d/c prior to surgery, consider use of esmolol intraop for pressure and rate control.
Buspirone	Continue to prevent development of withdrawal symptoms.	
Calcium channel blockers	Continue in patients with normal or slightly impaired heart function. Continue with caution in patients with LV dysfunction.	
Cholestyramine, colestipol		d/c day before surgery.
Cilostazol		d/c 3–5 d before surgery.
Clonidine	Continue to prevent withdrawal symptoms	Consider use of transdermal system 3–4 d

	such as rebound hypertension, tachycardia, tremors, and anxiety.	before surgery with overlap oral dose for 2–3 d if the patient is NPO perioperatively. Observe for hypotension and bradycardia. Resume oral dose 8 h after transdermal patch removal.
Clopidogrel	In patients with: Drug-eluting stents <12 mo Bare metal stents <1 mo if asymptomatic Bare metal stents <12 mo if symptomatic + any others	d/c 5–10 d before surgery. d/c at least 24 h before urgent on-pump CABG if possible, and urgent off-pump CABG may be considered within 24 h of clopidogrel administration.
Clozapine	Continue to prevent withdrawal symptoms and disease relapse. May need psychiatric consult	
Corticosteroids	Continue. May need to supplement due to surgical stress.	
Dabigatran		d/c 1–2 d prior to surgery in patients with good renal function and 3–5 d prior to surgery in patients with CrCl < 50 mL/min. Consider d/c earlier for patients undergoing major surgery or neuraxial anesthesia. Check aPTT morning of surgery. Reversible with idarucizumab
Digoxin	Continue (long half-life).	

Dipyridamole Diuretics (thiazides, loop)	Continue but monitor for volume depletion and ↓ K.	d/c 2–3 d before surgery
DMARDs (disease-modifying antirheumatic drugs)	Consider consult.	May increase immunosuppression and increase the risk of wound infection
DMARDs, methotrexate	Continue in patients with normal renal function.	
DMARDs, leflunomide		d/c 2 wk prior to surgery (long half-life)
DMARDs, sulfasalazine	Continue.	
DMARDs, azathioprine	Continue.	
DMARDs, hydroxychloroquine	Continue.	
Ezetimibe		d/c day before surgery.
Fibric acid derivatives (gemfibrozil, fenofibrate)		d/c day before surgery.
Fondaparinux		d/c 24 h prior to surgery. Start no sooner than 6 h postop. Not reversible with protamine
H2-blockers	Continue.	Consider iv formulations.
Heparin		d/c 4–6 h before surgery. Reversible with protamine. Restart postop 12–24 h after procedures with low risk of postop bleeding. Restart postop 48–72 h after procedures with high risk of postop bleeding.

		Monitor for HIT (heparin-induced thrombocytopenia). d/c at least 14 d prior to elective surgery due to potential CV effects and/or bleeding.
Herbals		
HIV therapy	Be cognizant of drug interactions between HIV medications and anesthetic agents.	
HMG Co-A reductase inhibitors (“statins”)	Continue—may be cardioprotective and anti-inflammatory	
HRT (e.g., estrogen)	Balance risk of VTE with risks of stopping HRT.	Consider VTE prophylaxis.
Inhaled pulmonary medications including beta-agonists, ipratropium, and steroids	Continue.	
Insulin		Reduce evening dose of insulin (give about 2/3 dose). Adjust or omit morning dose of insulin. Use a sliding scale replacement.
Insulin pump		Decrease dose on evening of surgery. Switch to insulin infusion except for short, simple cases.
Leukotriene inhibitors (montelukast, zafirlukast)	Continue.	
Lipid-controlling		d/c day before surgery

agents (cholestyramine, colestipol, gemfibrozil, niacin)		
Lithium	Continue with caution. May prolong blockade of NMBs	Monitor sodium and fluid status.
LMWH		d/c 12–24 h prior to surgery; 24 h if planning neuraxial anesthesia. Partially reversible with protamine. Restart 4 h after epidural catheter is removed.
MAOIs		d/c 10–14 d prior to surgery. Use MAOI-safe anesthetic agents.
Meds for myasthenia gravis	Risk for respiratory insufficiency. Use pyridostigmine evening before surgery.	Postop management with pyridostigmine im 1/10th dose and iv 1/30th dose continuous infusion at 2 mg/h.
Niacin		d/c day before surgery
Nitrates	Continue.	May use nitroglycerin 2% ointment to substitute for po nitrates
NSAIDs		Usually d/c 1–3 d before surgery depending on the medication
Opioids, chronic use	Continue to avoid withdrawal and pain exacerbation.	Review equianalgesic substitutions. Patient may need multimodal approach to temporarily control the surgical pain.
Oral antidiabetic		Hold sulfonylureas and

agents (in general)		nonsulfonylurea secretagogues evening before surgery. Short procedures—delay morning doses until po intake. Monitor for hypoglycemia especially in the longer-acting sulfonylureas.
Oral antidiabetic agents: metformin		Risk of lactic acidosis (rare). Recommendations vary from holding 48 h prior to surgery to continuing (in patients with normal renal function).
Oral contraceptives	Balance risk of VTE with risks of d/c.	Consider VTE prophylaxis.
Pentoxifylline		d/c at least 8 h prior to surgery
PPIs	Continue.	Consider iv formulation.
Prasugrel		d/c 7 d before surgery
Rivaroxaban		d/c at least 24 h prior to surgery or longer (3 d in patients with CrCl < 50 mL/min) if bleeding risk is of primary importance. Nonreversible with protamine. Check aPTT morning of surgery.
Selective estrogen receptor modulators (SERM)	Balance risk of VTE with risks of d/c.	Consider VTE prophylaxis.
SSRIs	Continue.	If it is decided to d/c, taper for 3 wk prior to surgery.

TCAs	Long half-life	If it is decided to d/c, taper for 3 wk prior to surgery. Norepinephrine is pressor of choice.
Theophylline		Hold morning of surgery. Toxic levels are associated with arrhythmias.
Thyroid medication	Medication has long half-life, so omitting a dose is unlikely to have clinical effect.	
Ticagrelor		d/c 5 d before surgery
Ticlopidine		d/c 10–14 d before surgery
Valproic acid	Continue.	Consider iv formulation.
Warfarin	Assess bleeding risk of procedure compared with risk of thromboembolism. Consider bridging therapy.	Hold warfarin 3–5 d prior to surgery and consider bridging therapy. Check INR morning of surgery. Consider low dose of vitamin K on day of surgery if INR remains above surgeon's target INR. If INR is significantly elevated and surgery is immediately necessary, consider a warfarin reversal protocol, which includes vitamin K, FFP, PCC, and factors.

APPENDIX G: SPECIAL CONSIDERATIONS FOR LATEX ALLERGY

Andrew J. Giustini, MD, PhD, Rebecca E. Claire, MD, Anita Honkanen, MD and Brenda Golianu, MD

Latex is the second most common cause of anaphylactic reactions under anesthesia (20%) after neuromuscular blocking agents. The incidence of intraop latex reactions has declined significantly as latex-containing materials have been removed from the OR. Latex gloves were the major source of latex proteins and were implicated in most cases of latex-mediated reactions. Latex exposure may occur through skin contact, mucous membrane exposure, inhalation of glove powder, ingestion, or parenteral injection. Reactions to latex are divided into three types: irritant contact dermatitis, type IV hypersensitivity (skin reaction similar to poison ivy), and type I, IgE-mediated hypersensitivity, which can lead to life-threatening anaphylaxis.

Populations at risk

- Patients with history of multiple surgical procedures, including those with myelomeningocele, spina bifida, and congenital genito-urinary tract anomalies.
- Patients with a history of myelomeningocele and spina bifida should be treated with a “latex precaution” regimen from birth, regardless of prior history of exposure or latex reactivity.
- Health care personnel with occupational exposure.
- Other individuals with occupational exposure such as rubber industry workers and hairdressers.
- Patients with history of atopy, hay fever, asthma, or eczema.
- Patients with history of food allergy to tropical fruits (such as avocado, kiwi, banana, mango) and chestnuts, which contain cross-reacting proteins with latex.
- Patients with history of rash or swelling following contact with balloons, rubber bands, erasers, gloves or other products which may contain latex.

It is currently estimated that as many as 17% of health care workers have been sensitized to latex. Occupational exposure can be minimized by avoiding powdered latex gloves and limiting the use of latex-containing gloves. Applying lotion to hands before using latex gloves facilitates the transfer of latex proteins to hands, and should be avoided.

In high-risk patients, latex-avoidance protocols are recommended as this may decrease

the incidence of subsequent intraop allergic reactions. Hospitals and ORs have decreased the use of products that contain latex to the extent that many are essentially latex free. Anesthesia carts can be assembled with latex-free products, reducing the risk of latex sensitization for all patients and negating the need for a special “latex-free cart.” The latex content of commonly used materials can be identified from external labeling, from package inserts, or directly from the manufacturers. Even minimal latex exposure (e.g., an injection through a latex port of iv tubing or opening a package of powdered latex gloves) has resulted in anaphylaxis.

Diagnosis of latex allergy is based on a focused history and physical examination with positive in vivo or in vitro test. In vitro serum tests for latex-specific IgE such as RAST are highly specific but have a high false-negative rate, up to 30%. Skin testing identifies patients with a high titer of IgE to latex but must be performed with appropriate safeguards because it may induce systemic anaphylaxis.

Pharmacologic prophylaxis in the acute setting is controversial for patients with documented latex allergy. Prophylaxis medications, such as diphenhydramine, ranitidine, and hydrocortisone, are not universally successful in preventing latex anaphylaxis. Some authors have argued that pretreatment may mask the early immune responses, leaving anaphylaxis as the first evidence of an allergic reaction.

To prepare a latex-safe environment

- Notify OR nurses, anesthesia technicians, and surgical staff of the need for a latex-free room.
- Place a sign on the door stating: “LATEX-FREE ROOM.”
- Schedule the latex-allergic patient as the first case of the day to minimize the presence of airborne latex particles.
- Set up the room with latex-free materials (e.g., bag, bellows, ECG electrodes, pulse oximeter clip, iv tubing without latex ports, vinyl gloves, vinyl BP cuffs, clear micropore tape).
- Avoid bouffant surgical hair caps and shoe covers that contain latex bands.
- Wrap Webril™ around latex tubing on stethoscope, BP cuff, or tourniquet with Webril™ or cotton gauze if they contain latex derivatives.
- Removal of rubber stoppers from drug vials (instead of withdrawing through the stopper) is controversial.

Diagnosis of anaphylaxis or latex allergy

- **Early Manifestation** Skin: hives or urticaria at site of contact with latex product, red eyes, angioedema of eyelids, nasal congestion, GI cramping, headache, anxiety.
- **Late Manifestation** Respiratory: bronchospasm, wheezing, ↑ PIP, ↓ O₂ sat,

upsloping of ETCO₂ tracing.

- Intraop cardiac: ↓ BP, ↑ HR, cardiac arrest. Generally, type I hypersensitivity may not occur immediately at the beginning of a surgical procedure but after exposure of peritoneum or mucous membranes to latex.

Treatment of anaphylaxis

- Remove latex.
- Notify surgical team.
- Call for help.
- 100% O₂; secure airway; manually ventilate if needed.
- Discontinue all anesthetic agents, ± Trendelenburg's.
- Epinephrine 0.1–1 mcg/kg iv initially, increasing to 10 mcg/kg; repeated boluses may be needed to support BP; may require an epinephrine infusion (0.05–0.1 mcg/kg/min) and may be increased up to 1 mcg/kg/min in severe cases. IM dose of epinephrine is 0.01 mg/kg (maximum dose of 0.5 mg). If hypotension is refractory to epinephrine, consider the use of a vasopressin bolus of 0.01–0.05 units/kg iv; repeated boluses may be necessary. The use of an infusion can be considered after blood pressure is stabilized: 0.3 milliunits/kg/min to 2 milliunits/kg/min iv with a maximum of 40 milliunits/min (0.04 units/min).
- Administer iv fluid to ↑ preload and support BP.
- Review field to ensure all possible latex-containing equipment has been removed. (Instruct surgical staff to change to nonlatex gloves and scrutinize all equipment.)
- Administer steroids (hydrocortisone 100 mg iv or methylprednisolone 0.5 mg/kg iv (max 125 mg)), antihistamine (diphenhydramine 50 mg or 0.5 mg/kg iv; max = 50 mg), and H-2 blockers (ranitidine 1 mg/kg; max = 50 mg).
- Administer bronchodilators (albuterol) via nebulizer for bronchospasm.
- Continue steroids and diphenhydramine for 24–48 h or until symptoms resolve.
- Collect blood for mast cell tryptase (peaks at 30–60 min, t_{1/2} is 2 h) and histamine (peaks at <30 min).
- Inform pharmacy of latex sensitivity to ensure all products are prepared latex free.
- Refer patient to allergist for follow-up. Skin prick test or RAST can be performed (4–6 wk after the acute reaction resolves) to specifically test for latex allergy.
- Patients with a history of latex anaphylaxis should be advised to wear a Medic Alert bracelet.

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APPENDIX H: PERIOPERATIVE ACUPUNCTURE

Brenda Golianu, MD, and Jeannie Seybold, MD

We gratefully acknowledge the past contributions of Emily F. Ratner, MD

INTRODUCTION

Acupuncture is a treatment modality that has been practiced in China for over 3 millennia. Initially transmitted as an oral tradition, it was first described in written form in the Huangdi Neijing or the Yellow Emperor's Inner Canon, the seminal text of ancient Chinese medicine dating back to the 3rd century BC. Acupuncture involves placing very thin needles in the skin to stimulate the flow of qi (pronounced "chee") in a complex network of meridians in the body. There are 12 principal and 8 curious acupuncture meridians that correspond to physiologic and anatomical organ functions. Qi is a dynamic form of physical and spiritual energy that flows within the universe and in all organisms. One of the basic tenets of Chinese medicine is that illness and pain are caused by the stagnation or blockage of qi flow and/or the invasion of pathologic influences—traditionally known as wind, heat, cold, dampness, dryness, or fire—that result in imbalances of yin and yang. When a point is needled, a heavy sensation known as "deqi," or a mild paresthesia, may be experienced by the patient. The practitioner may sense a gentle contraction of the connective tissue surrounding the needle or may observe a flare developing around the needle. Needling, electrical and laser stimulation, acupressure, or even herbal therapies (moxibustion or capsicum plaster) over specific points have all been documented to alleviate pain and pathologic states. In addition to body acupuncture, which developed in China, Japan, and Korea, many different traditions have been developed that focus on needling specific body parts, that is, the ear, scalp, or hand, as microsystems representing the entire body. Acupuncture has been used to provide analgesia during surgery since the 1950s in China. However, it was little more than a curiosity in the United States before 1971, when reporter James Reston went to China to report on the diplomatic efforts of Henry Kissinger and President Richard Nixon. While in Beijing, Reston required an emergency appendectomy and received acupuncture for postoperative ileus and pain control. The popularity of acupuncture in the United States exploded after he published his experiences in the New York Times. In 1997, the NIH released a consensus statement supporting the use of acupuncture for adult postoperative and chemotherapy-induced nausea and vomiting and postoperative dental pain. It also stated that acupuncture may

be a useful adjunctive treatment in addiction, stroke rehabilitation, headache, menstrual cramps, tennis elbow, fibromyalgia, myofascial pain, osteoarthritis, low back pain, carpal tunnel syndrome, and asthma. During the last 10 years, acupuncture has been increasingly studied and used to treat acute postoperative pain as well as in chronic pain clinics throughout the United States.

MECHANISMS

Several mechanisms for acupuncture analgesia have been proposed. The gate control theory by Melzack and Wall in 1965 postulated that stimulation of a-beta fibers inhibits a-delta and c-fiber transmission of pain signals. This may be a local mechanism of action. Other studies have shown that electroacupuncture at low (2–4 Hz) and high frequencies (100 Hz and greater) selectively induces endorphin and enkephalin release, respectively. Conflicting evidence exists regarding the ability of naloxone to antagonize the analgesic effects of acupuncture. Some studies show that naloxone reverses acupuncture-induced analgesia, while others dispute this. This suggests the analgesic mechanisms of acupuncture are more complex than the release of endorphins. Additional evidence suggests that the frequency and intensity of stimulation determine the degree of naloxone reversibility. A review of functional MRI and positron emission tomography studies has shown that electroacupuncture exerts effects over the hypothalamus, somatosensory motor cortex, and rostral anterior cingulate cortex, with nonspecific modulation of the limbic system and hypothalamus. Other mechanisms of action may include the local generation of plasma and nitric oxide and release of adenosine. Acupuncture may also have anti-inflammatory properties.

PERIOPERATIVE USE

Acupuncture is effective for reducing PONV. P6 (Neiguan), the most thoroughly studied acupoint, is located three fingerbreadths proximal to the wrist crease, between the flexor carpi radialis and palmaris longus tendons and directly over the median nerve. A Cochrane review of 26 randomized trials noted significant reduction in nausea and the need for rescue antiemetics with the use of P6 acupoint stimulation (Fig. H-1). Direct electrical stimulation of this point was shown to be as efficacious as a standard dose of ondansetron for PONV in adults and resulted in a 37% reduction in nausea in children after tonsillectomy. There is evidence that transcutaneous electrical acupoint stimulation is also effective in reducing PONV. Stimulation of P6 by twitch monitoring using a standard nerve stimulator (at 1 Hz, 0.2 ms, 50 mA) during general anesthesia for laparoscopic surgery significantly reduced PONV for 24 h with an efficacy similar to that of commonly used antiemetic drugs. A trained medical acupuncture practitioner

would likely integrate P6 with a combination of body and ear acupuncture points to minimize PONV. Acupuncture has not been shown to eliminate the need for anesthetic medications during surgery, but it may be a useful adjuvant for perioperative analgesia and anxiolysis. In a randomized controlled trial of perioperative acupuncture for abdominal surgery, Kotani et al. showed 50% reduction in postoperative morphine use, 20–30% reduction in postop nausea, and 30–50% reduction in plasma cortisol and epinephrine levels. A reduction in postoperative pain and analgesic requirements was also seen in studies of acupuncture in patients having gynecologic, abdominal, thoracic, and orthopedic surgeries. Acupuncture is known to produce deep relaxation and sedation and may be useful for preoperative anxiolysis or for postoperative weaning of opioid medications in opioid-tolerant patients. Two recent studies demonstrated cardioprotective effects of electrical acupuncture in adult and pediatric cardiac surgical patients, as indicated by decreased postoperative serum cTnI (cardiac troponin I), decreased duration of postoperative mechanical ventilation, and shorter ICU stay. The risks of acupuncture are rare. The most common are minor bruising, limited capillary bleeding, pain, or local infection at the needling site. Anesthesiologists can be trained to provide acupuncture treatment for PONV and anxiolysis. More comprehensive perioperative treatment should be performed by or under the supervision of a trained medical acupuncturist.

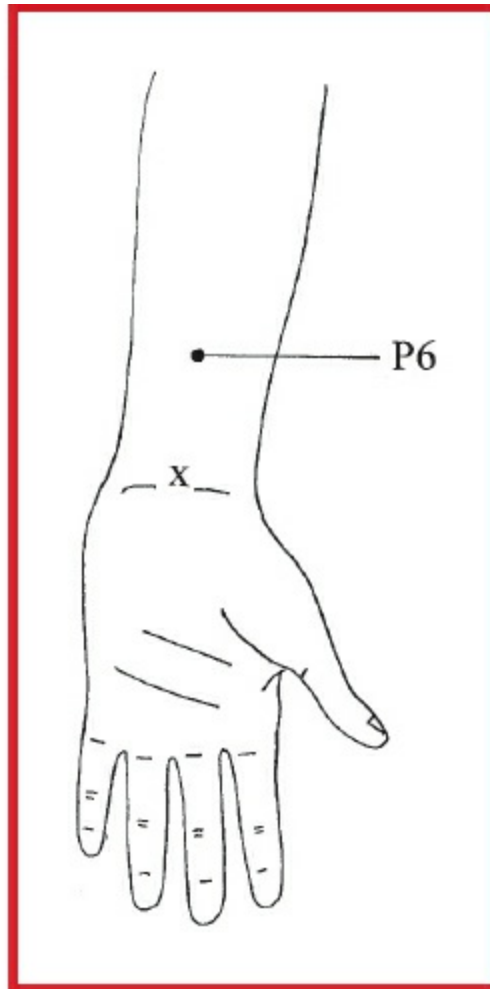


Figure H-1 P6 acupoint. P6 is located three fingerbreadths proximal to the wrist crease directly over the median nerve and between the tendons of palmaris longus and flexor carpi radialis. For PONV prophylaxis, the two nerve stimulator electrodes can be placed over the median nerve (1) at a point proximal to the wrist crease (marked with an x) 2 cm proximal to P6 and (2) on P6 or 1 cm distal to the P6 point. Stimulus parameters: 1 Hz, 0.2 ms, 50 mA during anesthesia.

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APPENDIX I: MANAGEMENT OF THE DIFFICULT AIRWAY

C. Philip Larson Jr., MD, CM

Management of the airway, particularly the difficult airway, effectively and safely is a skill that every anesthesia provider must have to be qualified to provide clinical anesthesia. Competency in airway management is mandatory before undertaking independent clinical practice of anesthesia.

DIFFICULT AIRWAY VERSUS DIFFICULT VENTILATION

Difficult airway	5–10% of patients depending on type of practice Tip-offs include prior history of difficult airway/intubation, small mouth opening, large tongue, prominent teeth, large neck, limited range of motion of head, short thyromental distance (<3 cm), Mallampati grade >2, upper-lip bite test below vermillion. Management: multiple options (see below)
Difficult ventilation	Rare, <1% of patients Causes obvious: large beard, thick redundant jowls, oropharyngeal mass, fractured larynx, mediastinal mass compressing trachea, lingual tonsillar hyperplasia (easily ventilated with LMA) Management: Consider LMA and Plan C (see below) or awake fiberoptic intubation (FOI).

MANAGEMENT OF DIFFICULT AIRWAY

Caution

Always determine that ventilation is possible with bag-mask or LMA before inducing full general anesthesia and paralysis. If it is not possible and ET tube is needed, consider awake intubation.

Patient positioning	Have head at top edge of table; bed height at umbilical level of operator. Do not bend over. Can
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	reposition patient for surgical needs after airway established.
Direct laryngoscopy (plan A)¹	Stand sideways facing right wall; keep left arm at side with ET tube in the left axilla (the only place where operator does not have to look away once the glottis identified). Put patient's head in sniff position by directing laryngoscope along the plane of handle (do not lift or pull handle toward oneself). With proper positioning, intubation should be a leg activity with the arm only acting as a fulcrum. Do not do more than two direct laryngoscopies. Multiple laryngoscopies result in inability to ventilate as well as intubate.
Endotracheal tube selection	Always use the smallest appropriate tube (7.0 men, 6.5–7.0 women). Small tubes are easier to insert than large tubes. If larger tube needed later, can always change tube using airway exchange catheter (AEC). Choice of tubes: standard, wire-reinforced (for prone surgery), EMG no. 6 (for nerve monitoring), laser, MLT tube (for microlaryngeal surgery)
Choice of LMAs	Many choices available, each with specific benefits/problems. Either classic or disposable suitable for routine care. Use classic when using as a conduit for FOI or Aintree catheter. Use LMA no. 4 for men and no. 3 or no. 4 for women. Avoid no. 5 when doing FOI as it blocks the glottic opening by invaginating the epiglottis. Intubation LMA; success rate 90–95%; less when intubation difficult. Make certain balloon on reusable silicone tube does not leak. Loosen 15-mm connector before inserting tube. Determine that tube is in trachea. Do not drop 15-mm connector while inserting tube holder and removing LMA. Fix connector firmly before utilizing ET tube.
Video laryngoscopes (plan B)³	GlideScope™ or C-MAC™ gives the best picture. Use the largest blade for most adults. Use either special rigid stylet provided or shape another stylet to same

Intubating catheters (plan B)¹

curvature as blade. Watch the tube enter the mouth before looking at the screen to avoid soft tissue trauma. Extract stylet slightly once tube at glottis opening if tube does not advance easily.

Cook (Frova) catheters: two types, styleted and unstyleted. 70-cm long and styleted has two adaptors (15 mm for O₂ insufflation, one for attaching syringe or gas sampling line). Tip rounded so it will not perforate soft tissue. Needs some shaping before use. Must see the epiglottis to use effectively; cannot be advanced beyond the carina. Use ET tube that closely approximates catheter size (6.5–7.5). Can use intubating catheter together with video laryngoscope if catheter bent to conform to the shape of the laryngoscope blade.

Eschmann: 60 cm long, easily wound in circle for transport. Loses shape and stiffness after a few washings

Aintree: 60-cm-long, hollow tube blunt at both ends; has snap-on 15-mm adaptor for O₂ insufflation. Use classic LMA or Tudor-Williams airway as conduit for insertion; may not advance through disposable LMAs. Must be experienced at FOI as all control and direction done with handle. Use 7.0–7.5 tube for insertions (6.5 will not advance).

FIBEROPTIC INTUBATION PROTOCOL; AWAKE AND ASLEEP²⁻⁴

Awake	Preferred technique when ventilation by bag-mask or LMA not possible
Premedication	If not contraindicated, patients should receive mild-to-moderate sedation with dexmedetomidine 1 mcg/kg over 10 min or fentanyl 0.3–0.5 mcg/kg and/or midazolam 2–4 mg iv.
Topical anesthesia	When premedication has been established, for awake FOI, the oropharynx is sprayed vigorously ~6

times over a span of 10 min with lidocaine 4% solution using a disposable EZ spray unit powered with oxygen. Initially, the spray is directed at the front of the tongue; gradually, it is directed further back in the throat, until the entire oropharynx is numb. Having the patient inhale deeply while spraying enhances anesthetic spread into the airway. In reality, the lateral recesses of the oropharynx need not be anesthetized topically because both fiberoptic laryngoscope (FOL) and ETT are confined to the midline of the mouth.

Tracheal anesthesia

With patient breathing oxygen from anesthesia circuit or transport mask, transtracheal injection of cocaine or lidocaine 4% (4 mL) is made through the cricothyroid membrane, using a 5-mL syringe and a 23-ga, 3/4-inch needle. So that this injection can be made as rapidly as possible, it is important to use a small syringe, making certain that the connection between syringe and needle is tight. It is important that operator's hand be fixed firmly against the patient's upper chest to assure that needle movement is minimized should the patient start to cough and that the full injection is made into the trachea. When the injection is complete, the patient is urged to cough vigorously.

Laryngoscopy

After the mouth and trachea are anesthetized with local anesthetic, an oral airway with a central orifice (e.g., Tudor-Williams, Patel, Ovassapian airway) is placed in the midline of the mouth. A 7-mm orotracheal tube, without connector attached, is placed over an FOL. With the operator at the patient's side near the waist, the FOL is introduced through the hole in the airway and advanced to the end of the airway. At this point, the epiglottis should be visible. The tip of the fiberoptic scope is flexed toward the operator about 15–20°, which should bring arytenoid cartilages and laryngeal opening into view. The scope is advanced into the larynx so that

tracheal rings can be visualized. Often, the carina also can be visualized. The laryngoscopist also can place the scope in the airway by darkening the room and using the scope as a light wand, directing the light externally to the sternal notch and advancing it down the trachea. If a wire-reinforced tube is desired, use a Patel or Ovassapian airway instead of the Tudor-Williams airway for guidance, because these airways can be removed with the tube connector in place.

Intubation

The scope is placed on the patient's chest and, holding it so that it not advanced further, the orotracheal tube is advanced gently into the trachea. To facilitate passage of the orotracheal tube past the arytenoid cartilages and into the larynx, it is necessary to rotate the tube counterclockwise 90°—or even as much as 180°—several times as it is being advanced.

Using rotational movement, the operator should never need to push hard on the tube to position it in the larynx. After the tube is in place, the FOL and oral airway are removed, and the 15-mm connector is reattached to the tube. ETCO₂ confirms proper placement. The orotracheal tube is then firmly taped in place at one side of the mouth. If manual ventilation of the lungs is possible, it is often beneficial for both patient and anesthesiologist to perform FOI with the patient anesthetized. This is particularly true for patients with cervical spine injuries where manipulation of the head during direct laryngoscopy or coughing during topicalization of the airway or insertion of the tube during awake FOI may aggravate the spinal cord injury. Asleep FOI involves inducing GA, inserting an LMA, and placing the patient on controlled ventilation. Then, proceed with plan C (see Ref 6). In brief, a 6.0 uncuffed tube is mounted on the FOI, the FOI is inserted into the trachea using the LMA as a guide, and the 6.0 tube is

lubricated and inserted into the laryngeal opening. The FOI is removed and the lungs are ventilated via the 6.0 tube. A medium-sized airway exchange catheter (AEC) is lubricated on one end and inserted into the trachea via the 6.0 tube. The AEC usually meets resistance at the distal end of the 6.0 tube. If this occurs, rotate the 6.0 tube while pushing on the AEC. Once the AEC is a few centimeters beyond the tip of the 6.0 tube, the tube and LMA are removed en bloc leaving the AEC in the airway. With the AEC in the far right corner of the mouth, thread the final endotracheal tube on the AEC and gently advance it through the mouth into the larynx, rotating it counterclockwise to avoid obstruction from the right arytenoid cartilage. Then, remove the AEC and connect the ETT to the anesthetic circuit.

Asleep (plan C)¹

Preferred technique when ventilation possible by bag-mask or LMA. Easy to determine by administering very low-dose propofol and initiating assisted/controlled ventilation with sevoflurane. Major induction agent is sevoflurane. Establish excellent anesthesia and paralysis. Insert classic LMA. Disposable LMAs have internal ridge that makes advancement of scope and/or tube impossible. In emergency, ridge can be cut out before insertion in the mouth. Mount 6.0 uncuffed ET tube with very tight 15-mm connector onto the scope. Focus FOL, maximize light source, minimize room light, and insert FOL via center struts of LMA into the glottis and trachea. Lubricate no. 6 tube and advance into the trachea; remove FOL; attach tube to circuit and ventilate lungs. Need high gas flow because of leak between the tube and LMA. Lubricate medium airway exchange catheter (AEC) and insert via no. 6 tube about 35–37 cm (5–7 cm beyond tube tip); remove ET tube and LMA together; insert final tube (6.5 or 7.0) rotating gently counterclockwise; remove AEC

	and palpate balloon on tube for final positioning above the sternal notch. Technique can be practiced safely in normal patient so, when needed, operator is comfortable and confident performing it.
Failed plans A–C (plan D)¹	Plans A–C very rarely fail, but if they should, plan D would present two choices: Proceed with a surgical airway (cricothyrotomy, tracheostomy) or awaken patient and bring back another day for awake FOI.

NASAL FIBEROPTIC

Awake	Requires adequate sedation and excellent topical anesthesia of the nose preferably with cocaine 4% pledgets or cotton applicators and oral spray with lidocaine 4% via EZ spray. Dilate nasal cavity with progressively larger nasal airways. Mount the smallest acceptable nasal RAE on the scope, insert the scope in the nose keeping it in the midline and asking an assistant to pull up on the jaw to open the oropharynx. Do not advance nasal RAE until the scope is well into the trachea.
Asleep	Preferred to awake as it is much more pleasant for both patient and anesthesia provider. Despite general anesthesia and paralysis, topical anesthesia of the nose with cocaine 4% and gradual nasal dilation are essential to avoid nasal bleeding. Have assistant do jaw thrust to open oropharynx.

CERVICAL SPINE INJURY: INTUBATION OPTIONS

Awake FOL	Preferred technique by many despite serious drawbacks: namely, coughing and bucking during oropharyngeal topicalization; during transtracheal injection (if used); and during insertion of fiberoptic scope and/or tube into the trachea. Should rarely be used now, and only in circumstance where bag-mask
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	ventilation is not possible.
In-line traction	Commonly used in emergency rooms in combination with direct laryngoscopy with patient either sedated or anesthetized. Multiple problems with this technique: (1) There is no established standard for how in-line traction should be applied, and no documentation of its effectiveness or safety. (2) In-line traction requires the use of excessive leverage on the laryngoscope to visualize the laryngeal opening. (3) In-line traction greatly compromises operator's ability to visualize glottis opening, even in patients who are not classified as difficult airways on preop assessment. ^{5,6}
Plan C (asleep)	Strongly preferred technique since there is no movement of the head during the procedure and there is no coughing/bucking since the patient is anesthetized and paralyzed. The only prerequisite for use is that bag-mask or LMA ventilation is possible. This is easily determined by administering a sedative dose of propofol in conjunction with low-dose inhaled sevoflurane and seeing if assisted ventilation is possible. If as the depth of anesthesia is increased with sevoflurane the patient becomes more difficult to ventilate, administration of a neuromuscular blocking drug and insertion of an LMA will usually resolve the problem. This technique can be used with a patient in a cervical collar or halo traction.
Difficult pediatric airway	Airway difficulties in infants and small children are generally due to congenital abnormalities of the face, mouth, chin, or upper airway. These can usually be handled effectively and safely by a combination of sedation or GA and insertion of a fiberoptic scope directly into the mouth or via a conduit such as an LMA.
Intubation	The scope is placed on the patient's chest, and, holding it so that it is not advanced further, the orotracheal tube is advanced gently into the trachea.

To facilitate passage of the orotracheal tube past the arytenoid cartilages and into the larynx, it is necessary to rotate the tube counterclockwise 90°—or even as much as 180°—several times as it is being advanced.

EXTUBATION STRATEGY

Delayed extubation	Always have an extubation strategy. Delaying extubation and mechanically ventilating the lungs for a time postop is frequently a good option.
Postoperative extubation	Establish full recovery from GA and paralysis, with sustained, spontaneous ventilation. Consider deflating the balloon on the tube, occluding the ET tube and determining if ventilation is possible around the tube. This finding is reassuring but no guarantee of sustained airway patency when the tube is removed. Consider inserting an LTA spray down the tube to prolong tolerance to the tube. If there is any question regarding airway patency, insert an AEC 5–7 cm beyond the tube before removing it. As long as it does not touch the carina and is not moved, it will not cause coughing and will not affect voice. Remove AEC when sustained ventilation is evident. If reintubation is necessary, use a tube that closely approximates the AEC in diameter.

MANAGEMENT OF LARYNGOSPASM

Standard technique	Positive pressure ventilation with oxygen along with jaw thrust. Technique may not work until patient is sufficiently hypoxic that vocal cords relax. May need to administer succinylcholine 20–50 mg iv or im. If this fails and ET intubation is not possible, consider immediate cricothyrotomy.
Best technique	Learn to do the laryngospasm maneuver ² and apply

it immediately after every removal of ETT or LMA. Should be standard of care because if done correctly, it always relieves laryngospasm within 15–20 sec. Find “laryngospasm notch,” which is behind the pinna of the ears. It is bounded by the coronoid process of the mandible anteriorly, the mastoid process posteriorly, and the base of the skull superiorly. To be effective, operator must feel the base of the skull. With patient lying supine, very firm pressure is directed inward bilaterally and at the same time pulling upward (skyward) on the coronoid process. Place transport mask with oxygen flowing on patient’s face prior to starting laryngospasm maneuver. Continue maneuver until patient moves arms up toward head (Larson reflex). Once the patient can do this, laryngospasm will not recur. The maneuver corrects airway obstruction from the tongue because the forward displacement of the mandible pulls the tongue forward. Why it corrects laryngospasm is not known. It is not due to pain because pain elsewhere does not resolve laryngospasm.

CRICOTHYROTOMY

Indications

Every anesthesia provider should be trained to perform a cricothyrotomy and be prepared to do it when needed. The risks associated with doing it are minimal and the benefit is lifesaving. Indications: partial or complete airway obstruction and inability to ventilate lungs or perform tracheal intubation; massive oral/tracheal bleeding making mask ventilation or tracheal intubation impossible, combined with rapidly decreasing SaO_2

Technique

Identify cricothyroid membrane between the thyroid cartilage superiorly and cricoid cartilage (round, signet ring shape) inferiorly. With scalpel, penknife,

or scissors, make 1–1.5 inches horizontal or vertical incision at the level of the cricoid. Cut down quickly to the cricoid cartilage, cut through it, and insert a small tube (no. 4–6F) or an AEC followed by a small tube in the opening. One generally has little time before the patient goes into cardiac arrest from hypoxia, so the maneuver must be done quickly. There are no structures in the midline of the neck that will be harmed by the incision, and bleeding from the incision is generally minimal. Reluctance to perform the maneuver when indicated generally results in patient death. Needle cricothyrotomies are not recommended. They are harder and more time consuming to perform and far less effective in establishing an airway.

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APPENDIX J: ANTICOAGULATION GUIDELINES FOR NEURAXIAL PROCEDURES

Einar Ottestad, MD, Alex J. Butwick, MD, MS, FRCA, Brendan Carvalho, MBBCh, FRCA, MDCH, Jamie Kuo, PharmD, BCCCP, and Sharon Jung, PharmD

ANTICOAGULATION GUIDELINES FOR NEURAXIAL PROCEDURES

■ GUIDELINES TO MINIMIZE RISK FOR SPINAL HEMATOMA WITH NEURAXIAL PROCEDURES**

	PRIOR TO NEURAXIAL/NERVE PROCEDURE Minimum time between last dose of antithrombotic agent AND neuraxial injection or neuraxial/nerve catheter placement	WHILE NEURAXIAL/NERVE CATHETER IN PLACE Restrictions on use of antithrombotic agents while neuraxial/nerve catheters are in place and prior to their removal	AFTER NEURAXIAL/NERVE PROCEDURE Minimum time between neuraxial/nerve catheter removal AND next coagulation test or antithrombotic agent
ANTICOAGULANTS, INJECTABLE			
Heparin 5000 units SQ q8h or q12h	4–6 hours	May be given; preferred over alternatives	1 hour
Heparin 7500 units SQ q8h or q12h	12 hours verify normal aPTT	May be given; preferred over alternatives	1 hour
Heparin IV infusion	4–6 hours verify normal aPTT	May be given; preferred over alternatives	1 hour
Dalteparin (Fragmin)	≥12 hours	No data - use not recommended	4 hours

Enoxaparin (Lovenox) 40 mg SQ daily	≥12 hours	CONTRAINDICATED Risk of spinal/epidural hematomas	4 hours
Enoxaparin (Lovenox) 30 mg SQ q12h	≥12 hours		4 hours
Enoxaparin (Lovenox) 1.5 mg/kg SQ daily or 1 mg/kg SQ q12h	≥24 hours		4 hours
Fondaparinux (Arixtra)	36–42 hours	CONTRAINDICATED	6 hours

ANTICOAGULANTS, ORAL

Apixaban (Eliquis) 2.5 mg BID–5 mg BID	3 days	No data - use not recommended	6 hours
Rivaroxaban (Xarelto) 15–20 mg daily	3 days	No data - use not recommended	6 hours
Betrixaban (Bevyxxa) 80 mg daily	3 days	No data - use not recommended	5 hours
Edoxaban (Savaysa) 30–60 mg daily	3 days	No data - use not recommended	6 hours
Dabigatran (Pradaxa) 75–150 mg BID	CrCl < 30 mL/min: avoid NA CrCl 30–49 mL/min: 5 days CrCl 50–79 mL/min: 4 days CrCl ≥ 80 mL/min: 3 days Renal fxn unknown: 5 days	No data - use not recommended	6 hours
Warfarin	4–5 days	check INR daily;	remove wr

(Coumadin)	verify normal INR	assess risk vs benefit when INR 1.5-3	1.5 and/or hours of fir
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DIRECT THROMBIN INHIBITORS, INJECTABLE

Argatroban IV infusion	Avoid neuraxial techniques		
Bivalirudin (Angiomax) IV infusion	Avoid neuraxial techniques		

ANTIPLATELET AGENTS

Aspirin	May continue dosage	Avoid neuraxial techniques if early postoperative use of other anti-hemostatic drugs is anticipated	May continue dosage
NSAIDs	May continue dosage		May continue dosage
Aspirin/dipyridamole (Aggrenox)	24 hours	No data - use not recommended	6 hours
Cangrelor (Kengreal) IV infusion	3 hours	No data - use not recommended	8 hours
Clopidogrel (Plavix)	5–7 days	No data - use not recommended	Without LE immediate hours
Prasugrel (Effient)	7–10 days	No data - use not recommended	Without LE immediate hours
Ticagrelor (Brilinta)	5–7 days	No data - use not recommended	Without LE immediate hours
Abciximab (Reopro) IV infusion	24–48 hours	No data - use not recommended	Contraindic weeks after monitor ne status if gi neuraxial te
Tirofiban (Aggrastat) IV infusion	4–8 hours	No data - use not recommended	

Eptifibatide (Integrelin) IV infusion	4–8 hours	No data - use not recommended	
THROMBOLYTIC AGENTS			
Alteplase (TPA) Full dose for stroke, MI, etc	48 hours and normalization of clotting studies (e.g. fibrinogen)	No data - use not recommended	Normalization of fibrinogen

**Guidelines do not address risks with multiple therapies or comorbidities that interfere with coagulation. Consult appropriate expert(s) as needed.

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APPENDIX K: ENHANCED RECOVERY AFTER SURGERY

Richard A. Jaffe, MD, PhD, Andrew J. Giustini, MD, PhD, and Brenda Golianu, MD

GENERAL CONSIDERATIONS

Enhanced recovery after surgery (ERAS) was conceived as an enhanced fast-track approach to the management of patients undergoing colorectal surgery that incorporated the additional goals of improving postop functional status and decreasing morbidity. ERAS principles have since been expanded both in scope and in application to many other types of surgery. Studies have demonstrated significant benefits of ERAS protocols including decreased length of stay, improved patient satisfaction, decreased morbidity, and cost reduction. ERAS works best as a multispecialty team approach to perioperative patient care. Even in the absence of a formal team, the application of individual elements of ERAS (e.g., limiting NPO duration, multimodal pain management, etc.) is likely to be beneficial for most patients undergoing anesthesia and elective surgery. Importantly, the unique combination of elements most beneficial for any specific surgical procedure and patient cannot be arbitrarily defined.

In the following table, many of the individual elements of enhanced recovery protocols are briefly presented (further details can be found in the Suggested Reading list below). ERAS protocols continue to evolve as the relative importance of the individual elements of enhanced recovery are studied. When considering individual elements, it is essential to appreciate their interdependence. For example, reducing the NPO interval and achieving adequate preop hydration will facilitate intraop fluid management. Obviously, the implementation of any ERAS element must take into consideration patient condition, co-morbidities, and surgical requirements that could affect applicability.

Element	Recommendations
Preop counseling/education	Patients should be routinely counseled regarding what to expect from their surgical and anesthetic experience and be given ample opportunity to have their questions answered in appropriate detail. Discussion should include lines, drains and urinary catheters, along with plans for postop pain

	management emphasizing the use of regional analgesia along with peripheral nerve blocks where possible. Postop wound care, activity limitations, and physiotherapy plans should also be discussed. A procedure-specific enhanced recovery handout with preop checklist should be given to patients.
Prehabilitation program	A program of preop exercise activities designed to increase aerobic capacity may improve recovery in selected patients. For obese patients with time permitting, the benefits of weight loss to a BMI < 30 may be beneficial. For elective major surgery, consider preop physical therapy for patients with significant deconditioning. More generally, prehabilitation refers to any preop intervention that can improve postop recovery including the optimization of any comorbid conditions, and implementation of the interventions listed below.
Preop smoking and alcohol cessation	Abstinence from both alcohol and smoking should be recommended in preparation for surgery. Chronic alcoholics may require withdrawal precautions.
Preop nutritional status	The patient's nutritional status should be assessed in advance of the surgery and optimized if possible. Immunonutrients (most trials have used some combination of arginine, glutamine, omega-3 fatty acids, and ribonucleic acid as a dietary supplement) may have anti-inflammatory effects, boost immune function, and improve outcome in surgical patients. Further study will be necessary to identify effective formulations and specific patient populations most likely to benefit. Impact Advanced Recovery® immunonutrition drink (Nestlé Health Science) is an example of a commercially available preop supplement that has been shown to improve some surgery outcome measures.

Preop fasting and carbohydrate loading	For most procedures, preop fasting should be limited to 2 h for clear (see-through) liquids and 6 h for solid foods or opaque liquids. Preop carbohydrate loading should be encouraged and can be accomplished using oral rehydration sports drinks (e.g., regular Gatorade™ or equivalent) 2 h before surgery.
Antithrombotic prophylaxis	Assess VTE risk. The use of sequential pneumatic compression sleeves is recommended to prevent venous thromboembolism. The addition of anticoagulants (e.g. heparin 5000u sc 30–60 min preop) should be discussed with the surgeon and should take into account patient risk factors for VTE.
Antimicrobial prophylaxis, bowel prep, and skin cleansing	Patients will often be provided with an antimicrobial skin cleanser to use before surgery. Minimize surgical site shaving. Bowel prep is now rarely indicated. Prophylaxis with iv cefazolin 30–60 min before skin incision is recommended.
Regional anesthesia and nerve blocks	When possible, the use of bolus or continuous regional anesthesia and/or specific nerve blocks should be encouraged as a component of anesthetic and analgesic management, particularly for major open surgical procedures.
Surgical approach	Although not always an option, minimally invasive and endoscopic approaches are associated with less postop pain, improved recovery, and patient satisfaction.
Anesthetic protocol	There is insufficient evidence of improved recovery to support the use of TIVA vs. inhalational anesthesia or to support the choice of specific anesthetic agents for most surgical procedures. When appropriate, the use of N ₂ O will reduce the need for other anesthetics and facilitate rapid recovery from anesthesia. PONV concerns can be eliminated with prophylactic antiemetics, and

	effects on intraop bowel distention appear to be minimal. Optimal ventilation can usually be achieved using volume controlled, pressure mode with TV = 8 ml/kg IBW, PEEP at least 5 cmH ₂ O, and ETCO ₂ = 40mmHg (mild permissive hypercapnia).
Fluid management	Fluid balance can be assessed by PPV (if art line in place), TEE (if available), or PVI especially when used in combination with small-volume (100–250 mL) bolus tests. Noninvasive cardiac output monitoring may also be useful. For most minor surgeries, a restrictive (zero balance) strategy is appropriate (see section below). A goal-directed fluid management strategy is useful for many major surgeries (see section below). Vasopressors should be used in preference to volume in euvolemic patients to maintain MAP unless otherwise contraindicated.
Multimodal pain management emphasizing nonopioid analgesia (see p. K-5)	PO or IV acetaminophen (e.g., 1 g) is a useful component of multimodal analgesia. PO celecoxib (COX-2 inhibitor; 200–400 mg po qd) or ketorolac (COX-1 inhibitor; 15–30 mg iv q6h) can provide other modes of analgesia (although some surgeons may be concerned about impaired hemostasis). A ketamine bolus (0.5 mg/kg) and subsequent infusion (5–10 mcg/kg/min) after induction may be a useful intraop analgesic supplement in select patient populations. Gabapentin (300–600 mg po preop) and tramadol (50 mg po postop) have side effects (e.g., prolonged sedation, N/V, pruritus, respiratory depression) that may be unfavorable in some patients or following some surgical procedures. Regional analgesia should be used when possible.
PONV	Routine use of 5-HT ₃ antagonists (e.g., ondansetron 4 mg iv) and dexamethasone (4–8 mg) is recommended. The addition of preop

	aprepitant (40 mg po qd) is often reserved for procedures or patients with a high risk of PONV. Scopolamine and promethazine have side effects (e.g., sedation, dysrhythmias) that may make them undesirable as primary agents especially in older patients.
Avoiding hypothermia	In general, active heating (e.g., forced-air blanket, circulating water blanket, airway humidifier) should be used to maintain core normothermia.
Oxygenation	Liberal oxygen administration (resulting in $SpO_2 > 95\%$) may be detrimental (e.g., $\uparrow$mortality, absorption atelectasis, acute lung injury, etc.) and should be avoided throughout the postop period if possible.
Urinary drainage	Bladder catheters should be removed as early as possible (usually POD 0–1).
Chest tube and other surgical drainage	Active management is required to avoid retention of fluids. Drainage tubes should be removed as soon as it can be safely accomplished.
Postop delirium management	Postop delirium is thought to be underdiagnosed and may result in delayed recovery, extended hospital stay, and cognitive impairment. Medical, metabolic, and medication etiologies should be excluded or treated. Effective pain management, along with nonpharmacologic interventions (return of hearing aids/eyeglasses, reassurance/reorientation, minimizing physical restraints, presence of family/friends, etc.) should be emphasized. Avoiding ketamine and high-dose benzodiazepines in older adults may be helpful. If nonpharmacologic treatment has failed, consider treating hyperactive delirium with low-dose haloperidol (0.5–1 mg iv) if not otherwise contraindicated (onset of beneficial effects may require >15 min). A dexmedetomidine infusion (e.g., 0.1 mcg/kg/h) should be considered in patients for whom haloperidol has failed or is

	contraindicated (see section on Postoperative Delirium in Appendix B for more details).
Postop nutrition	Postop enteral nutrition should be initiated as soon as possible. Chewing gum (30 min tid) may speed up the return of postop bowel function.
Early mobilization	Extubate as soon as criteria are met. Encourage early mobilization, at least sitting up in chair starting POD1 if possible.
Physiotherapy	Immediate pulmonary physiotherapy is often beneficial after long GETA. A physical therapist may be needed to assist with early mobilization following major surgery.

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2. Joshi GP, Pennant JH, Kehlet H: Evaluation of nitrous oxide in the gas mixture for anesthesia (ENIGMA) studies: the tale of two large pragmatic randomized controlled trials. *Anesth Analg* 2017;124(6):2077–2079.
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ENHANCED RECOVERY AFTER SURGERY: FLUID MANAGEMENT

Intraoperative fluid management strategies for enhanced recovery

Intraop fluid management can be significantly affected by the patient's adherence to liberal NPO instructions including preop hydration. In the past, "NPO after midnight" ensured that patients would be dehydrated, uncomfortable, and in a catabolic state before surgery. The typical 70-kg healthy adult will have lost >800 mL of primarily extracellular fluid volume by the end of an 8-h fast (Fig. K-1). As a general rule, deep levels of anesthesia (e.g. BIS <40) should be avoided, and vasopressors should be used to treat anesthesia-related hypotension.

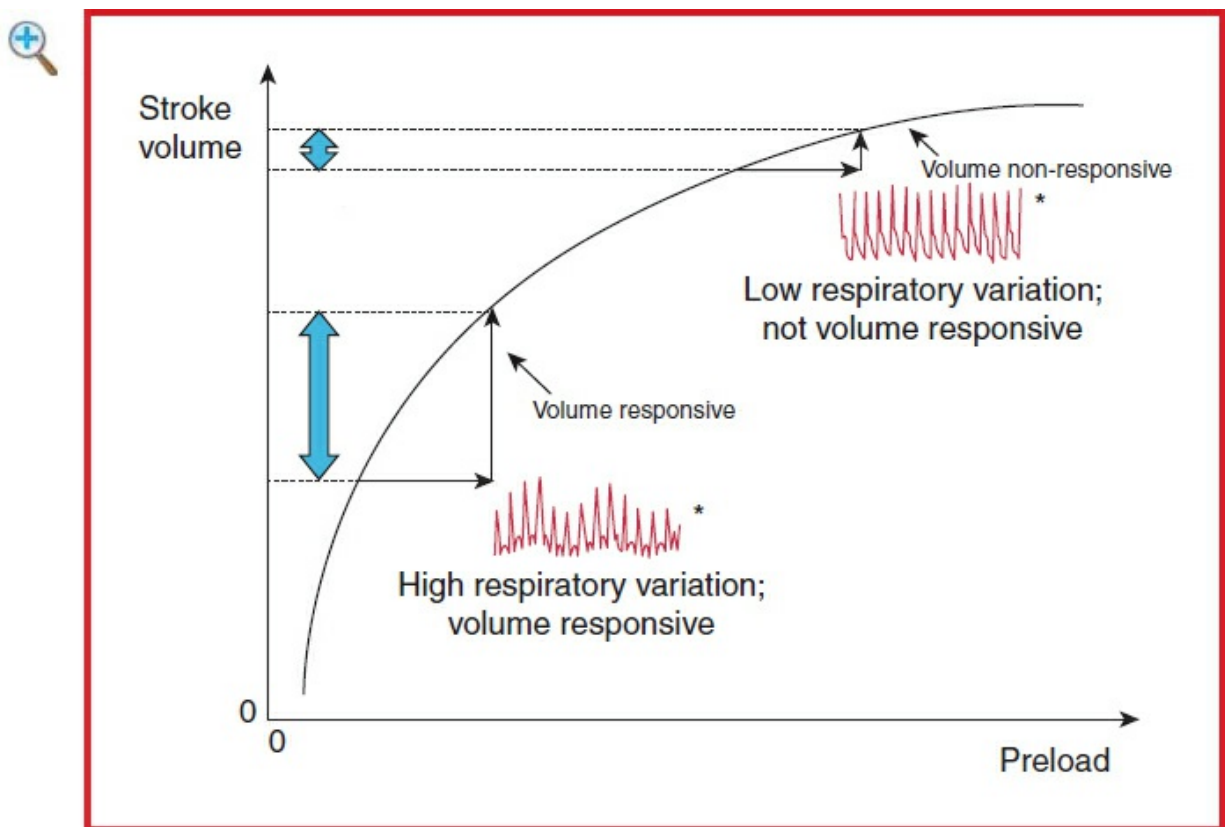


Figure K-1 Frank-Starling relationship with corresponding respiratory variations in the arterial pressure waveform. High respiratory variation in the arterial pressure or in the plethysmographic oximeter waveforms indicates that the patient is on the steep portion of the Frank-Starling relationship. Low

variation indicates that the patient is on the plateau. Cardiac output maximization concepts aim at increasing cardiac output until it reaches the plateau of the Frank-Starling relationship. This goal, as demonstrated on this figure, could be achieved by minimizing respiratory variation in arterial pressure or plethysmographic waveform. (Modified from Rinehart J, Liu N, Alexander B, et al: Closed-loop systems in anesthesia: is there a potential for closed-loop fluid management and hemodynamic optimization? *Anesth Analg* 2012; 114(1):130–43.)

Parameter	Goal-Directed Strategy	Restrictive (Zero Balance) Strategy
Patient population	Major surgery with potential for EBL $\geq$ 500 cc or significant fluid shift	Surgery in low-risk patient. EBL < 500 cc
IV fluids*	BSS @ 1–3 mL/kg/h + UO + 1.5 $\times$ EBL (or 1 $\times$ EBL if colloid used)	BSS @ 1–3 mL/kg/h + 1.5 $\times$ EBL
PPV (or SVV)	Give volume if PPV > 10–15% (TV: 8 cc/kg). Use 100–250 cc BSS bolus; repeat prn to keep PPV < 11%.	PPV usually not available (no art line)
PVI	May substitute for PPV (less reliable?)	← but probably better than nothing
TEE	Transgastric two-chamber (or midpapillary) view to assess LV cavity size. Normal end diastolic diameter = 45–50 mm $\pm$ 4 mm (F < M). Rx small EDV with 100–250 cc BSS bolus prn	Usually not necessary
CO/SV estimating devices	May substitute for PPV when PPV cannot be used	Usually not necessary
HR	Poor predictor of volume status	←
BP	Poor predictor of volume status; Rx $\downarrow$ BP 2° anesthesia	←

	with phenylephrine or ephedrine prn	
Urine output	Poor predictor of volume status Goal UO ≥ 0.3 mL/kg/h to avoid AKI	← expect/accept mild oliguria
CVP or PAOP	Poor predictor of volume status	← and usually not available
SvO₂ or ScvO₂	Poor predictor of volume status	← and usually not available

*assuming adequate preop hydration.

BSS, balanced salt solution (e.g., LR, Normosol®, PlasmaLyte®). Patients should be euvolemic and carb loaded preop per ERAS guidelines.

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ENHANCED RECOVERY AFTER SURGERY: MULTIMODAL ANALGESIA

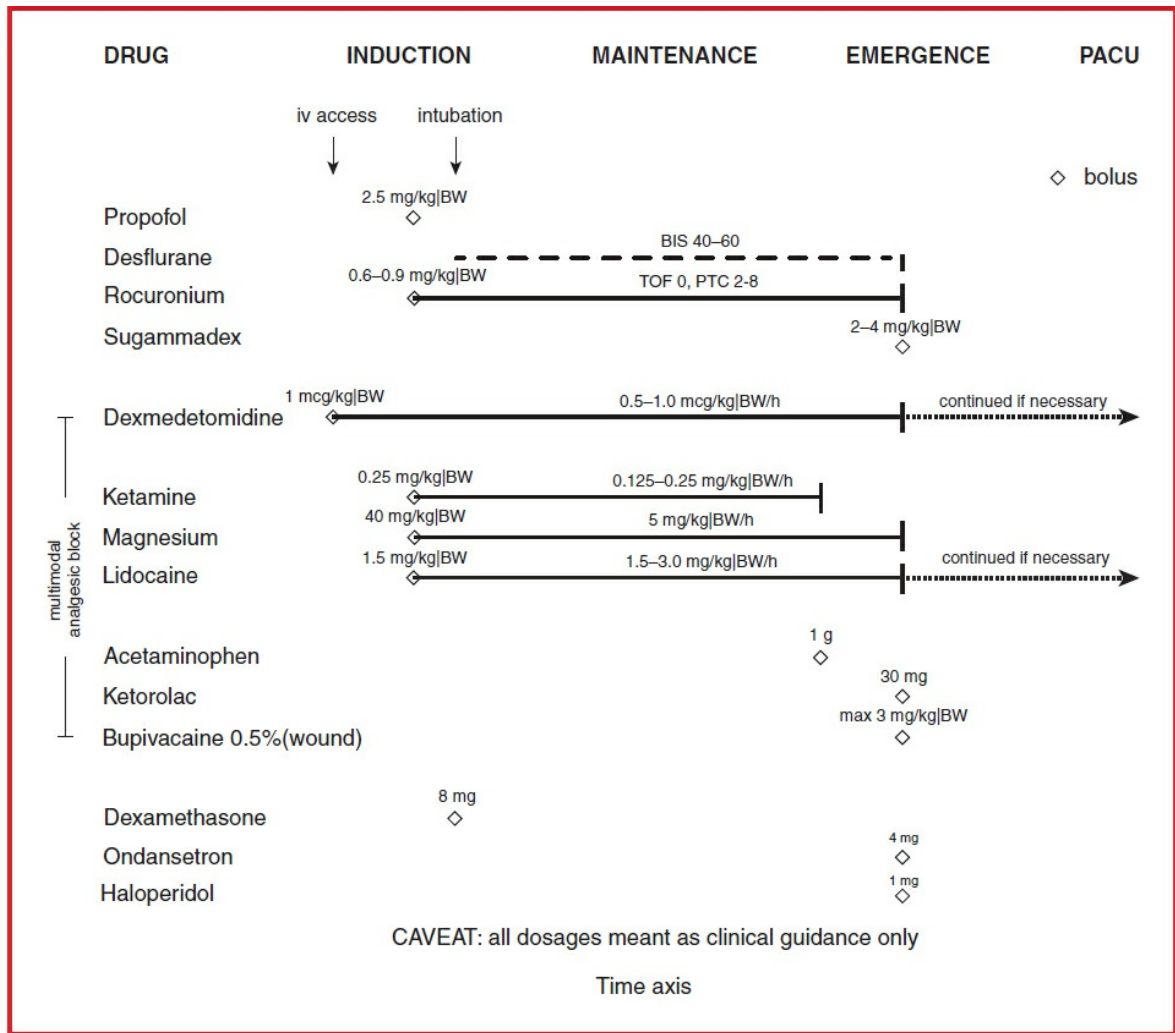


Figure K-2 An example of nonopioid multimodal analgesia/anesthesia. (Adapted from Mauermann E, Ruppen W, Bandschapp O. Different protocols used today to achieve total opioid-free general anesthesia without locoregional blocks. *Best Pract Res Clin Anaesthesiol* 2017; 31(4):533–45. Copyright © 2017 Elsevier. With permission.)

Multimodal analgesia options for a 70-kg patient without specific drug contraindications, undergoing major surgery (see Fig. K-2). Specific analgesic components and doses must be individualized for each patient. (See [Appendix C](#) for additional information on perioperative pain management.)

Multimodal Analgesia Options to Consider

Preinduction

Systemic

Acetaminophen 1000 mg
PO
Celecoxib 200 mg PO
Ibuprofen* 800 mg PO
Gabapentin 300–600 mg
PO

	Regional	Neuraxial analgesia (e.g., epidural or spinal opiates) or regional/peripheral nerve block
Preincision	Systemic	Ketamine 50 mg iv (especially in opioid-dependent patients) Fentanyl 100–250+ mcg iv
	Local	Local anesthetic infiltration of incision site
Intraop	Systemic	IV opiates (e.g., fentanyl, hydromorphone [low-dose], remifentanyl infusion @ 0.05–0.2 mcg/kg/min) Lidocaine infusion 1–2 mg/kg/h and/or ketamine infusion 0.25 mg/kg/h ± MgSO ₄ 2 g iv (may enhance opiate analgesia) ± Acetaminophen 1000 mg iv q6h
	Regional/local	Whenever possible: continuous regional anesthesia/analgesia Local anesthetic infiltration of incision at closing
Postop	Systemic	Acetaminophen 1000 mg PO q6h Ketorolac* 15–30 mg IV q6h or ibuprofen* 200–800 mg PO q6h or celecoxib 200 mg PO IV opiates (e.g., fentanyl)

Regional/local

bolus and PCA
Tramadol 50 mg PO q4h
for breakthrough pain
Continuous regional
analgesia

*May interfere with platelet function.

Suggested Readings

1. Mauermann E, Ruppen W, Bandschapp O: Different protocols used today to achieve total opioid-free general anesthesia without locoregional blocks. *Best Pract Res Clin Anaesthesiol* 2017; 31(4):533–45.
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ENHANCED RECOVERY AFTER SURGERY: PEDIATRIC SURGERY

ERAS pathways have become common for adult surgical procedures, especially in bowel, breast, and joint surgeries. These pathways typically employ standardized preop, intraop, and postop medications, fluid management strategies, surgical approaches, patient mobilization, and nutrition optimization to facilitate patient recovery, decrease morbidity, and decrease time to discharge from the hospital. ERAS pathways use opioid-sparing multimodal analgesics, antiemetics, optimal intraop fluid management, and preferred anesthetic method (including neuraxial and regional anesthetic techniques). In randomized controlled trials, these interventions have been shown to decrease morbidity, shorten length of stay, and reduce costs of care after various types of surgeries in adults. ERAS pathways have slowly become more common in pediatric surgical practice, with recent articles detailing their benefits.

Principles of ERAS can be applied to most types of surgical cases. While providing a very hemodynamically stable profile intraop, high-dose intraop and postop opioids represent one of the primary causes of prolonged hospital courses, including postop nausea and delayed return of bowel function after surgery. Therefore, many ERAS protocols emphasize the use of opioid-sparing techniques, including thoracic, lumbar and caudal epidurals; paravertebral catheters; and erector spinae plane catheters for thoracic and abdominal procedures. For unilateral upper and lower extremity procedures, single-shot or continuous nerve catheters are also often used. The use of adjuvants to intravenous opioids is also encouraged, including (when appropriate) intraop ketamine; preop or intraop acetaminophen that is continued postop; preop and

postop use of pregabalin or gabapentin; postop use of NSAIDS including ketorolac, ibuprofen, or celecoxib; and the use of low-dose lorazepam or diazepam to treat agitation or for muscle spasm after surgeries likely to cause spasm. When opioids are used, the goal should be to use the lowest effective dose of opioids to adequately control pain and to transition to oral analgesics as soon as the patient can tolerate.

The use of nonpharmacologic therapies at all phases of care, including child life consultation, distraction therapies, preop counseling where appropriate, and parental presence at induction, may help to decrease perioperative anxiety, decrease the patient's stress response, and reduce perioperative opioid requirements. Perioperative acupuncture has been shown to decrease postop pain, nausea, and vomiting and may help to decrease the inflammatory response perioperatively and thereby facilitate patient recovery.

Other principles of ERAS that are applicable to most pediatric surgeries include the optimization of nutrition and of fluid management. Nutrition is optimized by prescribing patients evidence-based diets before surgery or admitting patients the night before surgery to follow a standard feeding protocol, and then encouraging early enteral nutrition and discontinuation of parenteral fluids as soon as practical after surgery. Along with early feeding, it is especially important in postsurgical, hospitalized pediatric patients to start medications to reduce the risk of opioid-induced constipation, especially medications such as senna, bisacodyl, docusate, or polyethylene glycol. To prevent intravascular depletion as well as unnecessary fluids that can cause total body and bowel edema, careful assessment of the patient's fluid status is necessary before and during surgery, and general guidelines for fluid requirements for each surgery should be considered. Some pathways combine these aims by encouraging patients, for example, to continue drinking clear fluids containing electrolytes up until two hours prior to surgery and to minimize fasting to as little as is required for the safe performance of general anesthesia.

An evidence-based guide to antiemetic management is also important, as the two primary causes of unplanned admission after outpatient surgeries are uncontrolled pain and uncontrolled nausea.

The creation of ERAS pathways for specific types of surgery is typically done at each operating site by a careful review of the literature and then collaboration and agreement between surgeons, anesthesiologists, and other stakeholders including nursing staff, nutritionists, physical and occupational therapists, and patients. Inclusion and exclusion criteria of patients for each surgery are created, as are guidelines for both the anesthesiologist and surgeon, and perioperative counseling strategies for patients and

families. ERAS pathways include recommendations for preferred minimally invasive surgical techniques, multimodal analgesics, anesthetic techniques, protocols for maintenance of normothermia, hemodynamic goals, choice of antibiotics, fluid optimization, nutritional management, antiemetic choice, time to mobilization, and goals for discharge of patients. Another necessary part of ERAS pathways is the monitoring of metrics associated with surgical recovery to ensure that the ERAS pathway is enhancing recovery and to allow for further improvements.

Typical ERAS Elements Applicable to Pediatric Anesthetic Care

Preop

- Allow patients to drink clear, electrolyte-containing fluids until 2 h before surgery.
- Consider preop administration of oral acetaminophen, celecoxib, and/or gabapentin or pregabalin (see doses in table below).
- Assess volume status.
- Discuss with surgeon the use of neuraxial and regional techniques, when appropriate.
- Assess for how patient comorbidities will affect anesthetic and analgesic choices.
- Pharmacologic anxiolysis
 - Midazolam 0.5 mg/kg po (maximum 20 mg) or 0.05–0.1 mg/kg iv (maximum 2 mg)
 - Intranasal dexmedetomidine (2 mcg/kg)
- Nonpharmacologic anxiolysis options
 - Child life consultation
 - Distraction with music, videos, virtual reality
 - Preop psychological counseling
 - Perioperative acupuncture
 - Parental presence for induction

Intraop

- Attempt to maintain euvolemia with appropriate use of crystalloids and colloids.
- Use multimodal pain control. Can include the following:
 - Acetaminophen iv 10 mg/kg (<1 yr old) or 15 mg/kg (1 yr or older)
 - Dexmedetomidine iv (1 mcg/kg over 15 min if not used preop) or clonidine 1–2 mcg/kg
 - Ketamine iv (0.5 mg/kg at beginning of anesthetic and then additional 0.25 mg/kg every hour)

	• Ketorolac 0.5 mg/kg at end of surgery • Employ neuraxial and regional techniques when appropriate. • Use appropriate antiemetic treatments. • Employ techniques to maintain normothermia. • Change anesthetic plan based on best management of patient comorbidities.
Postop	• Continue multimodal analgesia. • Use medications to prevent constipation. • Encourage early enteral nutrition and patient mobilization, as appropriate.

Analgesic Adjuvants to Opioids

Indication or Category	Medication and Dosage	Maximum per Dose
Skeletal muscle spasm or anxiety. Use only one at a time.	Lorazepam (0.01 mg/kg) q6h iv/po Diazepam (0.1 mg/kg) q8h po	0.5 mg 5 mg
Neuropathic pain. Use only one at a time.	Pregabalin (1 mg/kg) q12h po Gabapentin (5 mg/kg) tid po	75 mg 300 mg
NSAIDs for general pain (do not use any of these in children <6 mo old). Use only one at a time.	Ibuprofen (10 mg/kg) q6h po Ketorolac (0.5 mg/kg) q6h iv Celecoxib (2 mg/kg) q12h po	600 mg 15 mg. Do not use for more than 3 d. 100 mg
Acetaminophen	Acetaminophen 10 mg/kg (<1 yr old) or 15 mg/kg (1 yr or older) q6h po/iv	650 mg
Opioid-induced pruritus	Nalbuphine (0.05 mg/kg) q4h iv prn	2.5 mg
Nausea (typically ondansetron as first-line therapy with one	Ondansetron (0.1 mg/kg) q8h iv/po prn Metoclopramide (0.1	4 mg 10 mg

alternative medication)	mg/kg) q6h iv/po prn Promethazine (0.25 25 mg mg/kg) q6h iv/po/pr prn
Constipation treatment and prevention	Bisacodyl 5 mg (3–11 yr old) or 10 mg (>11 yr old) qd pr prn Docusate 10 mg (<3 yr old), 25 mg (3–5 yr old), 50 mg (6–12 yr old), 100 mg (>12 yr old) bid po scheduled or prn Senna 4.3 mg (half tablet, for 2–5 yr old), 8.6 mg (one tablet, for 6–11 yr old), 19.2 mg (two tables, for >12 yr old) qd po scheduled or prn Polyethylene glycol 3350: 4.25 g (<5 yr old), 8.5 g (5–12 yr old), 17 g (>12 yr old) bid po

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4. Leeds IR, Boss EF, George JA, et al: Preparing enhanced recovery after surgery for implementation in pediatric populations. *J Pediatr Surg* 2016; 51:2126–9.
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APPENDIX L: HIGH-FLOW NASAL CANNULA (HFNC) AND TRANSNASAL HUMIDIFIED RAPID-INSUFFLATION VENTILATORY EXCHANGE (THRIVE)

Erin Bushell, MD

Background

1. Heated and humidified high-flow nasal cannulae (HFNC) have been employed in critical care areas for some time.
2. Use in the OR with apneic, anesthetized patients is more recent. Patel and Nouraei coined the acronym THRIVE when they began using the device in the OR on anesthetized patients and discovered that it significantly extended the safe apnea period.
3. HFNC may be delivered by the ventilator in the ICU or by a stand-alone device. When stand-alone devices such as the Optiflow® (Fisher & Paykel Healthcare) and the Hi-VNI® (Vapotherm®) are used on spontaneously breathing patients, this is usually referred to as HFNC, and when used on apneic patients, this is usually referred to as THRIVE.
4. Blenders are available for these devices and an $\text{FiO}_2 < 1$ may be used in spontaneously ventilating patients, but for apneic patients, an FiO_2 of 1 is standard.
5. The devices deliver a totally conditioned gas (37° at $44 \text{ mg H}_2\text{O/L} = 100\%$ saturation), which allows patients to tolerate high flows.
6. Flows can be titrated from 10 to 70 lpm. Even in dyspneic spontaneously breathing patients, this usually exceeds the peak inspiratory flow rate and thus provides noninvasive support for patients with hypoxemic respiratory failure.

Physiologic Principles

Five physiologic mechanisms are involved.

1. Apneic oxygenation:

- First described by Volhard in 1908 as diffusion ventilation, the mechanisms were expanded upon by Draper's notion of a "hemoglobin-oxygen pump" in 1947.
 - Given an oxygen consumption of ~200–250 mL/min, this amount of oxygen will move from the alveolus into the blood. However, due to buffering and the high solubility of carbon dioxide, only about 8–20 mL/min of additional CO₂ enter the alveoli. This results in a slight negative pressure in the alveoli and, provided the patient has a patent airway, net movement of oxygen (or whatever gas is in the pharynx) into the alveoli.
2. Positive airway pressure: When the mouth is closed, these devices provide ~1 cm H₂O of positive pressure for every 10 L of flow.
 3. Maintenance of FRC: The replacement of alveolar oxygen combined with positive airway pressure maintains FRC, thereby reducing atelectasis and minimizing desaturation due to shunting.
 4. Ventilatory exchange: In contrast to true apneic oxygenation where the PCO₂ rises ~2–3 mm Hg per min of apnea, when THRIVE is used, the observed increase in PCO₂ is only about 1 mm Hg per min of apnea. This may be due to synergy between high flow washout and cardiogenic oscillations.
 5. Dead space washout: The high flows employed wash out the pharyngeal space, effectively removing all supraglottic CO₂ and thus eliminating this dead space.

Applications/Indications

Included here is a list of the common current applications:

1. Preoxygenation of patients prior to intubation or induction of anesthesia and while securing difficult airways:
 - Three minutes of preoxygenation with HFNC has been shown to be as effective as 10 L via face

mask.

- In obese patients undergoing bariatric surgery, 8 ½ min of preoxygenation with HFNC was superior to face mask and equivalent (with a trend toward superior) to preoxygenation with 7 cm H₂O of CPAP.
 - In patients undergoing rapid sequence intubation, it allowed for longer apnea time.
 - It has been demonstrated to significantly improve oxygenation during intubation in the ICU and to optimize oxygenation and intubation conditions during awake fiberoptic intubation.
 - In contrast to face mask ventilation, it may be continued during efforts to secure an airway, thus further extending the safe apnea time. There are case reports in which it has provided lifesaving maintenance of oxygenation during 30-min periods of apnea while emergency surgical airways were secured.
2. During general anesthesia:
 - THRIVE has been successfully used as the sole means of oxygenation during general anesthesia for oral and laryngeal surgeries and ECT.
 3. During monitored anesthesia care:
 - The use of HFNC has been described for various procedures such as endoscopy, awake tracheostomies, and awake craniotomies.
 4. In critically ill patients:
 - HFNC has been employed in patients with stridor, acute epiglottitis, hypoxemic and hypercapnic respiratory failure, and heart failure.
 - HFNC has been used to prevent postextubation respiratory failure and reintubation.

Contraindications

1. Fire risk: THRIVE has been used successfully with laryngeal laser surgery as long as care is exercised to ensure all fuel sources are protected from the laser. THRIVE with an FiO₂ of 1 should not be used in areas near electrocautery. Instead, an FiO₂ of 0.3

- or lower should be used to reduce the risk of fire.
- 2. Recent sinus surgery or recent transsphenoidal hypophysectomy.
- 3. Patients who would not tolerate even a modest rise in PCO_2 (e.g., pulmonary hypertensives) should not undergo apneic oxygenation with THRIVE.

Other Considerations

- 1. Airway patency must be maintained: Laryngeal suspension, jaw thrust, or oral/nasal airways should be used to maintain oxygenation and allow for some ventilatory exchange.
- 2. Must have sufficient FRC at outset: THRIVE will maintain FRC but cannot expand atelectatic lungs. Therefore, it is best to use in slight head-up position.
- 3. Use in children: Although THRIVE is effective in extending the safe apneic period in children, early data indicate it may not offer the same degree of ventilatory exchange because the rate of rise of PCO_2 does not appear to be as reduced as it is in adults.

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ABBREVIATIONS

APPENDIX M: ABBREVIATIONS

~	approximately
↓	decrease (e.g. ↓BP = hypotension)
↑	increase (e.g. ↑BP = hypertension)
→	results in, leads to
⇐	see column to the left
2°	secondary to
A-a	alveolar-arterial
AAA	abdominal aortic aneurysm
ABC	argon beam coagulator
ACAS	Asymptomatic Carotid Atherosclerosis Study
ACE	angiotensin-converting enzyme
ACL	anterior cruciate ligament
ACLS	advanced cardiopulmonary life support
ACT	activated clotting time
ACTH	adrenocorticotrophic hormone
AD	autonomic dysreflexia
ADH	antidiuretic hormone (vasopressin)

AEC	airway exchange catheter
AED	antiepileptic drugs
AF	atrial fibrillation
AFL	atrial flutter
A/G	albumin/globulin
AH	autonomic hyperreflexia
AI	aortic insufficiency
AKI	acute kidney injury (formerly known as ARF)
ALCAPA	anomalous left coronary artery from pulmonary artery
ALI	acute lung injury
ALT	alanine aminotransferase (SGPT)
AOVM	angiographically occult vascular malformations
A-P	anteroposterior
AR	aortic regurgitation
ARDS	adult respiratory distress syndrome
AS	aortic stenosis
ASA	American Society of Anesthesiologists
ASD	atrial septal defect
AST	aspartate aminotransferase (SGOT)
ATG	antithymocyte globulin
ATLS	Advanced Trauma Life Support
ATN	acute tubular necrosis
AV	arteriovenous
A-V	atrioventricular
AVF	arteriovenous fistula
AVM	arteriovenous malformation
AVN	avascular necrosis
AVR	aortic valve replacement
BAEP	brainstem auditory evoked potential
BAER	brainstem auditory evoked response
BB	bronchial blocker
BBS	bilateral breath sounds

BD	balloon dilation
BE	base excess
β -HCG	β -human chorionic gonadotropin
BMI	body mass index
BMR	basal metabolic rate
BCNU	chemotherapy agent
BOR	branchio-oto-renal (syndrome)
BP	blood pressure
BPD	bronchopulmonary dysplasia
BPF	bronchopleural fistula
bpm	beats per minute
BRCA	breast cancer gene
BSA	body surface area
BSLT	bilateral sequential lung transplantation
BSO	bilateral salpingo-oophorectomy
BSS	balanced salt solution (e.g., LR, Normosol®, Plasma-Lyte®)
B-T	Blalock-Taussig (operation)
BUN	blood urea nitrogen
BVH	biventricular hypertrophy
Bx	biopsy

C-section	cesarean section
C-spine	cervical spine
Ca ⁺⁺	calcium
CABG	coronary artery bypass graft(ing)
CAD	coronary artery disease
CAH	congenital adrenal hyperplasia
CAJ	cricoarytenoid joint
CBC	complete blood count
CBF	cerebral blood flow
CBV	cerebral blood volume
CCAM	congenital cystic adenomatoid malformation
CCU	coronary care unit

CD	Crohn disease
CDDP	cisplatin
CDH	congenital diaphragmatic hernia
CEA	carotid endarterectomy
CHD	congestive heart disease
CHF	congestive heart failure
CI	cardiac index
CK	creatine kinase
CLO	congenital lobar overdistension
cm	centimeter
CMC	carpometacarpal (joint)
cmH ₂ O	centimeters of water
CMRO ₂	cerebral O ₂ consumption
CN	cranial nerve
CNS	central nervous system
CO	cardiac output
CO ₂	carbon dioxide
COPD	chronic obstructive pulmonary disease
CP	cerebral palsy
CPAP	continuous positive airway pressure
CPB	cardiopulmonary bypass
CPD	citrate-phosphate-dextrose
CPK	creatine phosphokinase
CPM	continuous passive motion
CPNB	continuous peripheral nerve block
CPP	cerebral perfusion pressure
Cr	creatinine
CRI	chronic renal insufficiency
CRPS	complex regional pain syndrome (formerly known as RSD)
CRT	cardiac resynchronization therapy
CSE	combined spinal-epidural
CSF	cerebrospinal fluid
CSI	cranial spinal irradiation

CT	computed tomography
CTR	carpal tunnel release
CTS	carpal tunnel syndrome
CUF	continuous ultrafiltration
CUSA	Cavitron ultrasonic surgical aspirator
CV	cardiovascular
CVA	cerebrovascular accident
CVP	central venous pressure
CXR	chest x-ray
d	day(s)
D5W	dextrose 5% in water
D&C	dilation and curettage
D&E	dilation and evacuation
DAPT	dual antiplatelet therapy
DBP	diastolic blood pressure
DC	direct current
d/c	discontinue
DCR	dacryocystorhinostomy
DDAVP	desmopressin acetate
DFT	defibrillation threshold
DHCA	deep hypothermic circulatory arrest
DI	diabetes insipidus
DIC	disseminated intravascular coagulation
DIP	distal interphalangeal (joint)
DJD	degenerative joint disease
DKA	diabetic ketoacidosis
dL	deciliter(s)
DL	direct laryngoscopy
DL _{CO}	carbon monoxide diffusion capacity
DLT	double-lumen tube
DM	diabetes mellitus
DO ₂	oxygen delivery

DOE	dyspnea on exertion
DORV	double-outlet right ventricle
DOS	day of surgery
DP	dorsalis pedis
DPG	diphosphoglycerate angiography
DTR	deep tendon reflex
DVT	deep venous thrombosis
d/w	discuss with
Dx	diagnosis
EA	esophageal atresia
EAC	endoaortic clamp catheter
EACA	epsilon-aminocaproic acid
EARC	endoaortic return cannula
EB	epidermolysis bullosa
EBL	estimated blood loss
EBV	estimated blood volume
EC-IC	extracranial-intracranial
ECG	electrocardiogram
echo	echocardiogram
ECMO	extracorporeal membrane oxygenation
ECT	electroconvulsive therapy
ED	emergency department
EDAS	encephaloduro-arteriosynangiosis
EEA	end-to-end anastomosis
EEG	electroencephalogram
EF	ejection fraction
EJ	external jugular
EMG	electromyogram
EMLA	eutectic mixture of local anesthetic
ENT	ear-nose-throat
EP	evoked potentials
EPS	electrophysiologic studies

EPV	endopulmonary vent
ER	emergency room
ERCP	endoscopic retrograde cholangiopancreatography
ERV	expiratory reserve volume
ESC	endocoronary sinus catheter
ESLD	end-stage liver disease
ESRD	end-stage renal disease
ESS	endoscopic sinus surgery
ET	endotracheal
ETCO ₂	end-tidal CO ₂
ETN ₂	end-tidal N ₂
ETOH	alcohol
ETT	endotracheal tube
EVAR	endovascular aneurysm repair
EVD	endovascular drain (extraventricular catheter)
FAP	familial adenomatous polyposis
FAST	focused assessment with sonography for trauma
FB	foreign bodies
FDP	flexor digitorum profundus (tendon)
FDS	flexor digitorum superficialis (tendon)
FDT	forced duction test
FEIBA	factor B inhibitor bypass activity
FESS	functional endoscopic sinus surgery
FEV ₁	forced expiratory volume (in 1 sec)
FFP	fresh-frozen plasma
FHR	fetal heart rate
FIGO	International Federation of Gynecology and Obstetrics
FiO ₂	fraction of inspired oxygen
FNA	fine needle aspiration
FOB	fiberoptic bronchoscopy
FOI	fiberoptic intubation
FOL	fiberoptic laryngoscopy

Fr	French (size)
FRC	functional residual capacity
FSP	fibrin-split products
FTSG	full-thickness skin graft
FTT	failure to thrive
FVC	forced vital capacity
g	gram(s)
ga	gauge
GA	general anesthesia
GCS	Glasgow Coma Scale
GDT	goal-directed therapy
GE	gastroesophageal
GERD	gastroesophageal reflux disease
GETA	general endotracheal anesthesia
GFR	glomerular filtration rate
GGTP	gamma-glutamyl transpeptidase (liver enzyme)
GH	growth hormone
GI	gastrointestinal
GIA	gastrointestinal anastomosis
GIFT	gamete intrafallopian transfer
GM-CSF	granulocyte-macrophage colony-stimulating factor
Gn-RH	gonadotropin-releasing hormone
GSW	gunshot wound
GT	gastric tube
GTD	gestational trophoblastic disease
GU	genitourinary
h	hour(s)
HA	headache
H&P	history and physical examination
Hb	hemoglobin
Hb/Hct	hemoglobin/hematocrit

HbA	adult hemoglobin
HbF	fetal hemoglobin
HCC	hepatocellular carcinoma
HCG	human chorionic gonadotropin
HCO ₃	bicarbonate
Hct	hematocrit
HELLP	hemolysis, elevated liver enzymes, and low platelet count
HFNC	high-flow nasal cannula
HFV	high-frequency ventilation
Hg	mercury
HIV	human immunodeficiency virus
HLHS	hypoplastic left heart syndrome
HM	hyoid myotomy
Ho	holmium-YAG laser
HPV	hypoxic pulmonary vasoconstrictive (reflex)
HR	heart rate
HSV	highly selective vagotomy
HTN	hypertension
HVA	homovanillic acid
Hx	history

IABP	intra-aortic balloon pump
IBD	inflammatory bowel disease
IC	inspiratory capacity
ICD	implantable cardioverter-defibrillator
ICP	intracranial pressure
ICU	intensive care unit
ID	inside diameter
IE	inspiratory/expiratory
IgE	immunoglobulin E
IHSS	idiopathic hypertrophic subaortic stenosis
II	image intensifier
IIC	intermediate intensive care

IJ	internal jugular (vein)
im	intramuscular
IMA	internal mammary artery
IMF	intermaxillary fixation
iMRI	interventional magnetic resonance imaging
INH	isonicotinic acid hydrazide (isoniazid)
INR	international normalized ratio
IOFB	intraocular foreign body
IOL	intraocular lens
IOP	intraocular pressure
IORT	intraoperative radiation therapy
IPAA	ileal pouch and anastomosis
IPG	implantable generator
IPPV	intermittent positive pressure ventilation
IRDS	infant respiratory distress syndrome
ISS	injury severity score
IT	iliotibial
ITP	idiopathic thrombocytopenic purpura
IUP	intrauterine pressure
iv or IV	intravenous
IVC	inferior vena cava
IVF	in vitro fertilization
IVH	intraventricular hemorrhage
IVP	intravenous pyelogram
IVS	intact ventricular septum

J	Joule
JET	junctional ectopic tachycardia
JI	jejunoileal
JVD	jugular venous distension

K ⁺	potassium
KCl	potassium chloride

kg	kilogram(s)
KVO	keep vein open
L	liter
LA	left atrium or local anesthetic
LAD	left anterior descending
LAE	left atrial enlargement
LAP	left atrial pressure
LAVH	laparoscopy-assisted vaginal hysterectomy
LBBB	left bundle branch block
LDH	lactate dehydrogenase
LE	lower extremity
LEEP	loop electrosurgical excision procedure
LFT	liver function test
LGL	Lown-Ganong-Levine (syndrome)
LH-RH	luteinizing hormone–releasing hormone
LHSV	laparoscopic highly selective vagotomy
LIMA	left internal mammary artery
LINAC	linear accelerator
LLETZ	large loop excision of the transformation zone
LLD	left lateral decubitus
LLQ	left lower quadrant
LMA	laryngeal mask airway
LMG	laser midline glossectomy
LMP	last menstrual period
LMWH	low molecular weight heparin
LP	lumbar puncture
LR	lactated Ringer (solution)
LRD	living related donor
LSC	laparoscope approach
LTA	lidocaine tracheal anesthesia
LUQ	left upper quadrant
LV	left ventricle/ventricular

LVAD	left ventricular assist device
LVEDP	left ventricular end diastolic pressure
LVEF	left ventricular ejection fraction
LVH	left ventricular hypertrophy
LVOT	left ventricular outflow tract
LVOTO	left ventricular outflow tract obstruction
LVRS	lung volume reduction surgery
m	meter(s)
M	molar
MAC	monitored anesthesia care
MAGPI	meatal advancement glanuloplasty
MAO-B	monoamine oxidase-B
MAOI	monoamine oxidase inhibitor (antidepressant)
MAP	mean arterial pressure
MAPCAs	major aortopulmonary collaterals
MCA	middle cerebral artery
MCKD	medullary cystic kidney disease
MDI	multidirectional instability
MEA	multiple endocrine adenopathy
MED	microendoscopic lumbar discectomy
MEN	multiple endocrine neoplasia
MEP	motor-evoked potential
mEq	milliequivalent
mEqK ⁺	milliequivalency of potassium
metHb	methemoglobin
mg	milligram(s)
Mg ⁺⁺	magnesium
MgSO ₄	magnesium sulfate
MH	malignant hyperthermia
MI	myocardial infarction
MICAB	minimally invasive coronary artery bypass
MICABG	minimally invasive coronary artery bypass graft(ing)

µg	microgram(s)
MIDCAB	minimally invasive direct coronary artery bypass
MIDCABG	minimally invasive direct coronary artery bypass graft(ing)
MIF	maximum inspiratory flow
min	minute(s)
mL	milliliter(s)
MLT	microlaryngeal tube
mm	millimeter(s)
MMEF _{25–75}	maximum midexpiratory flow
mm Hg	millimeters of mercury
MMPM	multimodal pain management
mo	month(s)
MOGA	mandibular osteotomy and genioglossal advancement
MR	mitral regurgitation
MRA	magnetic resonance angiography
MRI	magnetic resonance imaging
MRSA	methicillin-resistant <i>Staphylococcus aureus</i>
MRT	magnetic resonance–guided therapy
MS	mitral stenosis
MTP	massive transfusion protocol
MUF	modified ultrafiltration
MUGA	multigated acquisition (scan)
MUPIT	Martinez Universal Perineal Interstitial Template
MV	minute ventilation; mitral valve
MVA	motor vehicle accident
MVC	motor vehicle collision
MVD	microvascular decompression
MVO ₂	mixed venous oxygen content
MVP	mitral valve prolapse
Na ⁺	sodium
NaHCO ₃	sodium bicarbonate
NB	nota bene (note well)

NDMR	nondepolarizing muscle relaxant
Nd:YAG	neodymium-YAG laser
NEC	necrotizing enterocolitis
ng	nanogram(s)
NG	nasogastric (tube)
NIBP	noninvasive blood pressure
NICU	neonatal intensive care unit
NIF	negative inspiratory force
nl	normal
NLD	nasolacrimal duct
NMB	neuromuscular blocker
NMR	nondepolarizing muscle relaxant
NO	nitric oxide
NOAC	new oral anticoagulants
NOE	naso-orbital ethmoid
NORA	non–operating room anesthesia
NPH	neutral protamine Hagedorn (insulin)
npo	nothing by mouth
NS	normal saline
NSAID	nonsteroidal anti-inflammatory drug
NS/LR	normal saline/lactated Ringer (solution)
NSR	normal sinus rhythm
NTE	not to exceed
NTG	nitroglycerin
NTP	nitroprusside
N ₂ O	nitrous oxide
N/V	nausea and vomiting
OA	osteoarthritis
OATS	osteochondral autograft transfer system
OCR	oculocardiac reflex
OD	outside diameter
OEIS	omphalocele exstrophy imperforate anus spinal defect

OG	orogastric (tube)
OLV	one-lung ventilation
OPCAB	off-pump coronary artery bypass
OPCABG	off-pump coronary artery bypass graft(ing)
OPLL	ossification of the posterior longitudinal ligament
OR	operating room
ORIF	open reduction and internal fixation
ORR	oculorespiratory reflex
OSA	obstructive sleep apnea
O ₂	oxygen
O ₂ sat	oxygen saturation

P	pulse
PA	pulmonary artery
PaCO ₂	partial pressure of CO ₂ (arterial)
PACU	postanesthesia care unit
PAD	pulmonary artery diastolic
PADP	pulmonary artery diastolic pressure
PAK	pancreas after kidney (transplant)
PAOP	pulmonary artery occlusion pressure
PaO ₂	partial pressure of oxygen (arterial)
PAP	pulmonary artery pressure
PAR	postanesthesia room
PAWP	pulmonary artery wedge pressure
PCA	patient-controlled analgesia
PCC	prothrombin complex concentrate
PCEA	patient-controlled epidural analgesia
PCL	posterior cruciate ligament
PCO ₂	partial pressure of carbon dioxide
PCV	parietal cell vagotomy
PCWP	pulmonary capillary wedge pressure
PDA	patent ductus arteriosus
PDE	phosphodiesterase

PDPH	postdural puncture headache
PE	pulmonary embolus
PEEP	positive end-expiratory pressure
PEF	peak expiratory flow
PET	positron emission tomography
PetCO ₂	end-tidal CO ₂ partial pressure
PFNS	preservative-free normal saline
PFO	patent foramen ovale
PFT	pulmonary function test
PGE	prostaglandin E
PICC	peripherally inserted central catheter
PICU	pediatric intensive care unit
PID	pelvic inflammatory disease
PIH	pregnancy-induced hypertension
PIP	peak inspiratory pressure
PKP	penetrating keratoplasty
PLIF	posterior lumbar interbody fusion
Plt	platelet
PND	paroxysmal nocturnal dyspnea
po	by mouth
POC	product of conception
POD	postoperative day
POEM	peroral endoscopic myotomy
PONV	postoperative nausea and vomiting
POOP	perineal one-stage operative pull-through
POPE	postobstructive pulmonary edema
PO ₂	partial pressure of oxygen
PPH	postpartum hemorrhage
PPI	proton pump inhibitor
ppm	parts per million
PPTL	postpartum tubal ligation
PPV	positive pressure ventilation or pulse pressure variation
pr	per rectum

PRBC	packed red blood cell
prn	as needed
PS	pulmonic stenosis
PSARP	perineal sagittal anorectoplasty
PSSD	postsurgical stress disorder
PT	prothrombin time
PTCA	percutaneous transluminal coronary angioplasty
PTM	posterior tibial muscle
PTS	posttourniquet syndrome
PTSD	posttraumatic stress disorder
PTT	partial thromboplastin time
PUD	peptic ulcer disease
PUV	posterior urethral valve
PVA	polyvinyl alcohol
PVB	paravertebral block
PVC	premature ventricular contraction
PVD	peripheral vascular disease
PVI	pleth variability index
PVOD	pulmonary vascular occlusive disease
PVR	pulmonary vascular resistance

q	query
qd	every day
Q_p/Q_s	pulmonary-to-systemic flow ratio
QRS	QRS complex of ECG
qs	quantum satis (sufficient quantity)

RA	radial artery
RAD	right axis deviation
RAE	right atrial enlargement
RAP	right atrial pressure
RAST	radioallergosorbent test
RBBB	right bundle branch block

RBC	red blood cell
RC	rotator cuff
RCA	right coronary artery
RCP	retrograde cerebral perfusion
RCT	randomized controlled trial
RDS	respiratory distress syndrome
RF	radiofrequency
RFA	radiofrequency ablation
RFT	renal function test
RIND	reversible ischemic neurologic deficit
RLD	right lateral decubitus
RLQ	right lower quadrant
r/o	rule out
ROM	range of motion
ROP	retinopathy of prematurity
RR	respiratory rate
RSD	reflex sympathetic dystrophy (obsolete)
RSI	rapid-sequence induction
RSV	respiratory syncytial virus
RUE	right upper extremity
RUQ	right upper quadrant
RV	right ventricle
RVE	right ventricular enlargement
RVEDP	right ventricular end-diastolic pressure
RVESV	right ventricular end-systolic volume
RVH	right ventricular hypertrophy
RVOT	right ventricular outflow tract
RVOTO	right ventricular outflow tract obstruction
Rx	treatment
SA	sinoatrial (node)
SAB	subarachnoid block
SAD	subacromial decompression

SAH	subarachnoid hemorrhage
SAM	systolic anterior motion
SBE	subacute bacterial endocarditis
SBO	small bowel obstruction
SBP	systolic blood pressure
sc	subcutaneous
SCD	sequential compression device
SCFE	slipped capital femoral epiphysis
SCT	sacroccygeal teratoma
sec	second(s)
SGOT	serum glutamic-oxaloacetic transaminase
SGPT	serum glutamate-pyruvate transaminase
SI	sacroiliac
SIADH	syndrome of inappropriate antidiuretic hormone
SICU	surgical intensive care unit
sl	sublingual
SLAP	superior labral anterior to posterior
SLE	systemic lupus erythematosus
SLH	subtotal laparoscopic hysterectomy
SLT	single-lung transplant or single-lumen tube
SMA	superior mesenteric artery
SMV	superior mesenteric vein
SNP	sodium nitroprusside
SOB	short(ness) of breath
S/P	status post
SPK	simultaneous kidney/pancreas (transplant)
SpO ₂	oxygen saturation measured by pulse oximetry
SPV	systolic pressure variation
SSA	sickle cell anemia
SSEP	somatosensory evoked potential
SSI	surgical site infection
SSRI	selective serotonin reuptake inhibitor
SSS	sick sinus syndrome

S-T	wave of ECG
ST-T	ST-T wave of ECG
STA	superficial temporal artery
STP	sodium thiopental
STSG	split-thickness skin graft
SV	stroke volume
SVC	superior vena cava
S _v O ₂	venous O ₂ saturation
SVR	systemic vascular resistance
SVT	supraventricular tachycardia
Sx	signs and symptoms
Sz	seizure
T	temperature
T&C	type and cross-match
T&S	type and screen
T/A	tonsillectomy/adenoidectomy
TAAA	thoracoabdominal aortic aneurysm
TAB	therapeutic abortion
TACO	transfusion-associated circulatory overload
TAH	total abdominal hysterectomy
TAPVC	total anomalous pulmonary venous connection
TB	tuberculosis
TBI	total body irradiation
TBSA	total body surface area
TCA	tricyclic antidepressant
TCD	transcranial Doppler
TE	tangential excision
TEA	thromboendarterectomy
TECAB	totally endoscopic coronary artery bypass
TEE	transesophageal echocardiogram (graphy)
TEF	tracheoesophageal fistula
TEG	thromboelastogram

TET	tubal embryo transfer
TEVAR	thoracic endovascular aortic repair
TGA	transposition of the great arteries
TGV	transposition of the great veins
THRIVE	transnasal humidified rapid-insufflation ventilatory exchange
TIA	transient ischemic attack
tid	3 times per day
TIPS	transjugular intrahepatic portosystemic shunt
TIVA	total intravenous anesthesia
TKO	to keep open
TL	thoracolumbar
TLC	total lung capacity
TLH	total laparoscopic hysterectomy
TLIF	transforaminal lumbar interbody fusion
TMJ	temporomandibular joint
TMJD	temporomandibular joint dysfunction
TOF	tetralogy of Fallot, train of force
tPA	tissue plasminogen activator
TPN	total parenteral nutrition
TR	tricuspid regurgitation
TRALI	transfusion-related acute lung injury
TRAM	transverse rectus abdominis muscle (flap)
TSA	total shoulder arthroplasty
TSC	tethered spinal cord
TSH	thyroid-stimulating hormone
TT	thrombin time
TTE	transthoracic echocardiography
TTP	thrombotic thrombocytopenic purpura
TUMT	transurethral microwave thermotherapy
TUNA	transurethral needle ablation
TUR	transurethral resection
TURBT	transurethral resection of bladder tumor
TURP	transurethral resection of the prostate

TUVP	transurethral vaporization of the prostate
TV	tidal volume
TVOR	transvaginal oocyte retrieval
TVT	tension-free vaginal tape procedure
TW airway	Tudor-Williams airway
TWR	total wrist replacement
Tx	transplant
U	unit(s)
UA	urinalysis
UAC	umbilical artery catheter
UBC	unicameral bone cyst
UC	ulcerative colitis
UE	upper extremity
UO	urine output
UPJ	ureteropelvic junction
UPPP	uvulopalatopharyngoplasty
URI	upper respiratory infection
US	ultrasound
UTI	urinary tract infection
UVC	umbilical vein catheter
UW	University of Wisconsin (solution)
V&A	vagotomy and antrectomy
V&P	vagotomy and pyloroplasty
VA	ventriculoatrial
VAE	venous air embolism
VAGM	vein of Galen malformation
VALH	vaginally assisted laparoscopic hysterectomy
VAT	video-assisted thoracoscopy
VATS	video-assisted thoracoscopic surgery
VBG	vertical banded gastroplasty
VC	vital capacity

Vd	volume of distribution
VF	ventricular fibrillation
VIP	vasoactive intestinal peptide
VIPoma	vasoactive intestinal peptide-secreting tumor
VLAP	visual laser ablation of the prostate
VMA	vanillylmandelic acid
VO ₂	oxygen consumption
VP	ventriculoperitoneal
VPI	velopharyngeal insufficiency
V/Q	ventilation-perfusion ratio
VRE	vancomycin-resistant enterococci
VS	vital signs
VSD	ventricular septal defect
VT	ventricular tachycardia
VUR	vesicoureteral reflux
WBC	white blood cell
WPW	Wolff-Parkinson-White (syndrome)
W/U	workup
XRT	x-ray therapy
yr	year(s)

SUBJECT INDEX

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